

AZURE CLOUD SOLUTIONS DEPLOYMENT AND ADMINISTRATION PROJECT

PREPARED BY
DHIVYA LAKSHMI S



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 - Create an Action Group
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TASK 01: DISK SNAPSHOT MANAGEMENT AND VM PROVISIONING

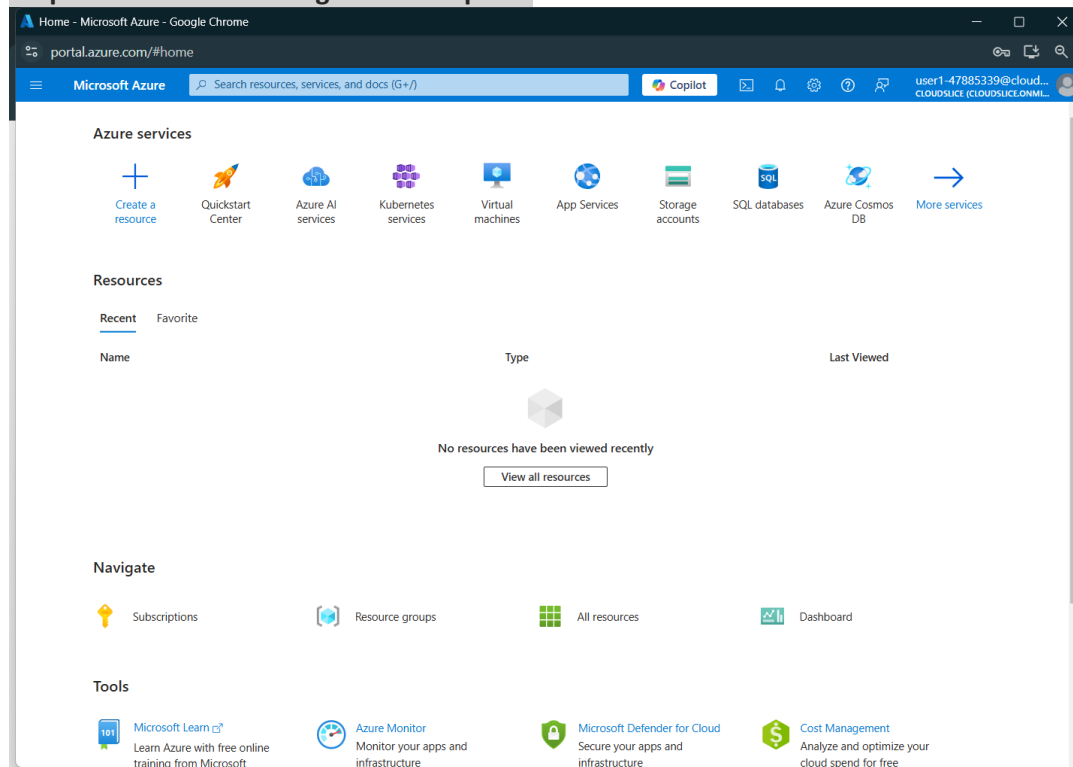
Objective:

Create and manage snapshots of Azure Managed Disks to enable point-in-time backups and demonstrate disaster recovery capabilities.

Steps Performed:

- Created a snapshot of an existing managed disk using the Azure Portal.
- Generated a new managed disk from the snapshot to replicate the original disk's state.
- Provisioned a virtual machine using the snapshot-based disk to simulate recovery.
- Verified that the VM booted successfully and data integrity was maintained.

Prepare to create a managed disk snapshot



VM1 - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/a57df68e-a26a-4473-91d2-97aff58ce4ea/resourceGroups/rg1lod47885339/providers/...

Microsoft Azure Search resources, services, and docs (G+/)

user1-47885339@cloud... CLOUDSLICE (CLOUDSLICEONM...

Home > Virtual machines >

VM1 Virtual machine

Search Help me copy this VM in any region

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Connect

Networking

Settings

Availability + scale

Security

Backup + disaster recovery

Operations

Monitoring

Automation

Help

Essentials

Resource group (move) rg1lod47885339

Status Running

Location East US

Subscription (move) Challenge Labs 04

Subscription ID a57df68e-a26a-4473-91d2-97aff58ce4ea

Tags (edit) Add tags

JSON View

Operating system Linux (ubuntu 16.04)

Size Standard DS1 v2 (1 vcpu, 3.5 GiB memory)

Public IP address 172.190.167.2

Virtual network/subnet VM1-vnet/subnet

DNS name Not configured

Health state -

Time created 1/31/2025, 12:26 AM UTC

Properties Monitoring Capabilities (7) Recommendations Tutorials

Virtual machine

Computer name VM1

Operating system Linux (ubuntu 16.04)

VM generation V1

VM architecture x64

Networking

Public IP address 172.190.167.2 (Network interface VM1-nic)

Public IP address (IPv6) -

Private IP address 10.0.0.4

Private IP address (IPv6) -

Auto-scale web application

Not secure 172.190.167.2

Gmail YouTube Maps News Translate Download Success [...] Home | Cloud: Data... Offer Details | Appli... Amazon High Volu... Perplexit

Auto-scale web application

VM1 - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/a57df68e-a26a-4473-91d2-97aff58ce4ea/resourceGroups/rg1lod47885339/providers/...

Microsoft Azure Search resources, services, and docs (G+)

user1-47885339@cloud... CLOUDSLICE (CLOUDSLICE.ONMICROSOFT.COM)

Home > Virtual machines >

VM1 Virtual machine

Search Help me copy this VM in any region

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Connect

Networking

Settings

Availability + scale

Security

Backup + disaster recovery

Operations

Monitoring

Automation

Help

Essentials

Resource group (move) rg1lod47885339

Status Stopped (deallocated)

Location East US

Subscription (move) Challenge Labs 04

Subscription ID a57df68e-a26a-4473-91d2-97aff58ce4ea

Tags (edit) Add tags

Operating system Linux

Size Standard DS1 v2 (1 vcpu, 3.5 GiB memory)

Public IP address VM1-ip

Virtual network/subnet VM1-vnet/subnet

DNS name Not configured

Health state -

Time created 1/31/2025, 12:26 AM UTC

JSON View

Properties Monitoring Capabilities (7) Recommendations Tutorials

Virtual machine

Computer name VM1

Operating system Linux

VM generation V1

VM architecture x64

Networking

Public IP address VM1-ip (Network interface VM1-nic)

Public IP address (IPv6) -

Private IP address 10.0.0.4

Private IP address (IPv6) -

Create a snapshot of an existing managed disk

rg1lod47885339 - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/a57df68e-a26a-4473-91d2-97aff58ce4ea/resourceGroups/rg1lod47885339/overview

Microsoft Azure Search resources, services, and docs (G+)

user1-47885339@cloud... CLOUDSLICE (CLOUDSLICE.ONMICROSOFT.COM)

Home > Resource groups >

rg1lod47885339 Resource group

Search Create Manage view Delete resource group Refresh Export to CSV Open query Assign tags Move

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Events

Settings

Cost Management

Cost analysis

Cost alerts (preview)

Budgets

Advisor recommendations

Monitoring

Automation

Help

Essentials

Subscription (move) Challenge Labs 04

Subscription ID a57df68e-a26a-4473-91d2-97aff58ce4ea

Location East US

Deployments 1 Succeeded

Tags (edit)

LODManaged : lod LabProfile : 57369 More (5)

Resources Recommendations

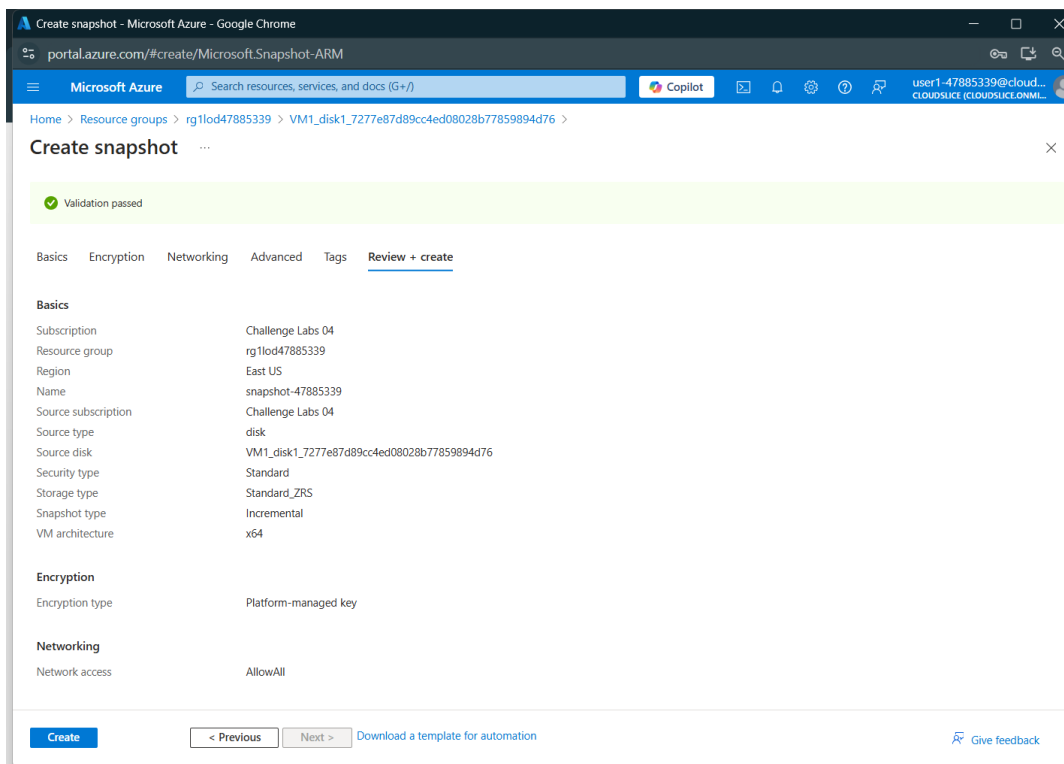
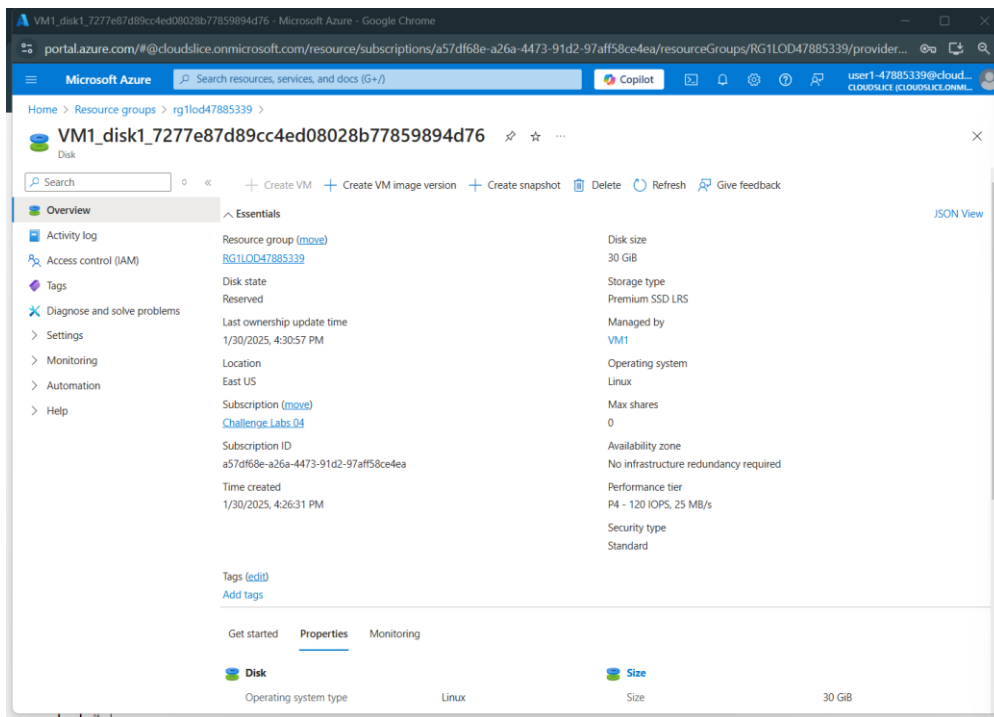
Filter for any field... Type equals all Location equals all Add filter

Showing 1 to 6 of 6 records. Show hidden types No grouping List view

Name	Type	Location
VM1	Virtual machine	East US
VM1-ip	Public IP address	East US
VM1-nic	Network interface	East US
VM1-nsg	Network security group	East US
VM1-vnet	Virtual network	East US
VM1_disk1_7277e67d89cc4ed08028b77859894d76	Disk	East US

< Previous Page 1 of 1 Next >

Give feedback



snapshot-47885339 - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/a57df68e-a26a-4473-91d2-97aff58ce4ea/resourceGroups/rg1lod47885339/providers/...

Microsoft Azure Search resources, services, and docs (G+)

user1-47885339@cloud... CLOUDSLICE (CLOUDSLICE.ONMICROSOFT.COM)

Home > Snapshot.snapshot-47885339-20250130163619 | Overview >

snapshot-47885339

Snapshot

Search

+ Create disk + Create VM image version Copy snapshot Delete Refresh Give feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Settings

Automation

Help

Essentials

Resource group (move) [rg1lod47885339](#)

Location East US

Subscription (move) [Challenge Labs 04](#)

Subscription ID a57df68e-a26a-4473-91d2-97aff58ce4ea

Snapshot state Unattached

Tags (edit) Add tags

Created 1/30/2025, 4:36:24 PM

Storage type Zone-redundant

Source VM1_disk1_7277e87d89cc4ed08028b77859894d76

Size 30 GiB

Encryption Platform-managed key

Get started Properties

Snapshot		Size	
Name	snapshot-47885339	Size	30 GiB
Snapshot type	Incremental	Storage type	Zone-redundant
VM generation	Gen 1		
Completion percent	100		
Provisioning state	Succeeded	Security type	Standard
VM architecture	x64		
Hibernation supported	-		

VM1 - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/a57df68e-a26a-4473-91d2-97aff58ce4ea/resourceGroups/rg1lod47885339/providers/...

Microsoft Azure Search resources, services, and docs (G+)

user1-47885339@cloud... CLOUDSLICE (CLOUDSLICE.ONMICROSOFT.COM)

Home > Virtual machines >

VM1

Virtual machine

Search

Help me copy this VM in any region

Connect Start Restart Stop Hibernate Capture Delete Refresh Open in mobile Feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Connect

Networking

Settings

Availability + scale

Security

Backup + disaster recovery

Operations

Monitoring

Automation

Help

Essentials

Resource group (move) [rg1lod47885339](#)

Status Running

Location East US

Subscription (move) [Challenge Labs 04](#)

Subscription ID a57df68e-a26a-4473-91d2-97aff58ce4ea

Operating system Linux

Size Standard D51 v2 (1 vcpu, 3.5 GiB memory)

Public IP address [172.190.166.112](#)

Virtual network/subnet [VM1-vnet/subnet](#)

DNS name [Not configured](#)

Health state -

Time created 1/31/2025, 12:26 AM UTC

Tags (edit) Add tags

Properties Monitoring Capabilities (7) Recommendations Tutorials

Virtual machine		Networking	
Computer name	VM1	Public IP address	172.190.166.112 (Network interface VM1-)
Operating system	Linux	Public IP address (IPv6)	-
VM generation	V1	Private IP address	10.0.0.4
VM architecture	x64		

Create a managed disk from a snapshot

snapshot-47885339 · Microsoft Azure · Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/a57df68e-a26a-4473-91d2-97aff58ce4ea/resourceGroups/rg1lod47885339/providers/...

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

user1-47885339@cloud...
CLOUDSLICE (CLOUDSLICE.ONMICROSOFT.COM)

Home > Snapshots >

snapshot-47885339

Snapshot

Search

«

+ Create disk + Create VM image version Copy snapshot Delete Refresh Give feedback

Overview

Activity log Access control (IAM) Tags Diagnose and solve problems Settings Automation Help

Essentials

Resource group (move)
rg1lod47885339

Location
East US

Subscription (move)
Challenge Labs 04

Subscription ID
a57df68e-a26a-4473-91d2-97aff58ce4ea

Snapshot state
Unattached

Tags (edit)
Add tags

Created
1/30/2025, 4:36:24 PM

Storage type
Zone-redundant

Source
VM1_disk1_7277e87d89cc4ed0802bb77859894d76

Size
30 GiB

Encryption
Platform-managed key

JSON View

Get started Properties

Snapshot

Name
snapshot-47885339

Snapshot type
Incremental

VM generation
Gen 1

Completion percent
100

Provisioning state
Succeeded

VM architecture
x64

Hibernation supported
-

Size

Size
30 GiB

Storage type
Zone-redundant

Security type

Security type
Standard

Create a managed disk - Microsoft Azure · Google Chrome

portal.azure.com/#create/Microsoft.ManagedDisk-ARM

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

user1-47885339@cloud...
CLOUDSLICE (CLOUDSLICE.ONMICROSOFT.COM)

Home > Snapshots > snapshot-47885339 >

Create a managed disk

Validation passed

Basics Encryption Networking Advanced Tags Review + create

Basics

Subscription
Challenge Labs 04

Resource group
rg1lod47885339

Region
East US

Disk name
VM2_OsDisk_1_47885339

Availability zone
1

Source type
Snapshot

Source subscription
Challenge Labs 04

Source snapshot
snapshot-47885339

OS type
Linux

Security type
Standard

VM generation
V1

VM architecture
x64

Size

Size
30 GiB

Performance tier
P4 - 120 IOPS, 25 MB/s (default)

Storage type
Premium SSD LRS

Encryption

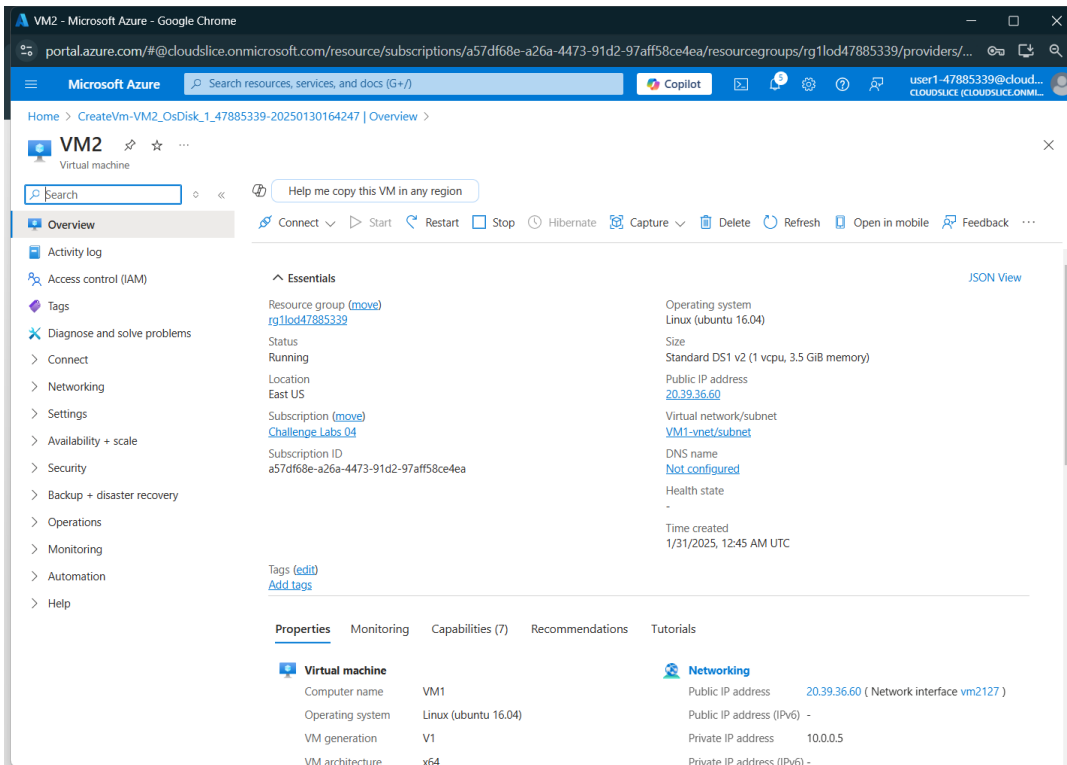
Create

< Previous

Next >

Download a template for automation

Give feedback



Outcome:

The process validated the effectiveness of Azure Managed Disk snapshots for disaster recovery and environment duplication. The successful boot confirmed that all configurations and data were preserved accurately, ensuring business continuity in case of failures.

TASK 02 – Configure a Virtual Machine Scale Set

Objective

Deploy and manage an Azure Virtual Machine Scale Set (VMSS) to **automatically scale virtual machines** based on workload demands, demonstrating elasticity and operational control.

Steps Performed:

- Created a VM scale set with a defined number of identical virtual machines.
- Increased the number of instances in the scale set to accommodate higher load.
- Deallocated a specific VM instance to optimize resource usage.

portal.azure.com/#view/Microsoft_Azure_Compute/AddNewNetworkInterfaceBladeV2/virtualMachineName/webappscaleret/location~/9%7B%27id%3A%27Fsubsc...

Microsoft Azure

Search resources, services, and docs (G+V)

Copilot

user1-47886387@cloud...
CLOUDSLICE (CLOUDSLICE.ONMML)

Home > Virtual machine scale sets > Create a Virtual Machine Scale Set (VMSS) >

Edit network interface

Network interface

Name *

vnet-eastus-nic01

Virtual network ⓘ

vnet-eastus

Subnet * ⓘ

snnet-eastus-1 (172.16.0.0/24)

NIC network security group ⓘ

☐ None

☒ Basic

☐ Advanced

Public inbound ports * ⓘ

☒ None

☐ Allow selected ports

Select inbound ports

Select one or more ports

1

All traffic from the internet will be blocked by default. You will be able to change inbound port rules in the VM > Networking page.

Public IP address ⓘ

OK

Cancel

Give feedback

Create a Virtual Machine Scale Set (VMSS) - Microsoft Azure - Google Chrome

portal.azure.com/#create/microsoft.vms

Microsoft Azure Search resources, services, and docs (G+)

Copilot

user1-47886387@cloud... CLOUDSLICE (CLOUDSLICE.ONML...)

Home > Virtual machine scale sets >

Create a Virtual Machine Scale Set (VMSS)

Validation passed

Basics Spot Disks Networking Management Health Advanced Tags Review + create

Basics

Subscription	Challenge Labs 06
Resource group	rg1lod47886387
Virtual machine scale set name	webappsscaleset
Region	East US
Orchestration mode	Uniform
Availability zone	None
Image	Ubuntu Server 24.04 LTS - Gen2
Size	Standard_DS1_v2 (1 vcpu, 3.5 GiB memory)
Scaling mode	Manually update the capacity
Instance count	2
Security type	Trusted launch virtual machines
Enable secure boot	Yes
Enable vTPM	Yes
Integrity monitoring	No
Enable Hibernation	No
Authentication type	Password
Username	azureuser

Spot

< Previous Next > Create

Download a template for automation Give feedback

webappsscaleset - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/d28df0a3-5e54-471f-b9ce-df15fa1a8ff3/resourcegroups/rg1lod47886387/providers/...

Microsoft Azure Search resources, services, and docs (G+)

Copilot

user1-47886387@cloud... CLOUDSLICE (CLOUDSLICE.ONML...)

Home > CreateVms-canonical.ubuntu-24_04-lts-server-20250130172532 | Overview >

webappsscaleset

Virtual machine scale set

Search

Move Start Restart Stop Hibernate Reimage Delete Refresh Feedback

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Instances
- Networking
- Settings
- Availability + scale
- Security
- Operations
- Monitoring
- Automation
- Help

Essentials

Resource group (move)	rg1lod47886387	Operating system	Linux
Status	2 out of 2 succeeded	Size	Standard_DS1_v2 (2 instances)
Location	East US	Public IP address	172.190.20.179
Subscription (move)	Challenge Labs 06	Public IP address (IPv6)	-
Subscription ID	d28df0a3-5e54-471f-b9ce-df15fa1a8ff3	Virtual network/subnet	vnet-eastus1/subnet-eastus-1
Tags (edit)	Add tags	Orchestration mode	Uniform
		Time created	1/31/2025, 1:31 AM UTC

Properties Monitoring Capabilities (6) Recommendations Tutorials

Virtual machine profile

Operating system	Linux
Capacity reservation group	-
Hibernation	Disabled

Azure Spot

Azure Spot	Disabled
------------	----------

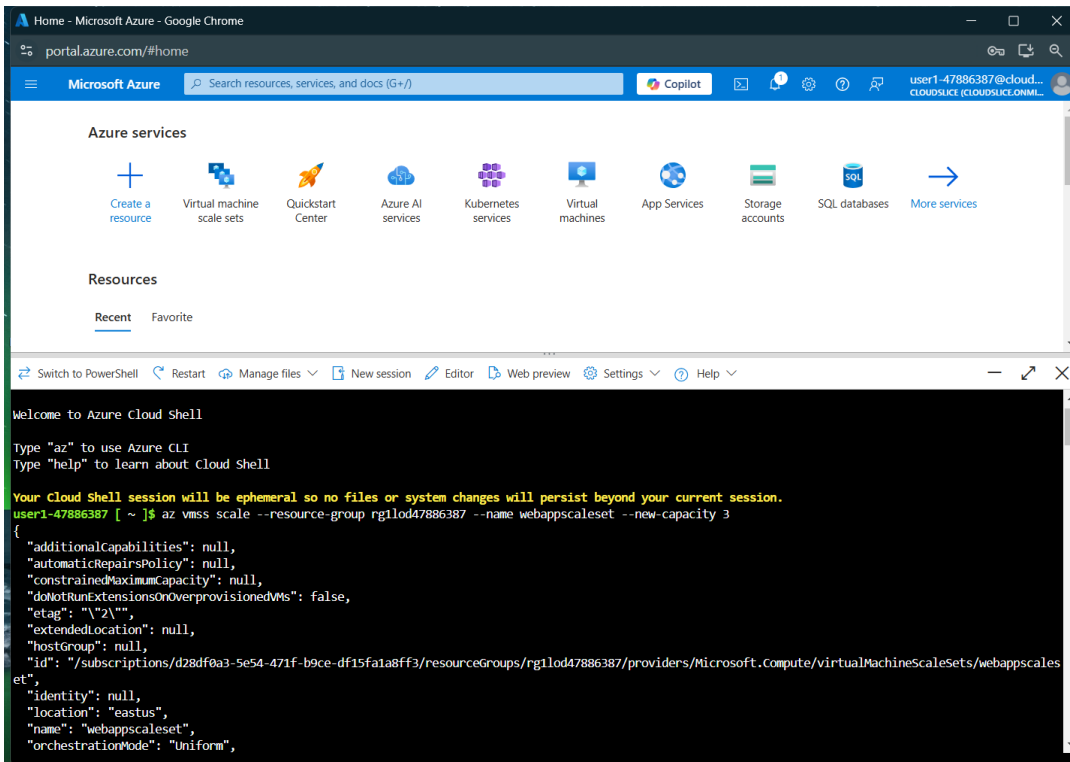
Status

Provisioning state	Succeeded
Power state	2 out of 2 running

Networking

Public IP address	172.190.20.179
Public IP address (IPv6)	-

Increase the number of virtual machine instances in a scale set

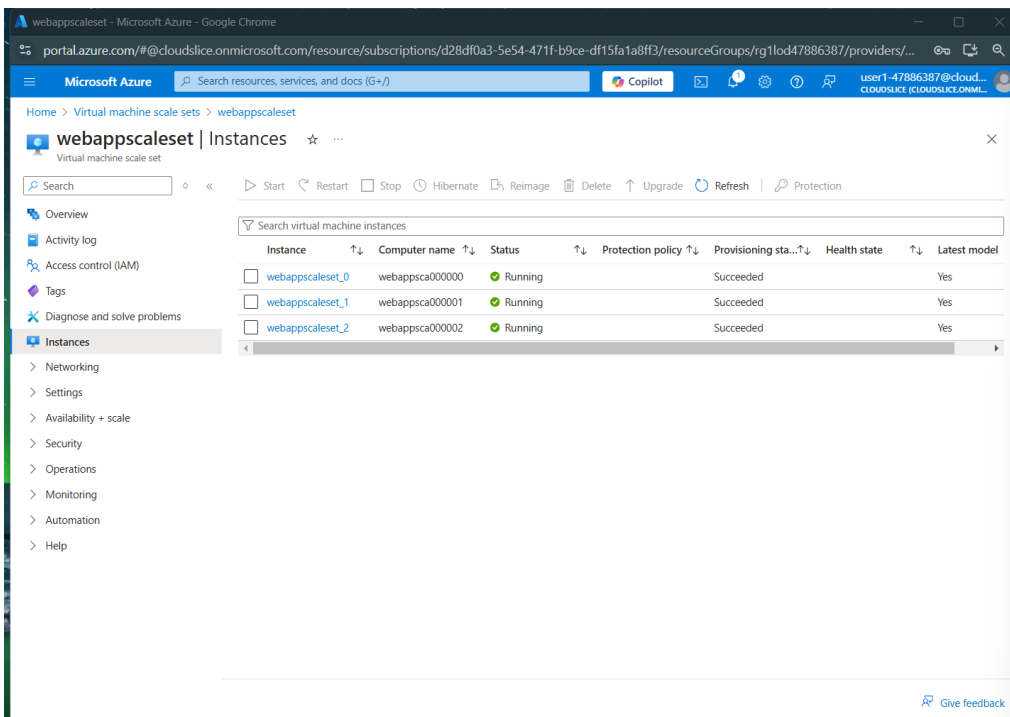


The screenshot shows the Microsoft Azure portal in Google Chrome. The top navigation bar includes the Azure logo, a search bar, and the user profile. Below the navigation bar, there are sections for 'Azure services' and 'Resources'. The 'Azure services' section contains icons for various services like Virtual machine scale sets, Quickstart Center, Azure AI services, Kubernetes services, Virtual machines, App Services, Storage accounts, and SQL databases. The 'Resources' section has tabs for 'Recent' and 'Favorite'. Below the portal, the Azure Cloud Shell is open, displaying a terminal window. The terminal shows the command to increase the capacity of a virtual machine scale set.

```
Welcome to Azure Cloud Shell

Type "az" to use Azure CLI
Type "help" to learn about Cloud Shell

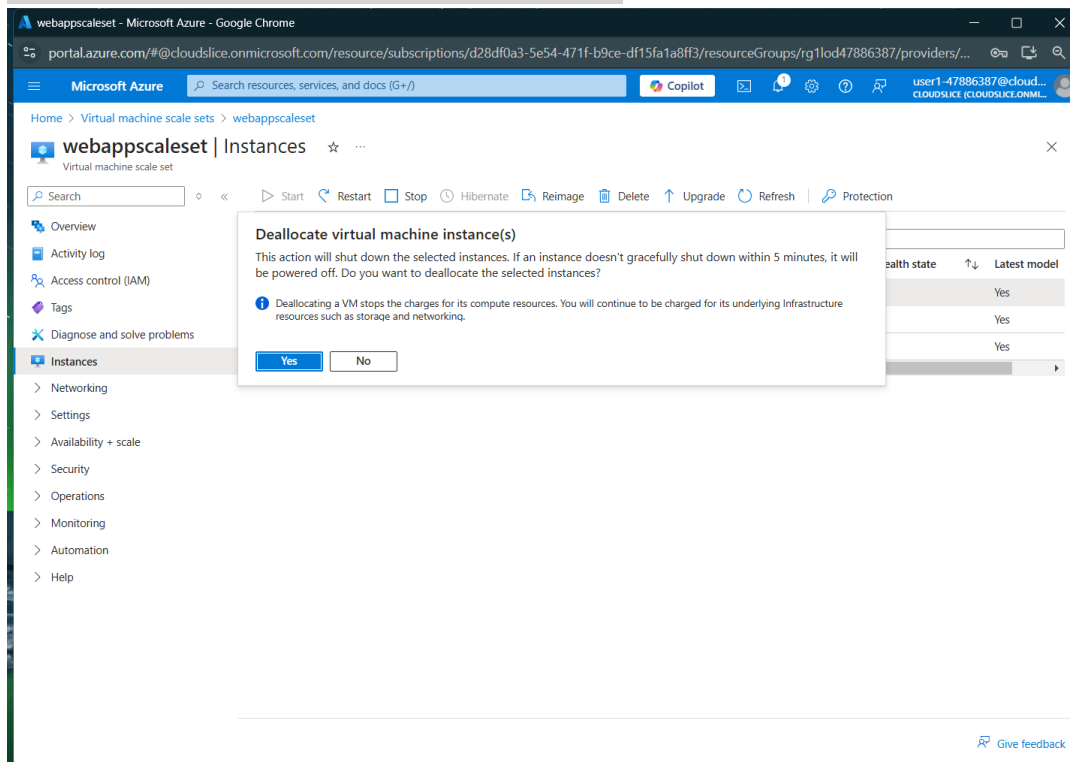
Your Cloud Shell session will be ephemeral so no files or system changes will persist beyond your current session.
user1-47886387 [ ~ ] $ az vmss scale --resource-group rg1lod47886387 --name webappsscaleset --new-capacity 3
{
  "additionalCapabilities": null,
  "automaticRepairsPolicy": null,
  "constrainedMaximumCapacity": null,
  "doNotRunExtensionsOnOverprovisionedVMs": false,
  "etag": "\"\"",
  "extendedLocation": null,
  "hostGroup": null,
  "id": "/subscriptions/d28df0a3-5e54-471f-b9ce-df15fa1a8ff3/resourceGroups/rg1lod47886387/providers/Microsoft.Compute/virtualMachineScaleSets/webappsscaleset",
  "identity": null,
  "location": "eastus",
  "name": "webappsscaleset",
  "orchestrationMode": "Uniform",
```



The screenshot shows the Azure portal interface for a virtual machine scale set named 'webappsscaleset'. The 'Instances' tab is selected, displaying a table of three instances. The table columns include Instance, Computer name, Status, Protection policy, Provisioning state, Health state, and Latest model. All three instances are in a 'Running' status with a 'Succeeded' provisioning state and a 'Yes' health state.

Instance	Computer name	Status	Protection policy	Provisioning state	Health state	Latest model
webappsscaleset_0	webappssca000000	Running		Succeeded	Yes	Yes
webappsscaleset_1	webappssca000001	Running		Succeeded	Yes	Yes
webappsscaleset_2	webappssca000002	Running		Succeeded	Yes	Yes

Deallocate a virtual machine instance in a scale set



webappscaleset - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/d28df0a3-5e54-471f-b9ce-df15fa1a8ff3/resourceGroups/rg1lod47886387/providers/...

Microsoft Azure Search resources, services, and docs (G+/I) Copilot user1-47886387@cloud... CLOUDSLICE (CLOUDSLICE.ONMICROSOFT.COM)

Home > Virtual machine scale sets > webappscaleset

webappscaleset | Instances ☆ ...

Virtual machine scale set

Search Start Restart Stop Hibernate Reimage Delete Upgrade Refresh Protection

Overview Activity log Access control (IAM) Tags Diagnose and solve problems Instances

Networking Settings Availability + scale Security Operations Monitoring Automation Help

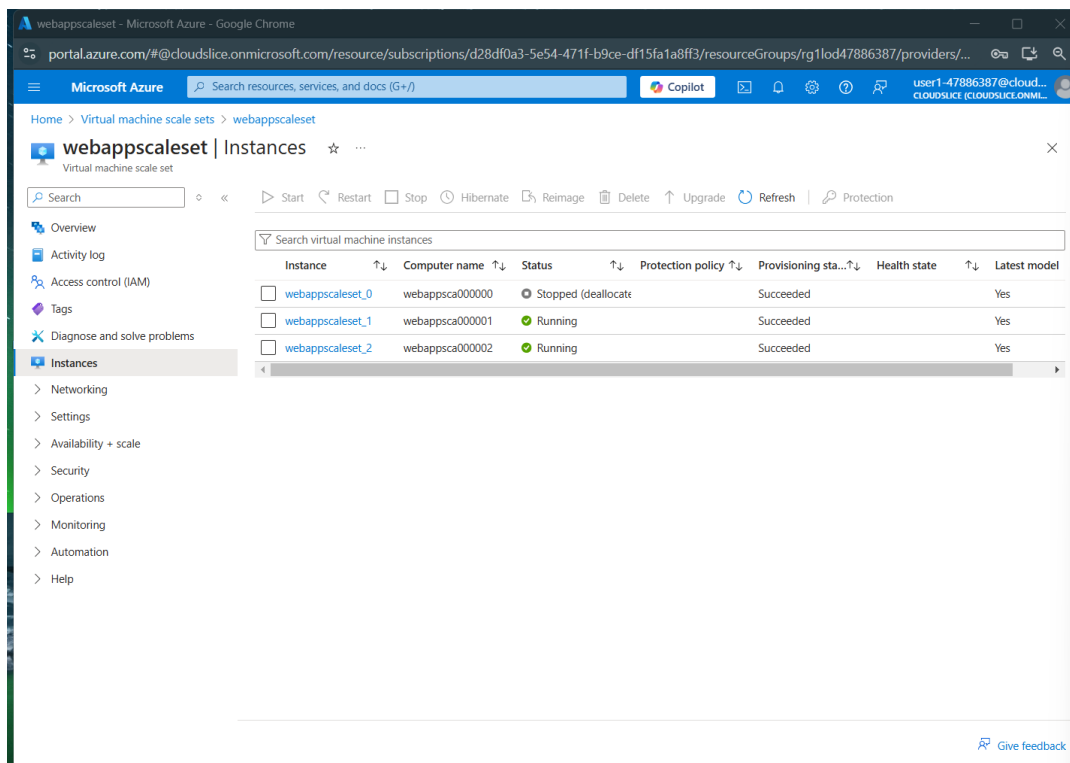
Deallocate virtual machine instance(s)

This action will shut down the selected instances. If an instance doesn't gracefully shut down within 5 minutes, it will be powered off. Do you want to deallocate the selected instances?

Deallocating a VM stops the charges for its compute resources. You will continue to be charged for its underlying Infrastructure resources such as storage and networking.

Yes No

Give feedback



webappscaleset - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/d28df0a3-5e54-471f-b9ce-df15fa1a8ff3/resourceGroups/rg1lod47886387/providers/...

Microsoft Azure Search resources, services, and docs (G+/I) Copilot user1-47886387@cloud... CLOUDSLICE (CLOUDSLICE.ONMICROSOFT.COM)

Home > Virtual machine scale sets > webappscaleset

webappscaleset | Instances ☆ ...

Virtual machine scale set

Search Start Restart Stop Hibernate Reimage Delete Upgrade Refresh Protection

Overview Activity log Access control (IAM) Tags Diagnose and solve problems Instances

Networking Settings Availability + scale Security Operations Monitoring Automation Help

Search virtual machine instances

Instance	Computer name	Status	Protection policy	Provisioning sta...	Health state	Latest model
☐ webappscaleset_0	webappscsca000000	Stopped (deallocate)		Succeeded	Yes	Yes
☐ webappscaleset_1	webappscsca000001	Running		Succeeded	Yes	Yes
☐ webappscaleset_2	webappscsca000002	Running		Succeeded	Yes	Yes

Give feedback

Outcome:

Successfully created a Virtual Machine Scale Set, demonstrated manual scaling by increasing instance count, and performed deallocation of an individual VM. This showcases the ability to dynamically adjust compute resources and manage workloads efficiently.

TASK 03 – Provision an Azure Container Instance

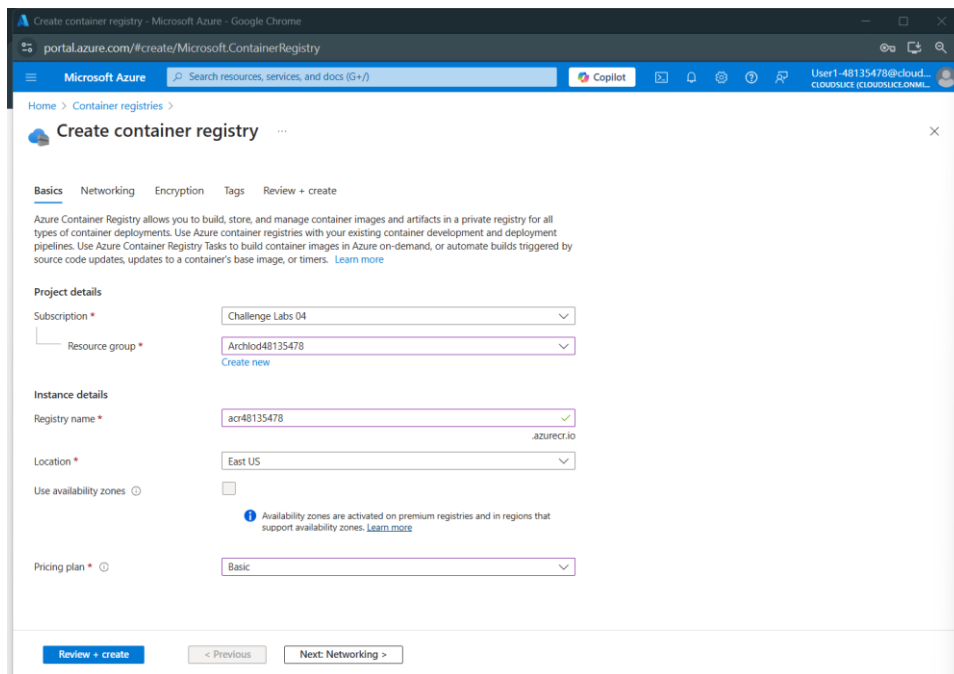
Objective:

Provision an Azure Container Registry and Azure Container Instance, then deploy a container-based web application to demonstrate modern containerized workloads without managing virtual machines.

Steps Performed:

- Created an Azure Container Registry as a secure repository for Docker images.
- Deployed an Azure Container Instance from the private registry, demonstrating rapid container deployment without managing VMs.
- Provisioned a web app powered by a container image to showcase a modern microservices architecture.

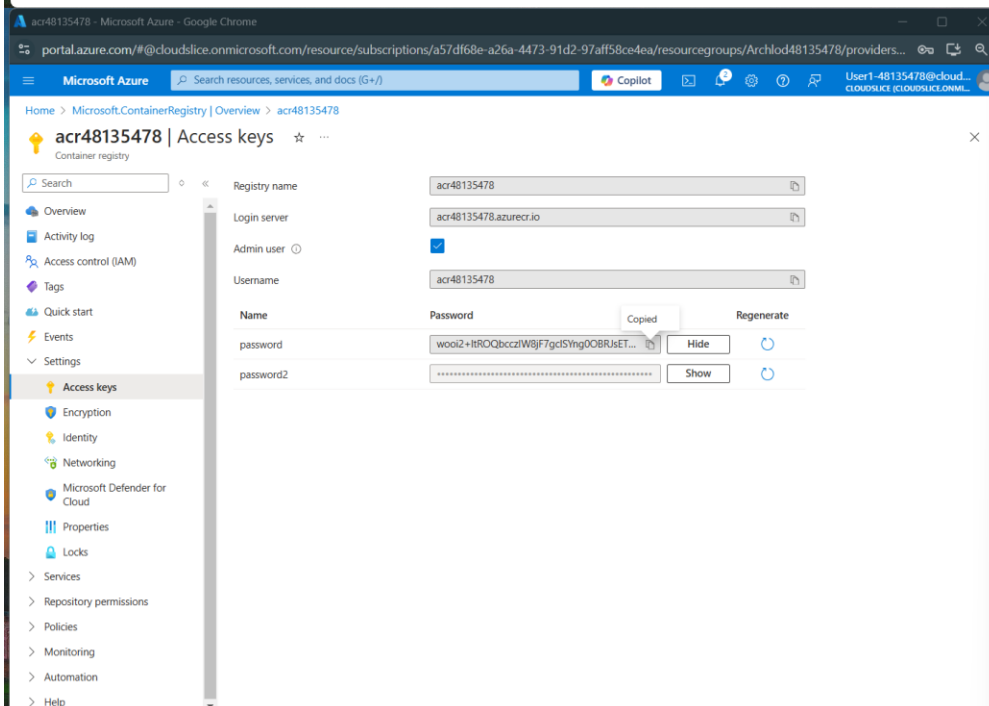
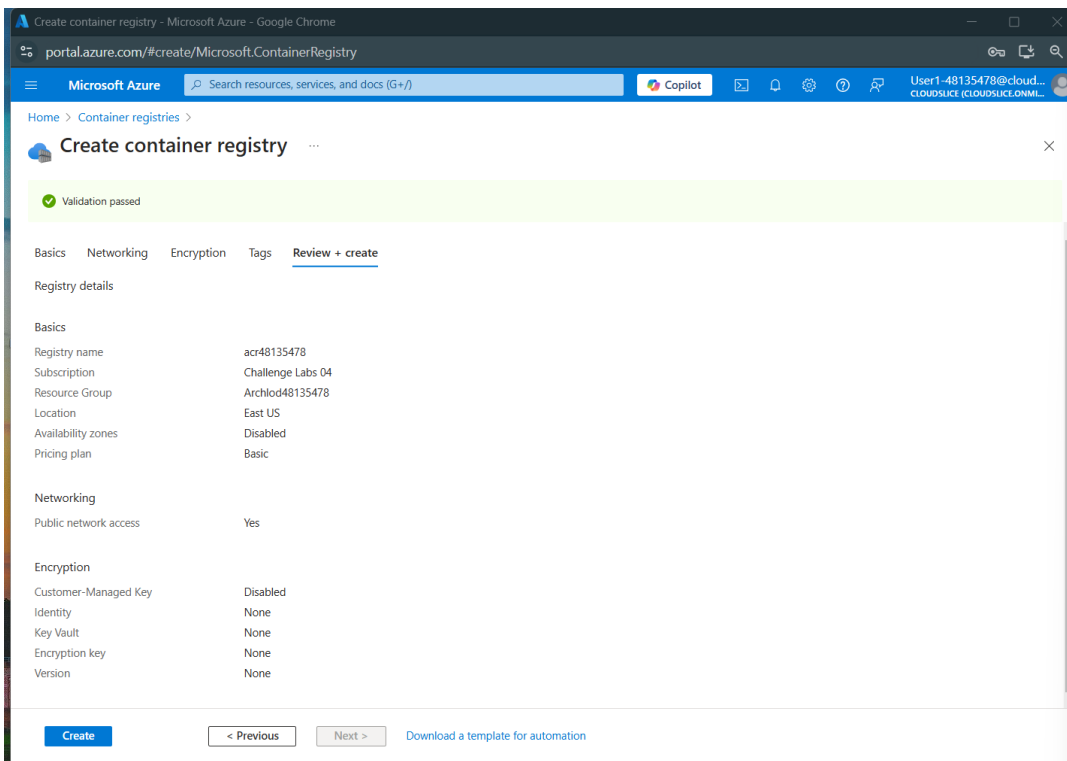
Provision an Azure Container Registry



The screenshot shows the 'Create container registry' page in the Microsoft Azure portal. The page is titled 'Create container registry' and has a breadcrumb trail 'Home > Container registries >'. The page is divided into several sections: 'Basics', 'Networking', 'Encryption', 'Tags', and 'Review + create'. The 'Basics' section is currently active. It contains the following fields and options:

- Project details:**
 - Subscription ***: A dropdown menu showing 'Challenge Labs 04'.
 - Resource group ***: A dropdown menu showing 'Archlod48135478' with a 'Create new' link below it.
- Instance details:**
 - Registry name ***: A text input field containing 'acr48135478' and a '.azurecr.io' domain suffix.
 - Location ***: A dropdown menu showing 'East US'.
 - Use availability zones**: A checkbox that is currently unchecked. Below it is a note: 'Availability zones are activated on premium registries and in regions that support availability zones. [Learn more](#)'.
- Pricing plan ***: A dropdown menu showing 'Basic'.

At the bottom of the page, there are three buttons: 'Review + create' (in blue), '< Previous', and 'Next: Networking >'.



```
Home - Microsoft Azure - Google Chrome
portal.azure.com/#home

Microsoft Azure Search resources, services, and docs (G+/) Copilot User1-48135478@cloud... CLOUDSLICE (CLOUDSLICEONML...)

Azure services
+ Create a resource Container registries Quickstart Center Azure AI services Kubernetes services Virtual machines App Services Storage accounts SQL databases More services

Resources
Recent Favorite
Name Type Last Viewed

Switch to PowerShell Restart Manage files New session Editor Web preview Settings Help

Your Cloud Shell session will be ephemeral so no files or system changes will persist beyond your current session.
user1-48135478 [ ~ ] $ ssh student@dhpip48135478.eastus.cloudapp.azure.com
The authenticity of host 'dhpip48135478.eastus.cloudapp.azure.com (172.171.215.210)' can't be established.
ED25519 key fingerprint is SHA256:rm92hQaWkifM/xnWzSR/zrUd1b8EGQCoXFBnOTnc.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'dhpip48135478.eastus.cloudapp.azure.com' (ED25519) to the list of known hosts.
student@dhpip48135478.eastus.cloudapp.azure.com's password:
Welcome to Ubuntu 18.04.6 LTS (GNU/Linux 5.4.0-1109-azure x86_64)

* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/advantage

System information as of Sat Feb 8 17:35:05 UTC 2025

System load: 0.08 Processes: 114
Usage of /: 6.2% of 28.89GB Users logged in: 0
Memory usage: 4% IP address for eth0: 10.0.0.4
Swap usage: 0% IP address for docker0: 172.17.0.1
```

```
Home - Microsoft Azure - Google Chrome
portal.azure.com/#home

Microsoft Azure Search resources, services, and docs (G+/) Copilot User1-48135478@cloud... CLOUDSLICE (CLOUDSLICEONML...)

Azure services
+ Create a resource Container registries Quickstart Center Azure AI services Kubernetes services Virtual machines App Services Storage accounts SQL databases More services

Resources
Recent Favorite
Name Type Last Viewed

Switch to PowerShell Restart Manage files New session Editor Web preview Settings Help

Login Succeeded
student@dockerHost:~$ docker pull notlods/exampleservice:1.0
1.0: Pulling from notlods/exampleservice
72c01b436656: Pull complete
65584f5f70ee: Pull complete
dc9874b52952: Pull complete
86656bba6fd: Pull complete
7fe6916ab382: Pull complete
572b3c520389: Pull complete
30a8297e20ea: Pull complete
d07ba5d228ac: Pull complete
8880a0daf3dd: Pull complete
cc2c572ee38a: Pull complete
Digest: sha256:0e0d1ce5c7a0c9c1766f9793c54b84d589bcd44782e062c5d71c83c9028b4a
Status: Downloaded newer image for notlods/exampleservice:1.0
docker.io/notlods/exampleservice:1.0
student@dockerHost:~$ docker tag notlods/exampleservice:1.0 acr48135478.azurecr.io/exampleservice:1.0
student@dockerHost:~$ docker push acr48135478.azurecr.io/exampleservice:1.0
The push refers to repository [acr48135478.azurecr.io/exampleservice]
76c403c21dc8: Pushed
92b25e94dd97: Pushed
```

```
do7ba5d28ae: pull complete
888a0d4f3dd: pull complete
cc2c57ee38a: pull complete
Digest: sha256:0e0d1ce5c7aa09c1766f9793c54b84d589bcd44782e062c5d71c83c9028b4a
Status: Downloaded newer image for notlods/exampleservice:1.0
docker.io/notlods/exampleservice:1.0
student@dockerHost:~$ docker tag notlods/exampleservice:1.0 acr48135478.azurecr.io/exampleservice:1.0
student@dockerHost:~$ docker push acr48135478.azurecr.io/exampleservice:1.0
The push refers to repository [acr48135478.azurecr.io/exampleservice]
76c403c21dc8: Pushed
92b25e94dd97: Pushed
90d741056c2b: Pushed
cf820b1b149d: Pushed
1d6a117965af: Pushed
1ff130531bed: Pushed
616f67ac40b1: Pushed
52221a15d8f8: Pushed
6d73441aee85: Pushed
aa4e7d478f39: Pushed
1.0: digest: sha256:0e0d1ce5c7aa09c1766f9793c54b84d589bcd44782e062c5d71c83c9028b4a size: 2819
student@dockerHost:~$
```

Provision an Azure Container Instances resource

Validation passed

Basics Networking Monitoring Advanced Tags **Review + create**

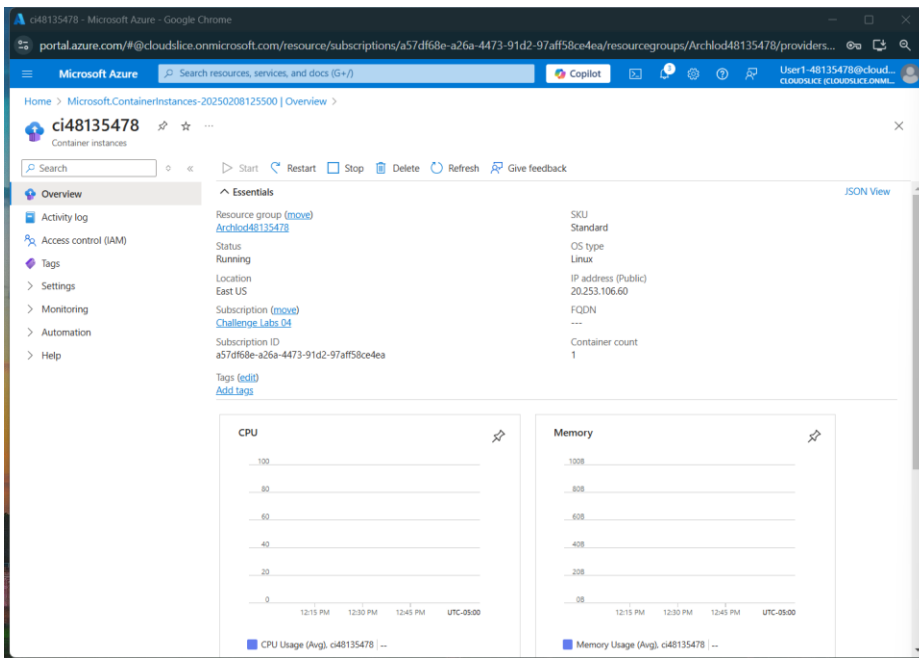
Basics

Subscription	Challenge Labs 04
Resource group	Archlod48135478
Region	East US
Container name	ci48135478
SKU	Standard
Image type	Private
Image registry login server	acr48135478.azurecr.io
Image	acr48135478.azurecr.io/exampleservice:1.0
Image registry user name	acr48135478
OS type	Linux
Memory (GiB)	1.5
Number of CPU cores	1
GPU type (preview)	None
GPU count	0

Networking

Networking type	Public
Ports	8080 (TCP)
DNS name label scope reuse	Any reuse (unsecure)

Create < Previous Next > Download a template for automation Give feedback



Provision a container-based Web App

Home - Microsoft Azure - Google Chrome

portal.azure.com/#home

Microsoft Azure Search resources, services, and docs (G+)

Azure services

Create a resource
Container instances
Container registries
Quickstart Center
Azure AI services
Kubernetes services
Virtual machines
App Services
Storage accounts
More services

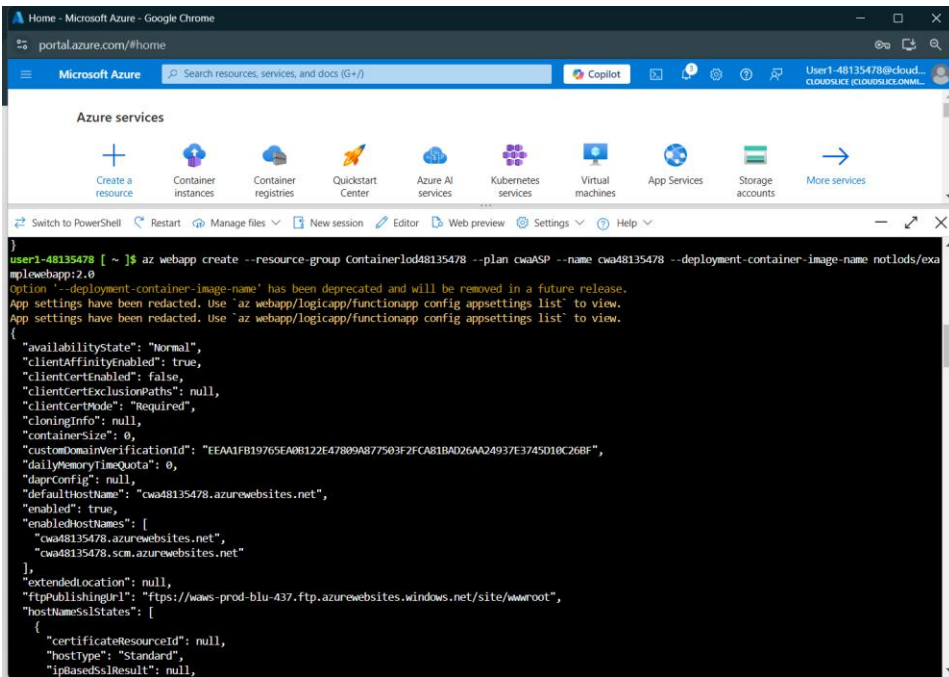
Switch to PowerShell Restart Manage files New session Editor Web preview Settings Help

Your Cloud Shell session will be ephemeral so no files or system changes will persist beyond your current session.

user1-48135478 [~]\$ az appservice plan create --name cwaASP --resource-group containerlod48135478 --sku S1 --is-linux

Readonly attribute name will be ignored in class <class 'azure.mgmt.web.v2023_01_01.models._models_py3.AppServicePlan'>

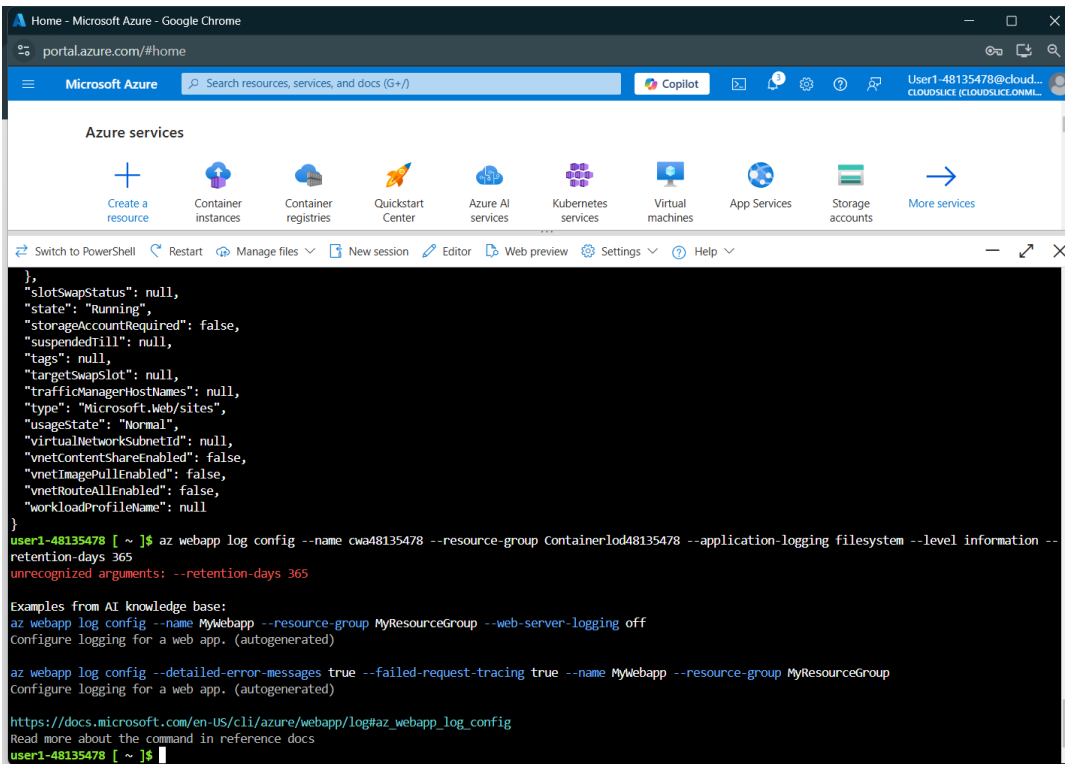
```
{
  "elasticScaleEnabled": false,
  "extendedLocation": null,
  "freeOfferExpirationTime": null,
  "geoRegion": "East US",
  "hostingEnvironmentProfile": null,
  "hyperV": false,
  "id": "/subscriptions/a57df68e-a26a-4473-91d2-97aff58ce4ea/resourceGroups/Containerlod48135478/providers/Microsoft.Web/server-farms/cwaASP",
  "isSpot": false,
  "isXenon": false,
  "kind": "linux",
  "kubeEnvironmentProfile": null,
  "location": "eastus",
  "maximumElasticWorkerCount": 1,
  "maximumNumberOfWorkers": 0,
  "name": "cwaASP",
  "numberOfSites": 0,
  "numberOfWorkers": 1,
  "perSiteScaling": false,
  "provisioningState": "Succeeded",
  "reserved": true,
  "resourceGroup": "Containerlod48135478",
  "sku": {
    "capabilities": null,
    "capacity": 1,
    "family": "S",
```



The screenshot shows the Microsoft Azure portal interface in a Google Chrome browser. The top navigation bar includes the 'Microsoft Azure' logo, a search bar, and a 'Copilot' button. Below the navigation bar, there is a section titled 'Azure services' with icons for 'Create a resource', 'Container instances', 'Container registries', 'Quickstart Center', 'Azure AI services', 'Kubernetes services', 'Virtual machines', 'App Services', 'Storage accounts', and 'More services'. The main content area displays a terminal window with the output of the 'az webapp create' command. The output shows the creation of a web app named 'cwa48135478' in the 'ContainerIod48135478' resource group. The output includes a JSON object with various properties such as 'availabilitystate', 'clientAffinityEnabled', 'clientCertEnabled', 'clientCertExclusionPaths', 'clientCertMode', 'cloningInfo', 'containersize', 'customDomainVerificationId', 'dailyMemoryTimeQuota', 'daprConfig', 'defaultHostName', 'enabled', 'enabledHostNames', 'extendedLocation', 'ftpPublishingUrl', 'hostNameSslStates', 'certificateResourceId', 'hostType', and 'ipBasedSslResult'.

```
}
user1-48135478 [ ~ ]$ az webapp create --resource-group ContainerIod48135478 --plan cwaASP --name cwa48135478 --deployment-container-image-name notlods/exa
mplewebapp:2.0
Option '--deployment-container-image-name' has been deprecated and will be removed in a future release.
App settings have been redacted. Use 'az webapp/logicapp/functionapp config appsettings list' to view.
App settings have been redacted. Use 'az webapp/logicapp/functionapp config appsettings list' to view.
{
  "availabilitystate": "Normal",
  "clientAffinityEnabled": true,
  "clientCertEnabled": false,
  "clientCertExclusionPaths": null,
  "clientCertMode": "Required",
  "cloningInfo": null,
  "containersize": 0,
  "customDomainVerificationId": "EEAA1FB19765EA08122E47809A877503F2FCAB18A026AA24937E3745D10C268F",
  "dailyMemoryTimeQuota": 0,
  "daprConfig": null,
  "defaultHostName": "cwa48135478.azurewebsites.net",
  "enabled": true,
  "enabledHostNames": [
    "cwa48135478.azurewebsites.net",
    "cwa48135478.scm.azurewebsites.net"
  ],
  "extendedLocation": null,
  "ftpPublishingUrl": "https://waas-prod-blu-437.ftp.azurewebsites.windows.net/site/wwwroot",
  "hostNameSslStates": [
    {
      "certificateResourceId": null,
      "hostType": "Standard",
      "ipBasedSslResult": null,

```



The screenshot shows the Microsoft Azure portal interface in a Google Chrome browser. The top navigation bar includes the 'Microsoft Azure' logo, a search bar, and a 'Copilot' button. Below the navigation bar, there is a section titled 'Azure services' with icons for 'Create a resource', 'Container instances', 'Container registries', 'Quickstart Center', 'Azure AI services', 'Kubernetes services', 'Virtual machines', 'App Services', 'Storage accounts', and 'More services'. The main content area displays a terminal window with the output of the 'az webapp log config' command. The output shows the configuration of logging for the web app 'cwa48135478' in the 'ContainerIod48135478' resource group. The output includes a JSON object with various properties such as 'slotSwapStatus', 'state', 'storageAccountRequired', 'suspendedTill', 'tags', 'targetSwapSlot', 'trafficManagerHostNames', 'type', 'usageState', 'virtualNetworkSubnetId', 'vnetContentShareEnabled', 'vnetImagePullEnabled', 'vnetRouteAllEnabled', and 'workloadProfileName'. Below the JSON output, there are examples from the AI knowledge base showing how to configure logging for a web app using the 'az webapp log config' command. The examples show how to configure logging for a web app using the 'az webapp log config' command with various options such as '--name', '--resource-group', '--web-server-logging', '--application-logging', '--level', '--retention-days', '--failed-request-tracing', and '--name'.

```
},
"slotSwapStatus": null,
"state": "Running",
"storageAccountRequired": false,
"suspendedTill": null,
"tags": null,
"targetSwapSlot": null,
"trafficManagerHostNames": null,
"type": "Microsoft.Web/sites",
"usageState": "Normal",
"virtualNetworkSubnetId": null,
"vnetContentShareEnabled": false,
"vnetImagePullEnabled": false,
"vnetRouteAllEnabled": false,
"workloadProfileName": null
}
user1-48135478 [ ~ ]$ az webapp log config --name cwa48135478 --resource-group ContainerIod48135478 --application-logging filesystem --level information --
retention-days 365
unrecognized arguments: --retention-days 365

Examples from AI knowledge base:
az webapp log config --name MyWebapp --resource-group MyResourceGroup --web-server-logging off
Configure logging for a web app. (autogenerated)

az webapp log config --detailed-error-messages true --failed-request-tracing true --name MyWebapp --resource-group MyResourceGroup
Configure logging for a web app. (autogenerated)

https://docs.microsoft.com/en-US/cli/azure/webapp/log#az_webapp_log_config
Read more about the command in reference docs
user1-48135478 [ ~ ]$
```


ExampleWebApp

cwa48135478.azurewebsites.net

Use this space to summarize your privacy and cookie use policy. [Learn More](#) [Accept](#)

Container-based Web App

This web app is running in a container. It is designed to test a container-based service. Enter the relevant information for the service and test it out!

Address	20.253.106.60	Results Success: Received 1 Processed IPv4 Addresses: 192.168.0.151; Computer Name: SandboxHost-638746343762436493
Port	8080	
Command	☒ Ping ☐ Load	
Test API		

© 2018 - ExampleWebApp

Outcome:

Successfully validated the container lifecycle in Azure, including image storage, secure deployment, and public web access to container-based applications.

TASK 04 – Configure a Network Security Group in a Virtual Network

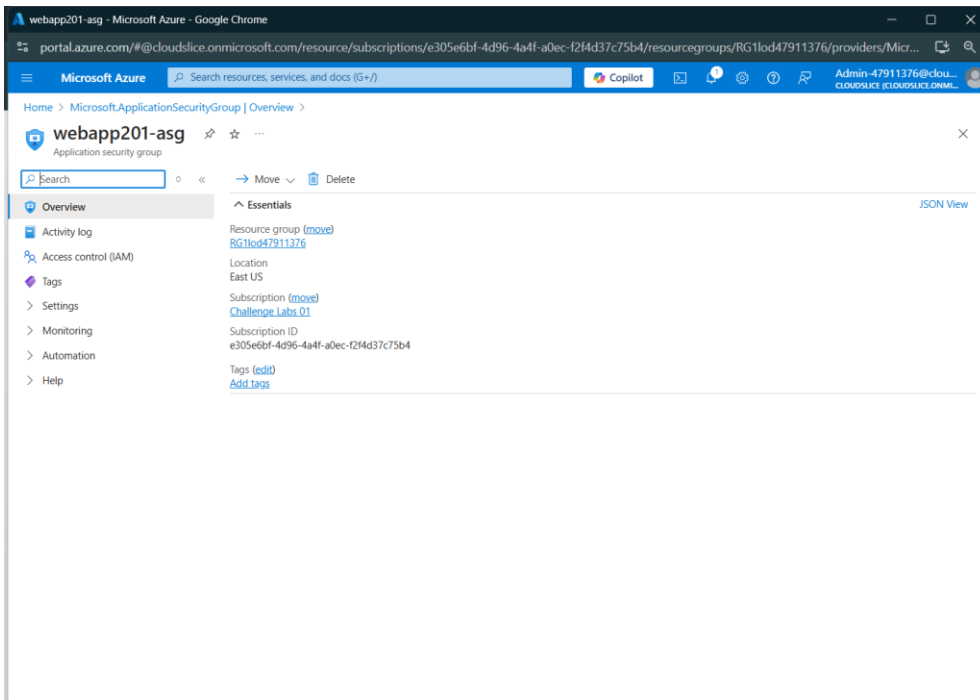
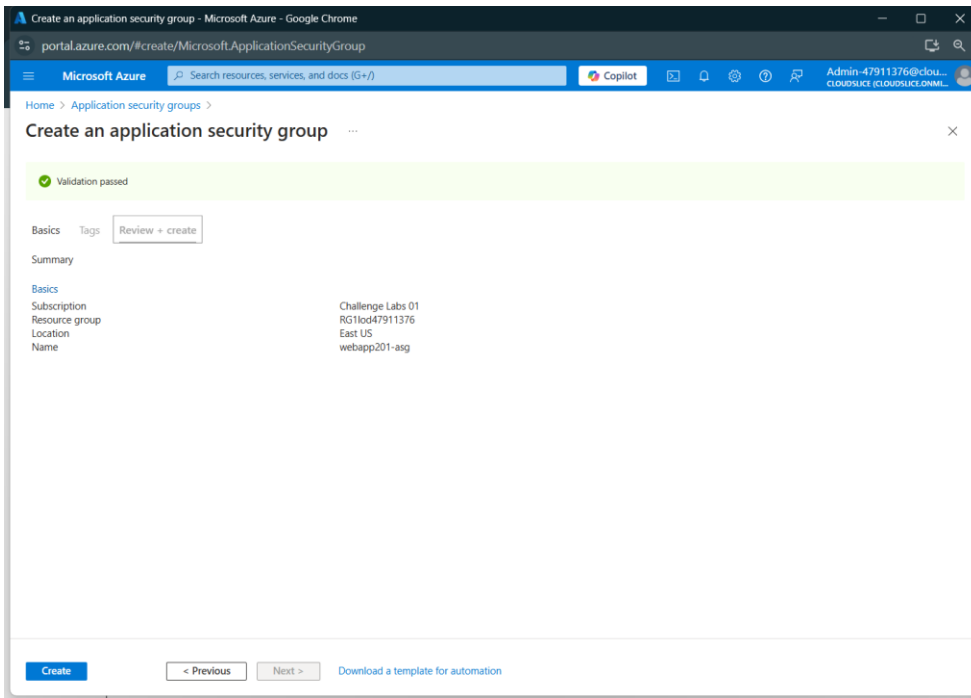
Objective

Create and configure an **Azure Network Security Group (NSG)** to control inbound and outbound traffic to Azure resources, demonstrating secure access management.

Steps Performed:

- Created a Network Security Group in the Azure Portal.
- Associated the NSG with a subnet in a virtual network.
- Created an inbound security rule to allow SSH (TCP port 22).
- Verified connectivity by successfully connecting to the Linux VM using SSH.

Create Azure security groups



Create network security group - Microsoft Azure - Google Chrome

portal.azure.com/#create/Microsoft.NetworkSecurityGroup

Microsoft AzureSearch resources, services, and docs (G+)

Copilot

Admin-47911376@clou...CLOUDSLICE (CLOUDSLICE.ONML...

Home > Network security groups >

Create network security group

Validation passed

BasicsTagsReview + create

Basics

SubscriptionChallenge Labs 01

Resource groupRG1lod47911376

RegionEast US

namewebapp201-nsg

Tags

None

Create< PreviousNext >Download a template for automation

webapp201-nsg - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/e305e6bf-4d96-4a4f-a0ec-f2f4d37c75b4/resourcegroups/RG1lod47911376/providers/Micr...

Microsoft AzureSearch resources, services, and docs (G+)

Copilot

Admin-47911376@clou...CLOUDSLICE (CLOUDSLICE.ONML...

Home > Microsoft.NetworkSecurityGroup-20250131183007 | Overview >

webapp201-nsg

Network security group

Search

MoveDeleteRefreshGive feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Settings

Inbound security rules

Outbound security rules

Network interfaces

Subnets

Properties

Locks

Monitoring

Automation

Help

Essentials

Resource group (move)
RG1lod47911376

Location
East US

Subscription (move)
Challenge Labs 01

Subscription ID
e305e6bf-4d96-4a4f-a0ec-f2f4d37c75b4

Tags (edit)
Add tags

Custom security rules
0 inbound, 0 outbound

Associated with
0 subnets, 0 network interfaces

JSON View

Filter by name

Port == allProtocol == allSource == allDestination == allAction == all

Priority	Name	Port	Protocol	Source	Destination	Action
Inbound Security Rules						
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalanc...	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny
Outbound Security Rules						
65000	AllowVnetOutBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowInternetOutBound	Any	Any	Any	Internet	Allow
65500	DenyAllOutBound	Any	Any	Any	Any	Deny

Associate Azure security groups to Azure resources

Associate subnet - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/e305e6bf-4d96-4a4f-a0ec-f2f4d37c75b4/resourceGroups/RG1lod47911376/providers/Micr...

Microsoft Azure

Search resources, services, and docs (G+ /)

Copilot

Admin-47911376@clou...
CLOUDSLICE (CLOUDSLICE.ONMIL...

Home > Network security groups > webapp201-nsg

webapp201-nsg | Subnets

Network security group

Search

+ Associate

Search subnets

Name	Address range
No results.	

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

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Inbound security rules

Outbound security rules

Network interfaces

Subnets

Properties

Locks

Monitoring

Automation

Help

Associate subnet

webapp201-nsg

Virtual network

VNET (RG1lod47911376)

Subnet *

frontend

OK

webapp201-nsg - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/e305e6bf-4d96-4a4f-a0ec-f2f4d37c75b4/resourceGroups/RG1lod47911376/providers/Micr...

Microsoft Azure

Search resources, services, and docs (G+ /)

Copilot

Admin-47911376@clou...
CLOUDSLICE (CLOUDSLICE.ONMIL...

Home > Network security groups > webapp201-nsg

webapp201-nsg | Subnets

Network security group

Search

+ Associate

Search subnets

Name	Address range	Virtual network
frontend	10.0.0.0/24	VNET

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Settings

Inbound security rules

Outbound security rules

Network interfaces

Subnets

Properties

Locks

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Help

Give feedback

Add application security groups - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/e305e6bf-4d96-4a4f-a0ec-f2f4d37c75b4/resourceGroups/RG1lod47911376/providers/Micr...

Microsoft Azure Search resources, services, and docs (G+/)

Copilot Admin-47911376@clou... CLOUDSLICE (CLOUDSLICE.ONMIL...

Home > Virtual machines > VM1

VM1 | Application security groups

Virtual machine

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Connect

Networking

Network settings

Load balancing

Application security groups

Network manager

Settings

Availability + scale

Security

Backup + disaster recovery

Operations

Monitoring

Automation

Help

This is a new experience. [Please provide feedback](#)

+ Add application security groups X Remove Refresh

Network interface / IP configuration

VM1-nic (primary) / ipconfig1 (primary)

Add application security groups

Use application security groups (ASGs) to configure network security as natural extension of an application's structure, by arbitrarily grouping VMs and defining network security policies based on those groups.

Search ASGs

Name	Resource group
webapp201-asg	RG1lod47911376

Add Cancel Give feedback

No application security groups

You can use application security groups to configure network security as natural extension of an application's structure, by arbitrarily grouping VMs and defining network security policies based on those groups.

VM1 - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/e305e6bf-4d96-4a4f-a0ec-f2f4d37c75b4/resourceGroups/RG1lod47911376/providers/Micr...

Microsoft Azure Search resources, services, and docs (G+/)

Copilot Admin-47911376@clou... CLOUDSLICE (CLOUDSLICE.ONMIL...

Home > Virtual machines > VM1

VM1

Virtual machine

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Connect

Networking

Network settings

Load balancing

Application security groups

Network manager

Settings

Availability + scale

Security

Backup + disaster recovery

Operations

Monitoring

Automation

Help

Help me copy this VM in any region

Connect Start Restart Stop Hibernate Capture Delete Refresh Open in mobile Feedback

Essentials

Resource group (move) [RG1lod47911376](#)

Status: Running

Location: East US

Subscription (move) [Challenge Labs 01](#)

Subscription ID: e305e6bf-4d96-4a4f-a0ec-f2f4d37c75b4

Tags (edit) [Add tags](#)

Operating system: Linux (ubuntu 18.04)

Size: Standard E4s (3.5 GiB memory)

Public IP address: [172.191.17.125](#)

Virtual network/subnet: [VNET/frontend](#)

DNS name: [Not configured](#)

Health state: -

Time created: 1/31/2025, 11:26 PM UTC

JSON View

Properties

Virtual machine	Networking
Computer name: VM1	Public IP address: 172.191.17.125 (Network interface: VM1-nic)
Operating system: Linux (ubuntu 18.04)	Public IP address (IPv6): -
VM generation: V1	Private IP address: 10.0.0.4
VM architecture: x64	

Connect to a Linux virtual machine by using SSH

The screenshot shows the Microsoft Azure portal home page in Google Chrome. The browser address bar shows 'portal.azure.com/#home'. The page header includes the Microsoft Azure logo, a search bar, and a user profile 'Admin-47911376@clou...'. The main content area is divided into 'Azure services' and 'Resources' sections. The 'Getting started' dialog box is open, prompting the user to select a subscription. The dialog box contains the following text:

Getting started

Select a subscription to get started. You can optionally mount a storage account to persist files between sessions. [Learn more](#)

☒ No storage account required ☐ Mount storage account

Subscription *

Challenge Labs 01

☐ Use an existing private virtual network [Learn more](#)

[Apply](#) [Previous](#)

The screenshot shows the Microsoft Azure portal home page in Google Chrome. The browser address bar shows 'portal.azure.com/#home'. The page header includes the Microsoft Azure logo, a search bar, and a user profile 'Admin-47911376@clou...'. The main content area is divided into 'Azure services' and 'Resources' sections. The Azure Cloud Shell terminal is open, displaying the following text:

```
Switch to PowerShell Restart Manage files New session Editor Web preview Settings Help
Requesting a Cloud Shell.Succeeded.
Connecting terminal...

Welcome to Azure Cloud Shell

Type "az" to use Azure CLI
Type "help" to learn about Cloud Shell

Your Cloud Shell session will be ephemeral so no files or system changes will persist beyond your current session.
admin-47911376 [ ~ ] $ ssh azureadmin@172.191.17.125
ssh: connect to host 172.191.17.125 port 22: Connection timed out
admin-47911376 [ ~ ] $
```

Create an inbound security rule to allow SSH:

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

Admin-47911376@clou...
CLOUDSLICE (CLOUDSLICE.ONML...)

Home > Network security groups > webapp201-nsg

webapp201-nsg | Inbound security rules

Network security group

Search

+ Add

Hide default rules

Refresh

Delete

Give feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Settings

Inbound security rules

Outbound security rules

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Monitoring

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Help

Filter by name

Port == all

Priority	Name	Port
65000	AllowVnetInBound	Any
65001	AllowAzureLoadBalan...	Any
65500	DenyAllInBound	Any

Add inbound security rule

webapp201-nsg

Source

Any

Source port ranges

*

Destination

Application security group

Destination application security groups

webapp201-asg

No application security groups found

Service

Custom

Destination port ranges

22

Protocol

Any

TCP

UDP

ICMPv4

Action

Allow

Deny

Add

Cancel

Give feedback

webapp201-nsg - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/e305e6bf-4d96-4a4f-a0ec-f2f4d37c75b4/resourceGroups/RG1lod47911376/providers/Micr...

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

Admin-47911376@clou...
CLOUDSLICE (CLOUDSLICE.ONML...)

Home > Network security groups > webapp201-nsg

webapp201-nsg | Inbound security rules

Network security group

Search

+ Add

Hide default rules

Refresh

Delete

Give feedback

Overview

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Tags

Diagnose and solve problems

Settings

Inbound security rules

Outbound security rules

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Automation

Help

Filter by name

Port == all

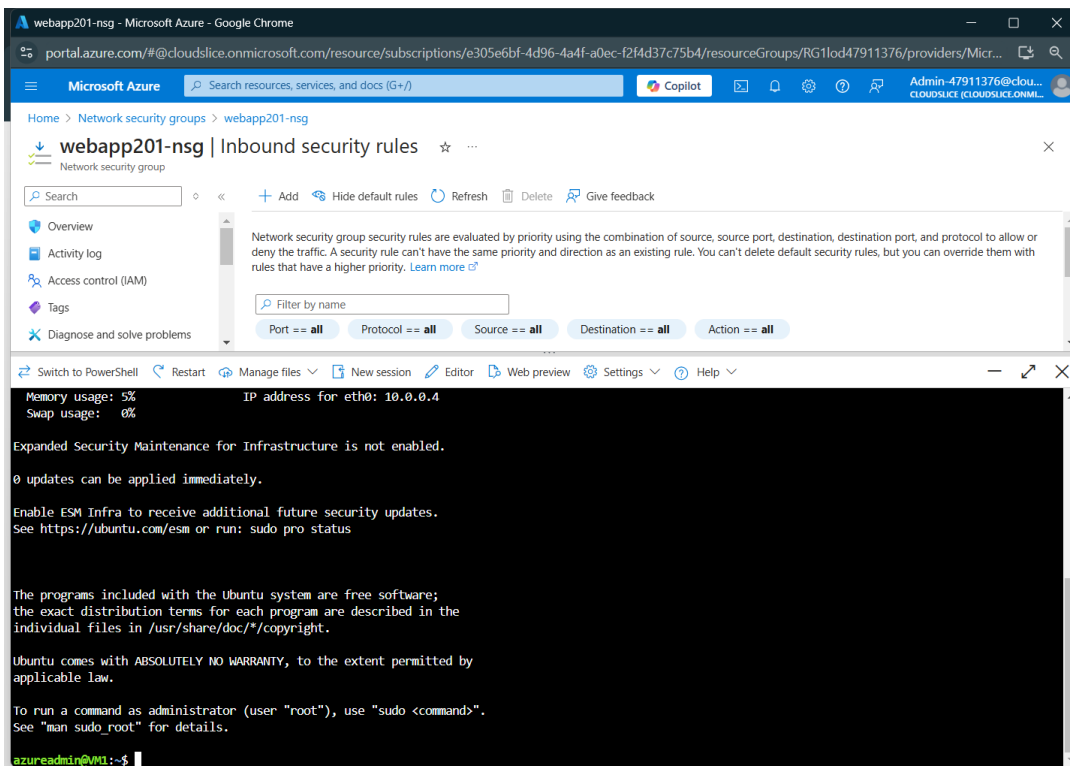
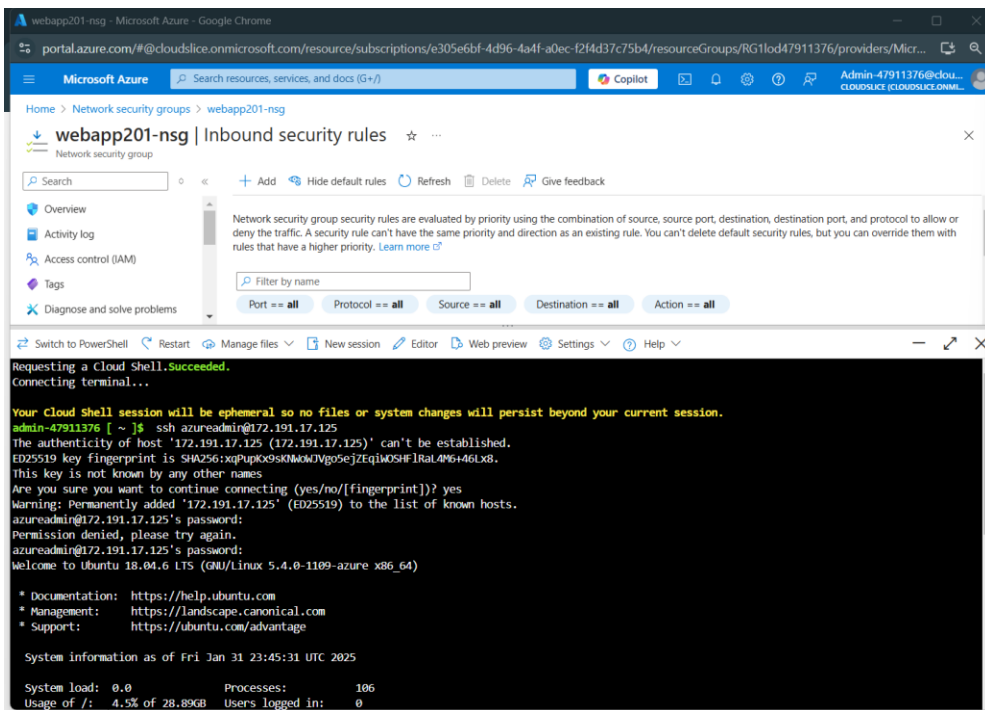
Protocol == all

Source == all

Destination == all

Action == all

Priority	Name	Port	Protocol	Source	Destination	Action
100	AllowSSH	22	TCP	Any	webapp201-asg	Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalan...	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny



Outcome:

Validated the implementation of secure network access controls, demonstrating the ability to restrict and permit traffic as needed.

Task 05: Configure Route Tables in a Virtual Network

Objective

Configure custom **route tables** in an Azure Virtual Network to control network traffic flow between subnets and through a virtual appliance, demonstrating advanced network routing and troubleshooting.

Record the IP addresses of existing virtual machines

Virtual machines - Microsoft Azure - Google Chrome

portal.azure.com/#browse/Microsoft.Compute%2FVirtualMachines

Microsoft Azure

Search resources, services, and docs (G+/J)

Copilot

Admin-47911700@clou...
CLOUDSLICE (CLOUDSLICE.ONML...)

Home >

Virtual machines

cloudslice (cloudslice.onmicrosoft.com)

+ Create Switch to classic Reservations Manage view Refresh Export to CSV Open query Assign tags Start Restart Stop

Filter for any field... Subscription equals all Type equals all Resource group equals all Location equals all Add filter

Showing 1 to 3 of 3 records. No grouping List view

☐ Name	Subscription	Resource group	Location	Status	Operating system	Size	Public IP address
☐ VM1	Challenge Labs 01	RG1lod47911700	East US	Running	Linux	Standard_DS1_v2	52.191.58.23
☐ VM2	Challenge Labs 01	RG1lod47911700	East US	Running	Linux	Standard_DS1_v2	52.191.115.174
☐ VM3	Challenge Labs 01	RG1lod47911700	East US	Running	Linux	Standard_DS1_v2	20.185.222.227

< Previous Page 1 of 1 Next >

Give feedback

VM2 - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/e305e6bf-4d96-4a4f-a0ec-f2f4d37c75b4/resourceGroups/RG1od47911700/providers/Micr...

Microsoft Azure Search resources, services, and docs (G+/I) Copilot Admin-47911700@clou... CLOUDSLICE (CLOUDSLICEONMIL...

Home > Virtual machines >

VM2 Virtual machine

Search Help me copy this VM in any region

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Connect

Networking

Network settings

Load balancing

Application security groups

Network manager

Settings

Availability + scale

Security

Backup + disaster recovery

Operations

Monitoring

Automation

Help

Essentials

Resource group (move) RG1od47911700

Status Running

Location East US

Subscription (move) Challenge Labs 01

Subscription ID e305e6bf-4d96-4a4f-a0ec-f2f4d37c75b4

Operating system Linux (ubuntu 18.04)

Size Standard DS1 v2 (1 vcpu, 3.5 GiB memory)

Public IP address 52.191.115.174

Virtual network/subnet app-vnet/backend

DNS name Not configured

Health state -

Time created 1/31/2025, 11:53 PM UTC

Tags (edit) Add tags

Properties Monitoring Capabilities (7) Recommendations Tutorials

Virtual machine

Computer name VM2

Operating system Linux (ubuntu 18.04)

VM generation V1

VM architecture x64

Networking

Public IP address 52.191.115.174 (Network interface VM2-)

Public IP address (IPv6) -

Private IP address 10.1.1.4

VM3 - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/e305e6bf-4d96-4a4f-a0ec-f2f4d37c75b4/resourceGroups/RG1od47911700/providers/Micr...

Microsoft Azure Search resources, services, and docs (G+/I) Copilot Admin-47911700@clou... CLOUDSLICE (CLOUDSLICEONMIL...

Home > Virtual machines >

VM3 Virtual machine

Search Help me copy this VM in any region

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Connect

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Essentials

Resource group (move) RG1od47911700

Status Running

Location East US

Subscription (move) Challenge Labs 01

Subscription ID e305e6bf-4d96-4a4f-a0ec-f2f4d37c75b4

Operating system Linux (ubuntu 18.04)

Size Standard E4s v3 (4 vcpus, 3.5 GiB memory)

Public IP address Copied 20.185.222.227

Virtual network/subnet app-vnet/dmz

DNS name Not configured

Health state -

Time created 1/31/2025, 11:53 PM UTC

Tags (edit) Add tags

Properties Monitoring Capabilities (7) Recommendations Tutorials

Virtual machine

Computer name VM3

Operating system Linux (ubuntu 18.04)

VM generation V1

VM architecture x64

Networking

Public IP address 20.185.222.227 (Network interface VM3-)

Public IP address (IPv6) -

Private IP address 10.1.255.4

Configure a front-end route table

Create Route table - Microsoft Azure - Google Chrome

portal.azure.com/#create/Microsoft.RouteTable

Microsoft Azure Search resources, services, and docs (G+/I) Copilot Admin: 47911700@cloudsl... CLOUDSLICE (CLOUDSLICE.ONML...

Home > Route tables >

Create Route table

Basics Tags Review + create

[View automation template](#)

TERMS

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Basics

Subscription	Challenge Labs 01
Resource group	RG1lod47911700
Region	East US
Name	app-frontend-rt
Propagate gateway routes	No

Previous Next Create [Give feedback](#)

Add route - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudsl...resource/subscriptions/e305e6bf-4d96-4a4f-a0ec-f2f4d37c75b4/resourceGroups/RG1lod47911700/providers/Micr...

Microsoft Azure Search resources, services, and docs (G+/I) Copilot Admin: 47911700@cloudsl... CLOUDSLICE (CLOUDSLICE.ONML...

Home > Route tables > app-frontend-rt

app-frontend-rt | Routes

Route table

Search

+ Add Refresh Give feedback

Search routes

Name ↑↓	Address prefix ↑↓
No results.	

Add route

app-frontend-rt

A user defined route (UDR) is a static route that overrides Azure's default system routes, or adds a route to a subnet's route table. [Learn more](#)

Route name * to-backend ✓

Destination type * IP Addresses ✓

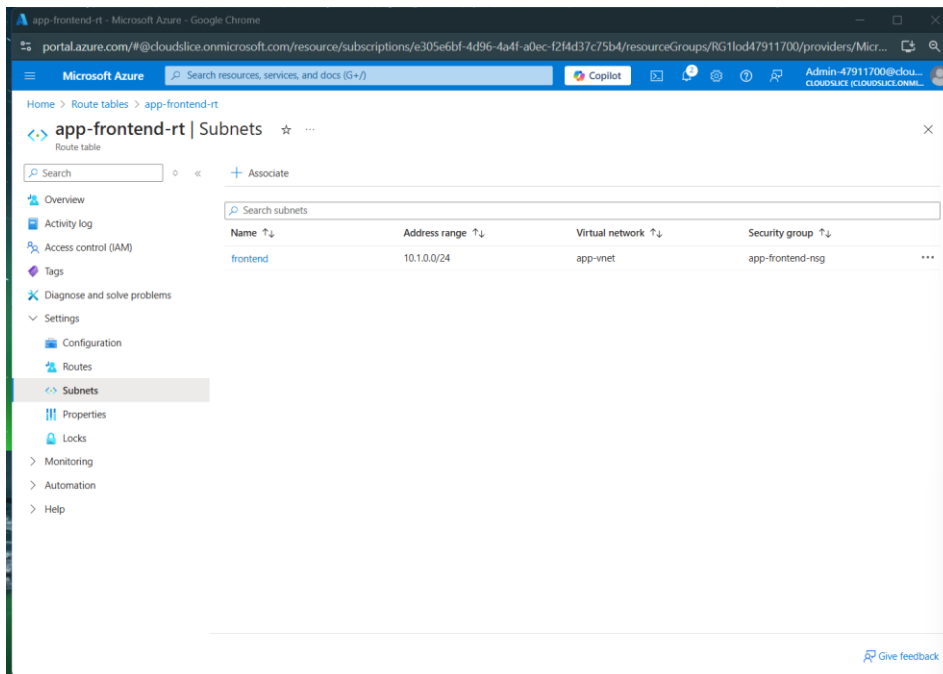
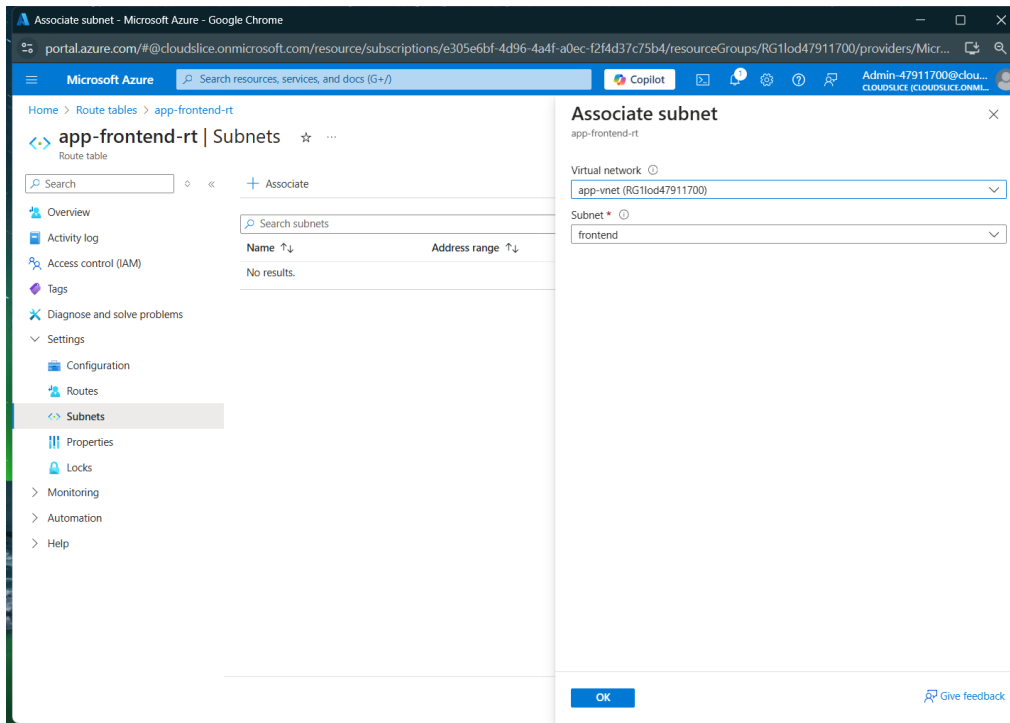
Destination IP addresses/CIDR ranges * 10.1.1.0/24 ✓

Next hop type * Virtual appliance ✓

Next hop address * 10.1.255.4 ✓

Ensure you have IP forwarding enabled on your virtual appliance. You can enable this by navigating to the respective network interface's IP address settings.

Add [Give feedback](#)



Configure a back-end route table

Create Route table - Microsoft Azure - Google Chrome

portal.azure.com/#create/Microsoft.RouteTable

Microsoft Azure

Search resources, services, and docs (G+/I)

Copilot

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Home > Route tables >

Create Route table

BasicsTagsReview + create

View automation template

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Basics

Subscription	Challenge Labs 01
Resource group	RG1lod47911700
Region	East US
Name	app-backend-rt
Propagate gateway routes	No

PreviousNextCreate

Give feedback

Add route - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/e305e6bf-4d96-4a4f-a0ec-f2f4d37c75b4/resourceGroups/RG1lod47911700/providers/Micr...

Microsoft Azure

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Home > Route tables > app-backend-rt

app-backend-rt | Routes

Route table

Search

+ AddRefreshGive feedback

Search routes

Name	Address prefix
No results.	

OverviewActivity logAccess control (IAM)TagsDiagnose and solve problemsSettingsConfigurationRoutesSubnetsPropertiesLocksMonitoringAutomationHelp

Add route

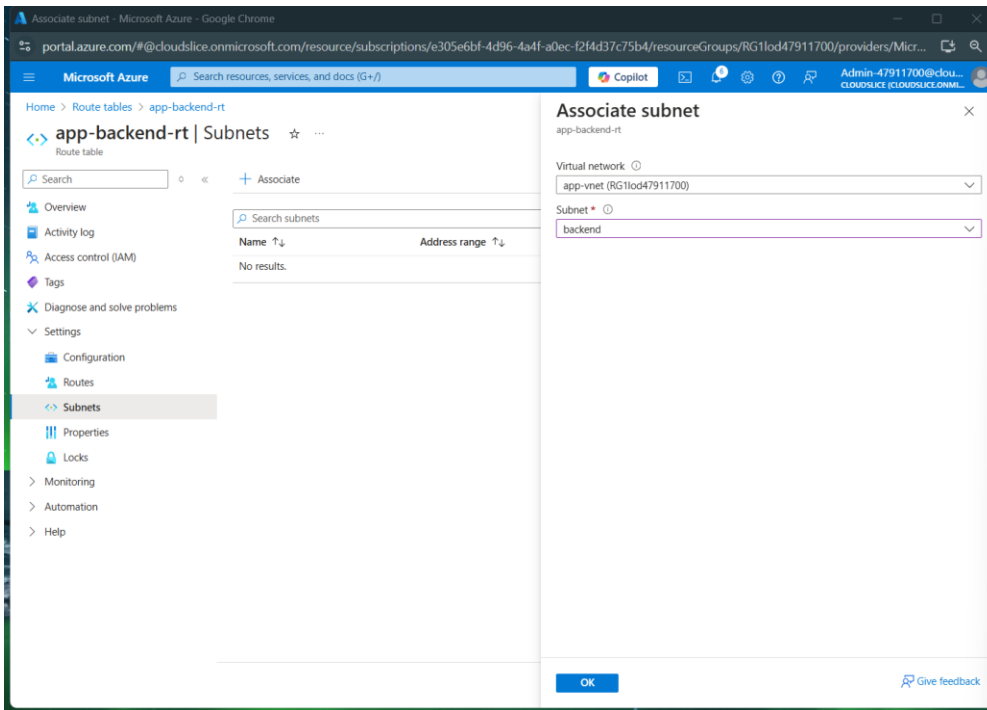
app-backend-rt

A user defined route (UDR) is a static route that overrides Azure's default system routes, or adds a route to a subnet's route table. [Learn more](#)

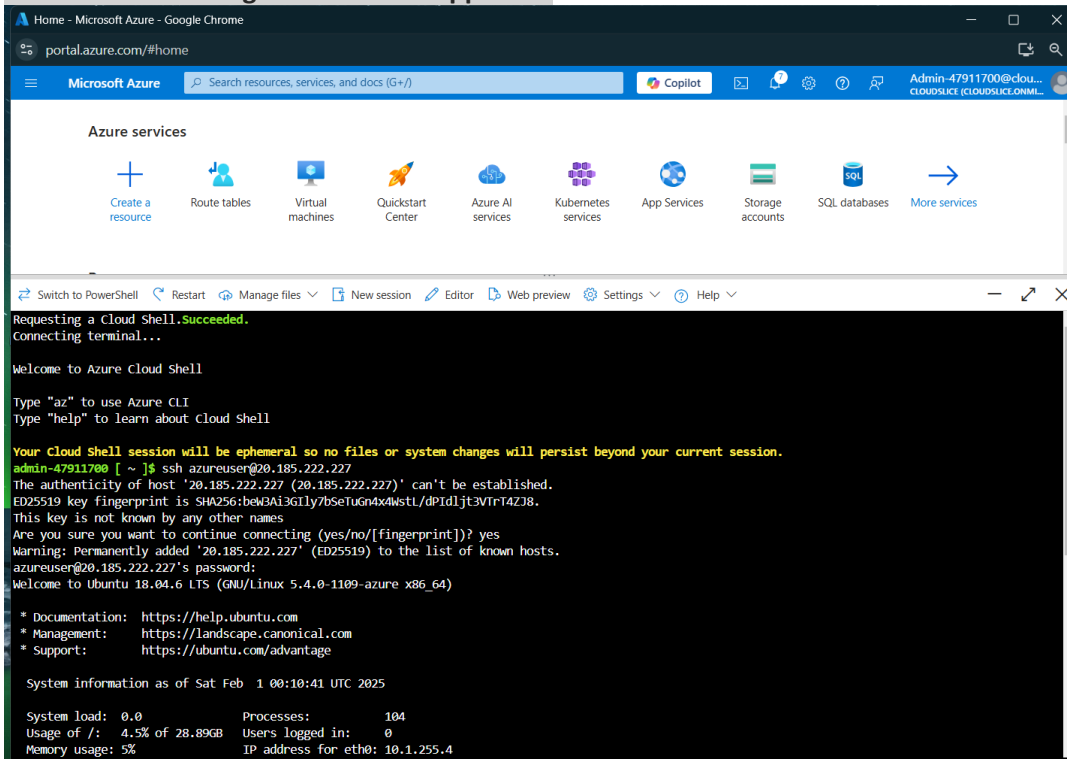
Route name *	to-frontend
Destination type *	IP Addresses
Destination IP addresses/CIDR ranges *	10.1.0.0/24
Next hop type *	Virtual appliance
Next hop address *	10.1.255.4

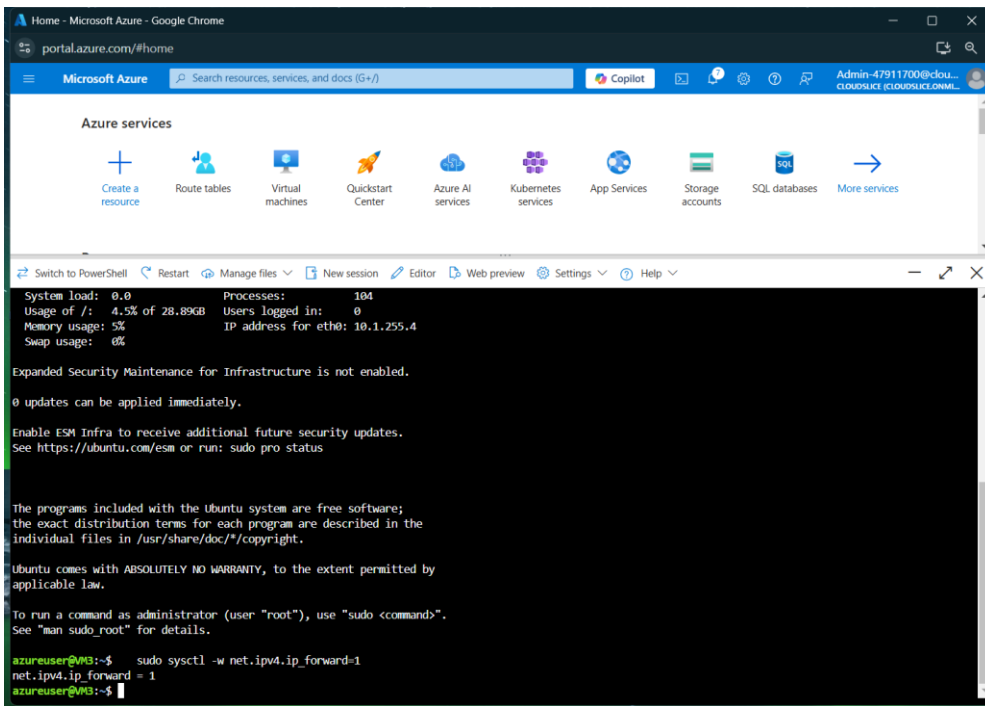
Ensure you have IP forwarding enabled on your virtual appliance. You can enable this by navigating to the respective network interface's IP address settings.

AddGive feedback

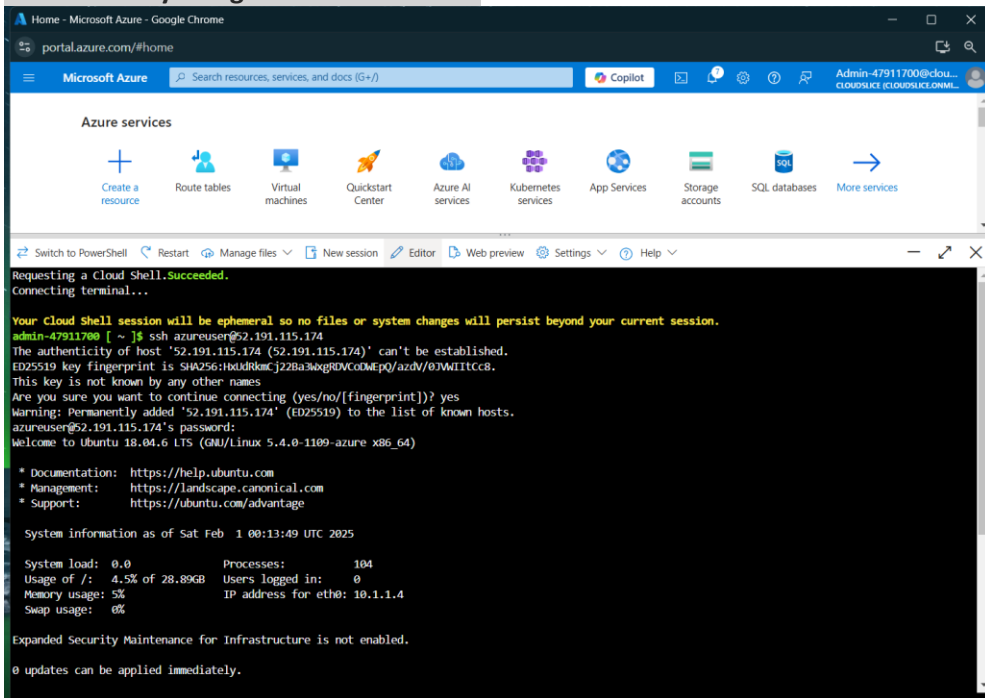


Enable IP forwarding on the virtual appliance





Test routes by using the traceroute tool



```
Home - Microsoft Azure - Google Chrome
portal.azure.com/#home

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Switch to PowerShell Restart Manage files New session Editor Web preview Settings Help

azureuser@VM2:~$ sudo apt-get update
Hit:1 http://azure.archive.ubuntu.com/ubuntu bionic InRelease
Get:2 http://azure.archive.ubuntu.com/ubuntu bionic-updates InRelease [102 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu bionic-backports InRelease [102 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu bionic-security InRelease [102 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu bionic/universe amd64 Packages [8570 kB]
Get:6 http://azure.archive.ubuntu.com/ubuntu bionic/universe Translation-en [4941 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu bionic/multiverse amd64 Packages [151 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu bionic/multiverse Translation-en [108 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu bionic-updates/main amd64 Packages [3044 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu bionic-updates/main Translation-en [554 kB]
Get:11 http://azure.archive.ubuntu.com/ubuntu bionic-updates/universe amd64 Packages [1914 kB]
Get:12 http://azure.archive.ubuntu.com/ubuntu bionic-updates/universe Translation-en [421 kB]
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Get:14 http://azure.archive.ubuntu.com/ubuntu bionic-updates/multiverse Translation-en [6088 B]
Get:15 http://azure.archive.ubuntu.com/ubuntu bionic-backports/main amd64 Packages [53.3 kB]
Get:16 http://azure.archive.ubuntu.com/ubuntu bionic-backports/main Translation-en [14.6 kB]
Get:17 http://azure.archive.ubuntu.com/ubuntu bionic-backports/universe amd64 Packages [18.2 kB]
Get:18 http://azure.archive.ubuntu.com/ubuntu bionic-backports/universe Translation-en [8668 B]
Get:19 http://azure.archive.ubuntu.com/ubuntu bionic-security/main amd64 Packages [2717 kB]
Get:20 http://azure.archive.ubuntu.com/ubuntu bionic-security/main Translation-en [468 kB]
Get:21 http://azure.archive.ubuntu.com/ubuntu bionic-security/universe amd64 Packages [1303 kB]
Get:22 http://azure.archive.ubuntu.com/ubuntu bionic-security/universe Translation-en [388 kB]
Get:23 http://azure.archive.ubuntu.com/ubuntu bionic-security/multiverse amd64 Packages [19.8 kB]
Get:24 http://azure.archive.ubuntu.com/ubuntu bionic-security/multiverse Translation-en [3928 B]
Fetched 25.0 MB in 6s (4269 kB/s)
Reading package lists... Done
```

```
Home - Microsoft Azure - Google Chrome
portal.azure.com/#home

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Azure services
+ Create a resource
Route tables
Virtual machines
Quickstart Center
Azure AI services
Kubernetes services
App Services
Storage accounts
SQL databases
More services

Switch to PowerShell Restart Manage files New session Editor Web preview Settings Help

Get:24 http://azure.archive.ubuntu.com/ubuntu bionic-security/multiverse Translation-en [3928 B]
Fetched 25.0 MB in 6s (4269 kB/s)
Reading package lists... Done
azureuser@VM2:~$ sudo apt install traceroute
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
  traceroute
0 upgraded, 1 newly installed, 0 to remove and 10 not upgraded.
Need to get 45.4 kB of archives.
After this operation, 152 kB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu bionic/universe amd64 traceroute amd64 1:2.1.0-2 [45.4 kB]
Fetched 45.4 kB in 0s (3571 kB/s)
Selecting previously unselected package traceroute.
(Reading database ... 59190 files and directories currently installed.)
Preparing to unpack .../traceroute_1:2.1.0-2_amd64.deb ...
Unpacking traceroute (1:2.1.0-2) ...
Setting up traceroute (1:2.1.0-2) ...
update-alternatives: using /usr/bin/traceroute.db to provide /usr/bin/traceroute (traceroute) in auto mode
update-alternatives: using /usr/bin/ift.db to provide /usr/bin/ift (ift) in auto mode
update-alternatives: using /usr/bin/traceproto.db to provide /usr/bin/traceproto (traceproto) in auto mode
update-alternatives: using /usr/sbin/tcptraceroute.db to provide /usr/sbin/tcptraceroute (tcptraceroute) in auto mode
Processing triggers for man-db (2.8.3-2ubuntu0.1) ...
azureuser@VM2:~$ traceroute VM1
traceroute to VM1 (10.1.0.4), 30 hops max, 60 byte packets
 1 vm3.internal.cloudapp.net (10.1.255.4)  2.607 ms  2.592 ms  2.580 ms
 2 vm1.internal.cloudapp.net (10.1.0.4)  4.499 ms  4.487 ms *
```

Outcome:

Successfully configured custom route tables and a virtual appliance to control traffic flow between subnets. Verified routing by testing traffic paths, demonstrating advanced network design and troubleshooting skills.

Task 06: Configure Global Virtual Network Peering

Objective:

Establish secure, private connectivity between virtual networks in different Azure regions by configuring global virtual network peering.

Steps Performed:

- Reviewed existing VNets, their address spaces, and regions to plan peering.
- Added peering links from the first VNet to the second VNet.
- Remediated overlapping address spaces to ensure unique, non-conflicting CIDRs.
- Configured the reverse peering from the second VNet to the first to establish bidirectional communication.

Review existing Azure resources

The screenshot displays the Microsoft Azure portal interface for a virtual machine named 'VM1'. The left-hand navigation pane shows various management options such as Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Connect, Networking, Settings, Availability + scale, Security, Backup + disaster recovery, Operations, Monitoring, Insights, Alerts, Metrics, Diagnostic settings, Logs, Workbooks, and Automation. The main content area is titled 'VM1' and includes a search bar and a 'Help me copy this VM in any region' button. Below this, there are action buttons: Connect, Start, Restart, Stop, Hibernate, Capture, Delete, Refresh, Open in mobile, and Feedback. The 'Essentials' section provides key information: Resource group (RG1lod48136516), Status (Running), Location (East US), Subscription (Challenge Labs 01), Subscription ID (e305e6bf-4d96-4a4f-a0ec-f2f4d37c75b4), Operating system (Linux (ubuntu 18.04)), Size (Standard F2), Public IP address (172.190.184.153), Virtual network/subnet (VNET1/subnet1), DNS name (Not configured), Health state (-), and Time created (2/8/2025, 6:31 PM UTC). A 'JSON View' link is also present. The 'Properties' tab is active, showing details for the Virtual machine (Computer name: VM1, Operating system: Linux (ubuntu 18.04), VM generation: V1, VM architecture: x64) and Networking (Public IP address: 172.190.184.153, Network interface: VM1-nic, Public IP address (IPv6): -, Private IP address: 10.1.0.4).

VNET1 - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/e305e6bf-4d96-4a4f-a0ec-f2f4d37c75b4/resourceGroups/RG1lod48136516/providers/Micr...

Microsoft Azure Search resources, services, and docs (G+)

Admin-48136516@cloudslice.onmicrosoft.com

Home > Virtual networks > VNET1

VNET1 Virtual network

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Settings

Address space

Connected devices

Subnets

Bastion

DDoS protection

Firewall

Microsoft Defender for Cloud

Network manager

DNS servers

Peerings

Service endpoints

Private endpoints

Properties

Locks

Essentials

Resource group (move) RG1lod48136516

Location (move) East US

Subscription (move) Challenge Labs 01

Subscription ID e305e6bf-4d96-4a4f-a0ec-f2f4d37c75b4

Address space 10.1.0.0/16

DNS servers Azure provided DNS service

Flow timeout Configure

BGP community string Configure

Virtual network ID c4c46e13-5fbd-49e7-b4de-eca0449b9035

Tags (edit) Add tags

Topology Properties Capabilities (5) Recommendations Tutorials

DDoS protection Configure additional protection from distributed denial of service attacks. Not configured

Azure Firewall Protect your network with a stateful L3-L7 firewall. Not configured

Peerings Seamlessly connect two or more virtual networks. Not configured

VNET1 - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/e305e6bf-4d96-4a4f-a0ec-f2f4d37c75b4/resourceGroups/RG1lod48136516/providers/Micr...

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Admin-48136516@cloudslice.onmicrosoft.com

Home > Virtual networks > VNET1

VNET1 | Subnets Virtual network

Search

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+ Subnet Refresh Manage users Delete

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

Search subnets

☐	Name ↑	IPv4	IPv6	Available IPs	Delegated to	Security group	Route table	
☐	subnet1	10.1.0.0/24	-	250	-	-	-	

Showing 1 subnet

Give feedback

Add a virtual network peering

Add peering - Microsoft Azure - Google Chrome

portal.azure.com/#view/Microsoft_Azure_Network/AddEditPeering.ReactView/IsAdd~/true/virtualNetworkId/%2Fsubscriptions%2Fe305e6bf-4d96-4a4f-a0ec-f2f4d3...

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

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Home > Virtual networks > VNET2 | Peerings >

Add peering

VNET2

Peering link name *

VNET1-to-VNET2

Virtual network deployment model

Resource manager

Classic

I know my resource ID

Subscription *

Challenge Labs 01

Virtual network *

VNET1 (RG1lod48136516)

Address space "10.1.0.0/16" overlaps with address space "10.1.0.0/16" of virtual network "VNET1". Virtual networks with overlapping address space cannot be peered. If you intend to peer these virtual networks, change address space "10.1.0.0/16".

Remote virtual network peering settings

Allow "VNET1" to access "VNET2"

☒

Allow "VNET1" to receive forwarded traffic from "VNET2"

☐

Allow gateway or route server in "VNET1" to forward traffic to "VNET2"

☐

Enable "VNET1" to use "VNET2's" remote gateway or route server

☐

Add

Cancel

Give feedback

VNET2 - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/e305e6bf-4d96-4a4f-a0ec-f2f4d37c75b4/resourceGroups/RG1lod48136516/providers/Micr...

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

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CLOUDSLICE (CLOUDSLICE.ONML...

Home > Virtual networks > VNET2

VNET2 | Address space

Virtual network

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The address space for a virtual network is composed of one or more non-overlapping address ranges that are specified in CIDR notation. IP Address Management (IPAM) is recommended to simplify address management and avoid overlapping address space. When not using IPAM, it is recommended to use an address range that is not globally routable, such as 172.16.0.0/12, or a range defined in RFC 1918 or RFC 6598. [Learn more](#)

Address space	Address range	Address count
10.1.0.0/16	10.1.0.0 - 10.1.255.255	65,536
10.2.0.0/16	10.2.0.0 - 10.2.255.255	65,536

Add additional address range

Updating the address space of a virtual network that has peers will cause the peered virtual networks to not be able to connect to this new address space until you perform a sync operation on the peerings. You can sync the peered virtual networks in the peerings tab, but requires you have contributor permissions on the peered virtual networks. [Learn more](#)

Peered virtual network address space

Peering name	Peered to	Address space	Address range
No results.			

Save

Cancel

Give feedback

38 | Page

Remediate overlapping address spaces

Add a subnet - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/e305e6bf-4d96-4a4f-a0ec-f2f4d37c75b4/resourceGroups/RG1od48136516/providers/Micr...

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Home > Virtual networks > VNET2

VNET2 | Subnets Virtual network

Search

+ Subnet Refresh

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

Search subnets

Showing 1 subnet

Add a subnet

Subnet purpose

Name *

IPv4

Include an IPv4 address space ☒

IPv4 address range
10.2.0.0 - 10.2.255.255

Starting address *

Size

Subnet address range

IPv6

Include an IPv6 address space ☐ This virtual network has no IPv6 address ranges.

Private subnet

Private subnets enhance security by not providing default outbound access. To enable outbound connectivity for virtual machines to access the internet, it is necessary to explicitly grant outbound access. A NAT gateway is the recommended way to provide outbound connectivity for virtual machines in the subnet. [Learn more](#)

Enable private subnet (no default outbound access) ☐

Security

Simplify internet access for virtual machines by using a network address translation gateway. Filter subnet traffic using a network security

[Add](#) [Cancel](#) [Give feedback](#)

VNET2 - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/e305e6bf-4d96-4a4f-a0ec-f2f4d37c75b4/resourceGroups/RG1od48136516/providers/Micr...

Microsoft Azure Search resources, services, and docs (G+/I) Copilot Admin-48136516@clou... CLOUDSLICE (CLOUDSLICE.ONML...

Home > Virtual networks > VNET2

VNET2 | Subnets Virtual network

Search

+ Subnet Refresh Manage users Delete

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

Search subnets

	Name	IPv4	IPv6	Available IPs	Delegated to	Security group	Route table
☐	subnet1	10.1.0.0/24	-	250	-	-	View
☐	subnet2	10.2.0.0/24	-	251	-	-	View

Showing 2 subnets

[Give feedback](#)

VM2-nic - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/e305e6bf-4d96-4a4f-a0ec-f2f4d37c75b4/resourceGroups/RG1od48136516/providers/Micr...

Microsoft Azure

Home > Resource groups > RG1od48136516 > VM2-nic

VM2-nic | IP configurations

Network interface

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Refresh

The virtual machine associated with this network interface will be restarted to utilize the new subnet.

IP Settings

Enable IP forwarding

Virtual network

Subnet *

subnet2 (10.2.0.0/24) 251 free IP addresses

251 free IP addresses

Private and public IP addresses can be assigned to a virtual machine's network interface controller. You can add as many private and public IPv4 addresses as necessary to a network interface, within the limits listed in the Azure limits article. [Learn more](#)

+ Add

Make primary

Delete

	Name	IP Version	Type	Private IP Address	Public IP Address
☐	ipconfig1	IPv4	Primary	10.1.0.4 (Dynamic)	40.65.219.190 (VM2-ip)

Apply

Discard changes

Give feedback

Delete Subnets - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/e305e6bf-4d96-4a4f-a0ec-f2f4d37c75b4/resourceGroups/RG1od48136516/providers/Micr...

Microsoft Azure

Home > Virtual networks > VNET2

VNET2 | Subnets

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Private endpoints

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+ Subnet

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Manage users

Delete

Create subnets to segment the virtual network address space into smaller pieces, each with its own IP address range. You can also remove an IP address from the subnet.

Search subnets

	Name	IP Version	IPv4	IPv6
☒	subnet1	10.1.0.0/24	-	-
☐	subnet2	10.2.0.0/24	-	-

Showing 2 subnets

Delete Subnets

The following subnets will be deleted from virtual network - 'VNET2'.
Subnets to be deleted (1)

subnet1

Delete confirmation

Deleting the selected resources is a permanent action and cannot be undone.

Delete

Go back

Enter 'delete' to confirm deletion *

delete

Delete

Cancel

Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/e305e6bf-4d96-4a4f-a0ec-f2fd37c75b4/resourceGroups/RG1lod48136516/providers/Mic...

Microsoft Azure Search resources, services, and docs (G+/)

Admin-48136516@cloudslice.onmicrosoft.com

Home > VNET2

VNET2 | Address space

Virtual network

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Settings

Address space

Connected devices

Subnets

Bastion

DDoS protection

Firewall

Microsoft Defender for Cloud

Network manager

DNS servers

Peerings

Service endpoints

Private endpoints

Properties

Locks

The address space for a virtual network is composed of one or more non-overlapping address ranges that are specified in CIDR notation. IP Address Management (IPAM) is recommended to simplify address management and avoid overlapping address space. When not using IPAM, it is recommended to use an address range that is not globally routable, such as 172.16.0.0/12, or a range defined in RFC 1918 or RFC 6598. [Learn more](#)

Address space	Address range	Address count
10.2.0.0/16	10.2.0.0 - 10.2.255.255	65,536
10.4.0.0/16	10.4.0.0 - 10.4.255.255	65,536

Add additional address range

Peered virtual network address space

Peering name	Peered to	Address space	Address range
No results.			

Save Cancel

Give feedback

Configure a global virtual network peering

VM2-nic - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/a59a8733-537b-465e-a883-4b94bfbf1069/resourceGroups/RG1lod47912124/provider...

Microsoft Azure Search resources, services, and docs (G+/)

Admin-47912124@cloudslice.onmicrosoft.com

Home > Resource groups > RG1lod47912124 > VM2-nic

VM2-nic | IP configurations

Network interface

Overview

Activity log

Access control (IAM)

Tags

Settings

IP configurations

DNS servers

Network security group

Properties

Locks

Monitoring

Automation

Help

Refresh

IP Settings

Enable IP forwarding ☐

Virtual network [VNET2](#)

Subnet * 250 free IP addresses

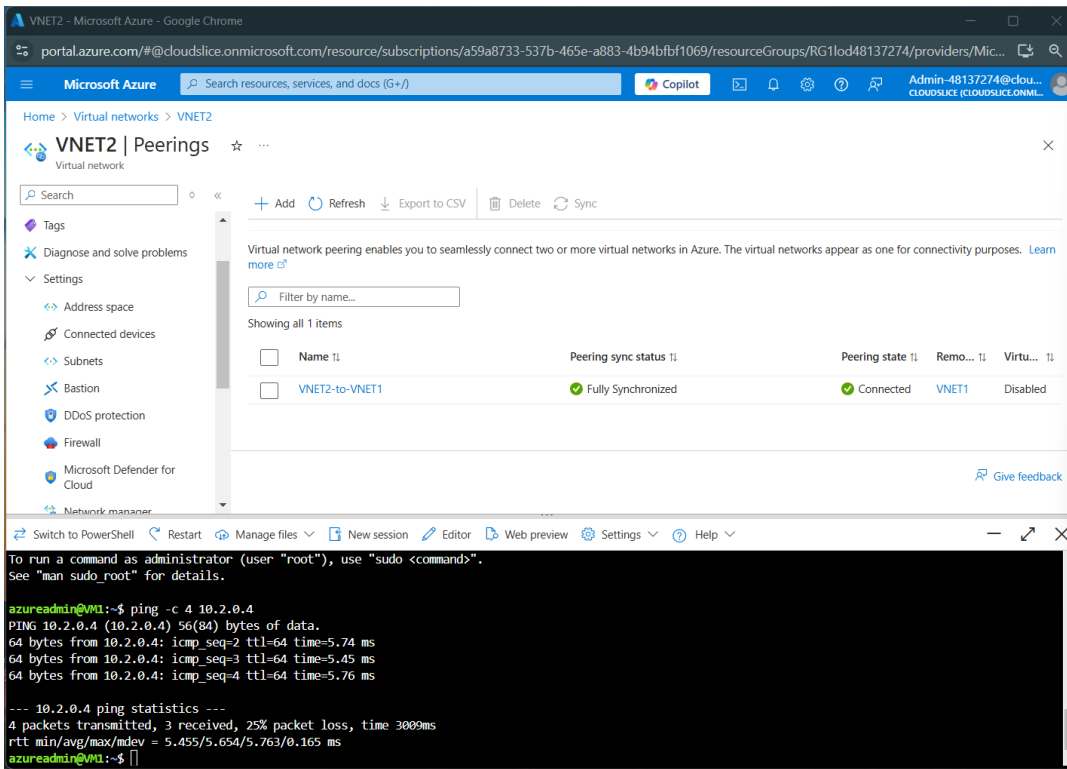
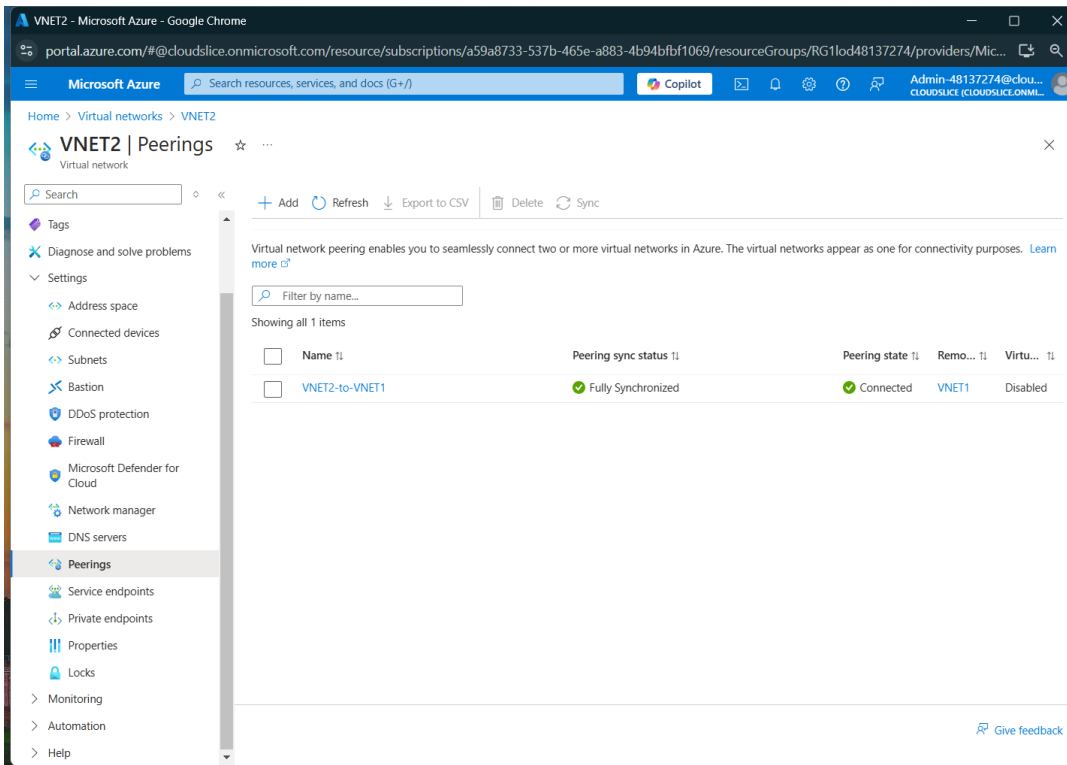
Private and public IP addresses can be assigned to a virtual machine's network interface controller. You can add as many private and public IPv4 addresses as necessary to a network interface, within the limits listed in the [Azure limits article](#). [Learn more](#)

+ Add Make primary Delete

Name	IP Version	Type	Private IP Address	Public IP Address
☐ ipconfig1	IPv4	Primary	10.2.0.4 (Dynamic)	40.84.28.64 (VM2-ip)

Apply Discard changes

Give feedback



Outcome:

Successfully configured global virtual network peering across Azure regions, ensuring secure, private, and low-latency connectivity between VNets. Remediated overlapping address spaces to enable the peering relationship.

Task 07: Enable High Availability Sets by Using Availability Sets

Objective:

Deploy multiple Azure virtual machines in an availability set and configure a load balancer to provide high availability and traffic distribution.

Steps Performed:

- Created an availability set to ensure fault domain and update domain distribution.
- Deployed multiple virtual machines to the availability set.
- Configured a public Azure Load Balancer with a health probe and load balancing rule to distribute web traffic across the VMs.

Create an availability set

Create availability set - Microsoft Azure - Google Chrome

portal.azure.com/#create/Microsoft.AvailabilitySet-ARM

Microsoft Azure Search resources, services, and docs (G+)

Home > Create a resource > Marketplace > Availability Set >

Create availability set

Enjoy more features at the same price with a VM Scale Set (VMSS). While availability sets offer high availability only, a flexible VMSS provides automatic scaling, optimization, and recovery, along with simple, central group management for up to 2,000 VMs-at no added cost. Pay only for capacity you use, while mixing VM sizes, zones, and fault domains. [Click to learn more]

Basics Advanced Tags Review + create

An Availability Set is a logical grouping capability for isolating VM resources from each other when they're deployed. Azure makes sure that the VMs you place within an Availability Set run across multiple physical servers, compute racks, storage units, and network switches. If a hardware or software failure happens, only a subset of your VMs are impacted and your overall solution stays operational. Availability Sets are essential for building reliable cloud solutions. [Learn more about availability sets.](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Challenge Labs 04

Resource group * corp-datalod48018539 [Create new](#)

Instance details

Name * AVSet-48018539

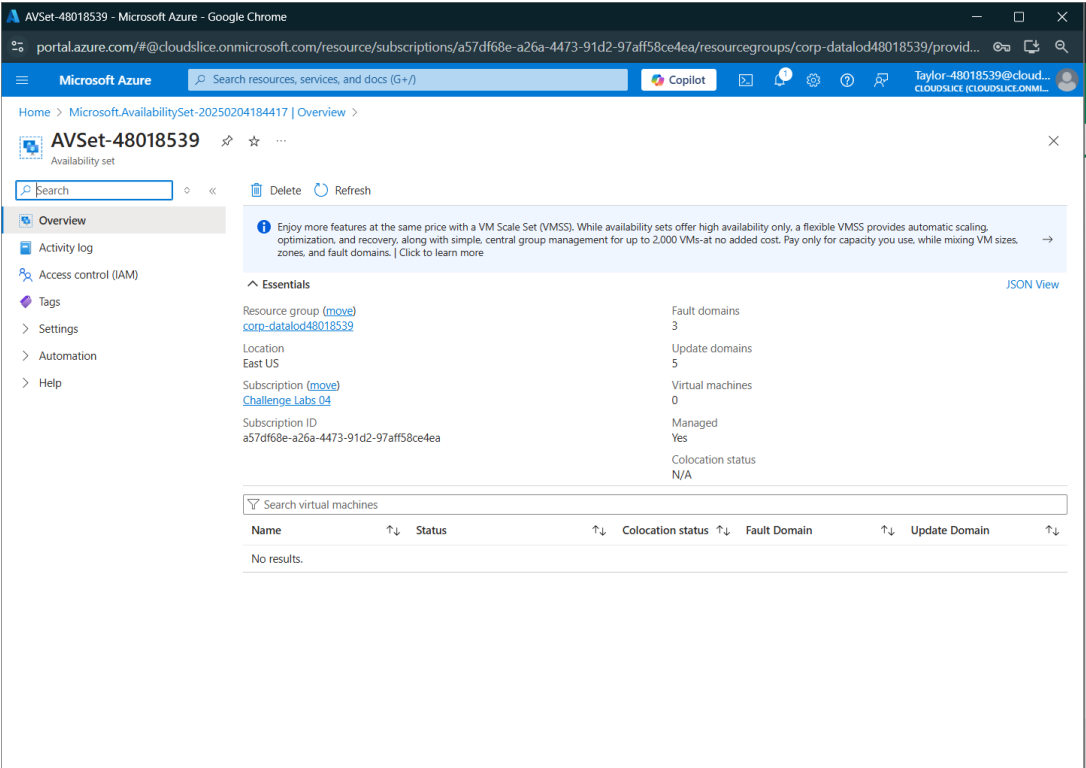
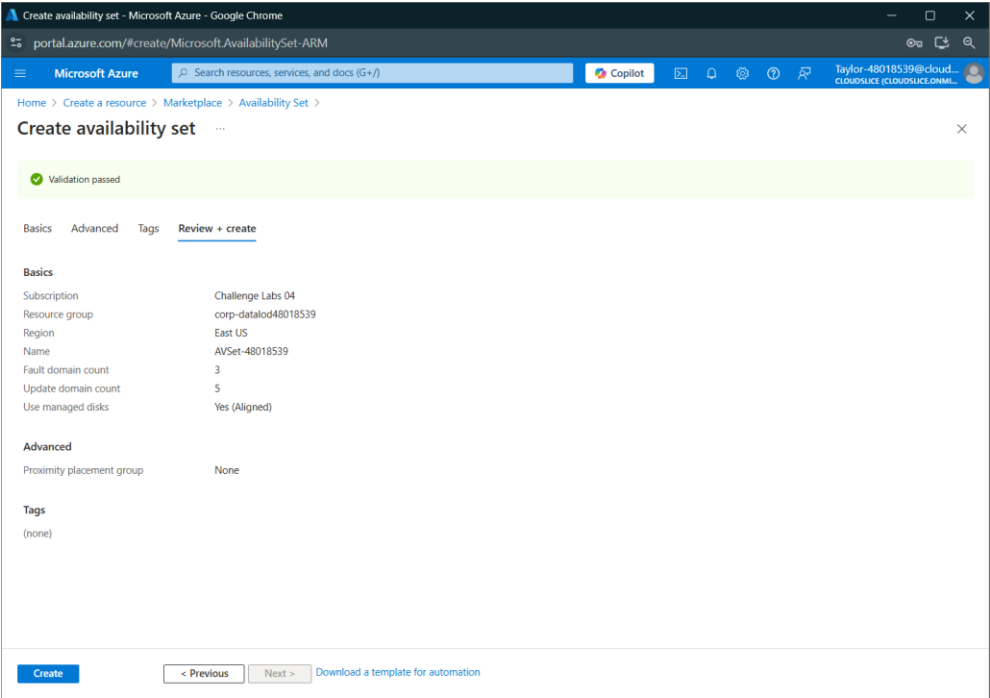
Region * (US) East US

Fault domains 3

Update domains 5

Use managed disks No (Classic) Yes (Aligned)

[Review + create](#) < Previous Next: Advanced >



Create virtual network - Microsoft Azure - Google Chrome

portal.azure.com/#create/Microsoft.VirtualNetwork-ARM

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

Taylor-48018539@cloud...
CLOUDSLICE (CLOUDSLICE.ONML)

Home > Create a resource > Marketplace > Virtual network >

Create virtual network

BasicsSecurityIP addressesTagsReview + create

[View automation template](#)

Basics

Subscription

Challenge Labs 04

Resource Group

corp-datalod48018539

Name

AVSet-Vnet

Region

East US

Security

Azure Bastion

Disabled

Azure Firewall

Disabled

Azure DDoS Network Protection

Disabled

IP addresses

Address space

10.0.0.0/16 (65,536 addresses)

Subnet

default (10.0.0.0/24) (256 addresses)

Tags

PreviousNextCreate

[Give feedback](#)

AVSet-Vnet

Virtual network

Search

MoveDeleteRefreshGive feedback

Overview

Activity log

Access control (IAM)

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Diagnose and solve problems

Settings

Address space

Connected devices

Subnets

Bastion

DDoS protection

Firewall

Microsoft Defender for Cloud

Network manager

DNS servers

Peerings

Service endpoints

Private endpoints

Properties

Locks

Essentials

Resource group (move)

corp-datalod48018539

Location (move)

East US

Subscription (move)

Challenge Labs 04

Subscription ID

a57df68e-a26a-4473-91d2-97aff58ce4ea

Address space

10.0.0.0/16

DNS servers

Azure provided DNS service

Configure

Flow timeout

Configure

BGP community string

Configure

Virtual network ID

b3a8d37f-9577-4c74-be51-afaabd7c712

Tags (edit)

Add tags

TopologyPropertiesCapabilities (5)RecommendationsTutorials

DDoS protection

Configure additional protection from distributed denial of service attacks.

Not configured

Azure Firewall

Protect your network with a stateful L3-L7 firewall.

Not configured

Peerings

Seamlessly connect two or more virtual networks.

Not configured

Deploy Azure virtual machines to an availability set

Create a virtual machine - Microsoft Azure - Google Chrome

portal.azure.com/#create/Microsoft.VirtualMachine

Microsoft Azure

Search resources, services, and docs (G+/I)

Copilot

Taylor-48018539@cloud...
CLOUDSLICE (CLOUDSLICE.ONML...

Home > Create a resource >

Create a virtual machine

Help me create a low cost VM Help me create a VM optimized for high availability Help me choose the right VM size for my workload

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Challenge Labs 04

Resource group * corp-datalod48018539
[Create new](#)

Instance details

Virtual machine name * VMFE1-48018539

Region * (US) East US

Availability options Availability set

Based on your input, you might want to consider creating this resource as a virtual machine scale set, which allows you to manage, configure and scale load balanced virtual machines. [Create as VMSS](#)

Availability set * AVSet-48018539
[Create new](#)

Security type Standard

Image * Windows Server 2019 Datacenter - x64 Gen2
[See all images](#) | [Configure VM generation](#)

This image is compatible with additional security features. [Click here to learn more.](#)

< Previous Next : Disks > Review + create

Give feedback

Create a virtual machine - Microsoft Azure - Google Chrome

portal.azure.com/#create/Microsoft.VirtualMachine

Microsoft Azure

Search resources, services, and docs (G+/I)

Copilot

Taylor-48018539@cloud...
CLOUDSLICE (CLOUDSLICE.ONML...

Home > Create a resource >

Create a virtual machine

Validation passed

Help me create a low cost VM Help me create a VM optimized for high availability Help me choose the right VM size for my workload

Basics

Subscription Challenge Labs 04

Resource group corp-datalod48018539

Virtual machine name VMFE1-48018539

Region East US

Availability options Availability set

Zone options Self-selected zone

Availability set AVSet-48018539

Security type Standard

Image Windows Server 2019 Datacenter - Gen2

VM architecture x64

Size Standard B2s (2 vcpus, 4 GiB memory)

Enable Hibernation No

Username AzureAdmin

Public inbound ports RDP

Already have a Windows license? No

Azure Spot No

Disks

OS disk size Image default

OS disk type Standard HDD LRS

< Previous Next > Create

[Download a template for automation](#) [Give feedback](#)

Create a virtual machine - Microsoft Azure - Google Chrome

portal.azure.com/#create/Microsoft.VirtualMachine-ARM

Microsoft Azure

Search resources, services, and docs (G+/f)

Copilot

Taylor-48018539@cloud...
CLOUDSLICE (CLOUDSLICE.ONML...)

Home > Virtual machines >

Create a virtual machine

Validation passed

Help me create a low cost VM

Help me create a VM optimized for high availability

Help me choose the right VM size for my workload

Details

Subscription

Resource group

Virtual machine name

Region

Availability options

Zone options

Availability set

Security type

Image

VM architecture

Size

Enable Hibernation

Username

Public inbound ports

Already have a Windows license?

Azure Spot

Challenge Labs 04

corp-datalod48018539

VMFE2-48018539

East US

Availability set

Self-selected zone

AVSet-48018539

Standard

Windows Server 2019 Datacenter - Gen2

x64

Standard B2s (2 vcpus, 4 GiB memory)

No

AzureAdmin

RDP

No

No

Disks

OS disk size

OS disk type

Use managed disks

Image default

Premium SSD LRS

Yes

< Previous

Next >

Create

Download a template for automation

Give feedback

Virtual machines - Microsoft Azure - Google Chrome

portal.azure.com/#view/HubsExtension/BrowseResource/resourceType/Microsoft.Compute%2FVirtualMachines

Microsoft Azure

Search resources, services, and docs (G+/f)

Copilot

Taylor-48018539@cloud...
CLOUDSLICE (CLOUDSLICE.ONML...)

Home >

Virtual machines

cloudslice (cloudslice.onmicrosoft.com)

+ Create

Switch to classic

Reservations

Manage view

Refresh

Export to CSV

Open query

Assign tags

Start

Restart

Stop

...

Filter for any field...

Subscription equals all

Type equals all

Add filter

Showing 1 to 2 of 2 records.

No grouping

List view

	Resource group	Location	Status	Operating system	Size	Public IP address	Disks	Update status
04	corp-datalod48018539	East US	Running	Windows	Standard_B2s	172.190.162.92	1	Enable periodic assess... ***
04	corp-datalod48018539	East US	Running	Windows	Standard_B2s	52.168.150.31	1	Enable periodic assess... ***

< Previous

Page 1 of 1

Next >

Give feedback

Configure an Azure load balancer

Add frontend IP configuration - Microsoft Azure - Google Chrome

portal.azure.com/#create/Microsoft.LoadBalancer-ARM

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

Taylor-48018539@cloud...
CLOUDSLICE (CLOUDSLICE.ONML)

Home > Create a resource > Marketplace > Load Balancer >

Create load balancer

BasicsFrontend IP configurationBackend poolsInbound rulesOutbound rulesTagsRe

A frontend IP configuration is an IP address used for inbound and/or outbound communication as defined within lo

+ Add a frontend IP configuration

Name ↑↓	IP address
Add a frontend IP to get started	

Add frontend IP configuration

Name *

LBIP-48018539

IP version

☒ IPv4

☐ IPv6

IP type

☒ IP address

☐ IP prefix

Public IP address *

Select public IP address

Create new

Gateway Load balancer ⓘ

None

Review + create

< Previous

Next: Backend pools >

Download a template for automation

Save

Cancel

Give feedback

Create load balancer - Microsoft Azure - Google Chrome

portal.azure.com/#create/Microsoft.LoadBalancer-ARM

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

Taylor-48018539@cloud...
CLOUDSLICE (CLOUDSLICE.ONML)

Home > Create a resource > Marketplace > Load Balancer >

Create load balancer

Validation passed

BasicsFrontend IP configurationBackend poolsInbound rulesOutbound rulesTagsReview + create

Basics

Subscription	Challenge Labs 04
Resource group	corp-dataod48018539
Name	LBFE-48018539
Region	East US
SKU	Standard
Tier	Regional
Type	Public

Frontend IP configuration

Frontend IP configuration name	LBIP-48018539
Frontend IP configuration IP address	To be created

Backend pools

None

Inbound rules

None

Create

< Previous

Next >

Download a template for automation

Give feedback

LBFE-48018539 - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/a57df68e-a26a-4473-91d2-97aff58ce4ea/resourcegroups/corp-datalod48018539/provid...

Microsoft Azure

Search resources, services, and docs (G+/f)

Copilot

Taylor-48018539@cloud...
CLOUDSLICE (CLOUDSLICE.ONML...

Home > Microsoft.LoadBalancer-20250205053704 | Overview >

LBFE-48018539

Load balancer

Search

Move Delete Refresh Give feedback

Overview

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Settings

Frontend IP configuration

Backend pools

Health probes

Load balancing rules

Inbound NAT rules

Outbound rules

Properties

Locks

Monitoring

Automation

Help

Essentials

Resource group (move)
corp-datalod48018539

Location
East US

Subscription (move)
Challenge Labs 04

Subscription ID
a57df68e-a26a-4473-91d2-97aff58ce4ea

SKU
Standard

Tags (edit)
Add tags

See more

Backend pool
-

Load balancing rule
-

Health probe
-

NAT rules
0 inbound

Tier
Regional

JSON View

Configure high availability and scalability for your applications

Create highly-available and scalable applications in minutes by using built-in load balancing for cloud services and virtual machines. Azure Load Balancer supports TCP/UDP-based protocols and protocols used for real-time voice and video messaging applications. [Learn more](#)

Balance IPv4 and IPv6 addresses

Native dual-stack endpoints help meet regulatory requirements and address the fast-growing number of devices in mobile and IoT. [Learn more](#)

Build highly reliable applications

Load Balancer improves application uptime by routing traffic to healthy nodes. [Learn more](#)

View health probes

Add backend pool - Microsoft Azure - Google Chrome

portal.azure.com/#view/Microsoft_Azure_Network/RegionalLBBackendPoolManage.ReactView/loadBalancerId/%2Fsubscriptions%2Fa57df68e-a26a-4473-91d2-9...

Microsoft Azure

Search resources, services, and docs (G+/f)

Copilot

Taylor-48018539@cloud...
CLOUDSLICE (CLOUDSLICE.ONML...

Home > LBFE-48018539 | Backend pools >

Add backend pool

LBFE-48018539

Name *

LBFE-48018539

Virtual network

AVSet-Vnet (corp-datalod48018539)

Backend Pool Configuration

NIC

IP address

IP configurations

IP configurations associated to virtual machines and virtual machine scale sets must be in same location as the load balancer and be in the same virtual network.

+ Add

Remove

Resource Name	Resource group	Type	IP configuration	IP Address	Availabi...
VMFE1-48018539	corp-datalod48018539	Virtual machine	ipconfig1	10.0.0.4	AVSET-4801
VMFE2-48018539	corp-datalod48018539	Virtual machine	ipconfig1	10.0.0.5	AVSET-4801

Save

Cancel

Give feedback

49 | Page

LBFE-48018539 - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/a57df68e-a26a-4473-91d2-97aff58ce4ea/resourceGroups/corp-datalod48018539/provid...

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

Taylor-48018539@cloud...
CLOUDSLICE (CLOUDSLICE.ONMIL...

Home > LBFE-48018539

LBFE-48018539 | Backend pools

Load balancer

Search

+ Add Refresh

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Backend pools

Health probes

Load balancing rules

Inbound NAT rules

Outbound rules

Properties

Locks

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Automation

Help

The backend pool is a critical component of the load balancer. The backend pool defines the group of resources that will serve traffic for a given load-balancing rule. [Learn more.](#)

Search

Add filter

Backend poolResource NameIP addressNetwork interf...Availability zoneRules countResource StatusAdmin state

LBBE-48018539 (2)

LBBE-48018539VMFE1-4801853910.0.0.4vmfe1-4801853976-0RunningNone

LBBE-48018539VMFE2-4801853910.0.0.5vmfe2-4801853994-0RunningNone

Give feedback

LBFE-48018539 - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/a57df68e-a26a-4473-91d2-97aff58ce4ea/resourceGroups/corp-datalod48018539/provid...

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

Taylor-48018539@cloud...
CLOUDSLICE (CLOUDSLICE.ONMIL...

Home > LBFE-48018539

LBFE-48018539 | Health probes

Load balancer

Search

+ Add Refresh Give feedback

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Save probe
Successfully saved probe 'LBPB-48018539'.

Type to start filtering ...

NameProtocolPortPathUsed By

LBPB-48018539Tcp80--

LBRL-48018539 - Microsoft Azure - Google Chrome

portal.azure.com/#view/Microsoft_Azure_Network/LoadBalancingRuleManage.ReactView/loadBalancerId/%2Fsubscriptions%2Fa57df68e-a26a-4473-91d2-97aff5...

Microsoft Azure Search resources, services, and docs (G+/)

Home > LBFE-48018539 | Load balancing rules

LBRL-48018539

LBFE-48018539

A load balancing rule distributes incoming traffic that is sent to a selected IP address and port combination across a group of backend pool instances. Only backend instances that the health probe considers healthy receive new traffic. [Learn more.](#)

Name * LBRL-48018539

IP version * ☒ IPv4 ☐ IPv6

Frontend IP address * LBIP-48018539 (51.8.210.252)

Backend pool * LBBE-48018539

Protocol ☒ TCP ☐ UDP

Port * 80

Backend port * 80

Health probe * LBPB-48018539 (TCP:80) [Create new](#)

Session persistence None

Session persistence specifies that traffic from a client should be handled by the

Save Cancel Give feedback

LBFE-48018539 - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/a57df68e-a26a-4473-91d2-97aff58ce4ea/resourceGroups/corp-datalod48018539/provid...

Microsoft Azure Search resources, services, and docs (G+/)

Home > LBFE-48018539

LBFE-48018539 | Load balancing rules

Load balancer

Search Add Refresh Delete

Overview

Activity log

Access control (IAM)

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Diagnose and solve problems

Settings

- Frontend IP configuration
- Backend pools
- Health probes
- Load balancing rules**
- Inbound NAT rules
- Outbound rules
- Properties
- Locks
- Monitoring
- Automation
- Help

A load balancer rule is used to define how incoming traffic is distributed to all the instances within the backend pool. A load-balancing rule maps a given frontend IP configuration and port to multiple backend IP addresses and ports. An example would be a rule created on port 80 to load balance web traffic. [Learn more.](#)

Filter by name...

Name	Protocol	Backend pool	Health probe	Health status
LBRL-48018539	TCP/80	LBBE-48018539	LBPB-48018539	View details

Give feedback

Outcome:

Successfully implemented a high-availability infrastructure, ensuring resilient workloads with minimized downtime and evenly balanced traffic.

Task 08: Configure Blob Storage with Public Access

Objective:

Create an Azure Storage Account with a blob container configured for public access and verify that uploaded files are publicly accessible.

Steps Performed:

- Created a storage account and a container with anonymous read permissions.
- Uploaded static files to the container.
- Verified public access by retrieving files via their URLs.

Create a storage account that has a blob container

The screenshot shows the 'Create a storage account' wizard in the Microsoft Azure portal. The 'Advanced' tab is selected, displaying security settings. The 'Security' section includes options for secure transfer, anonymous access, and key access, all of which are checked. The 'Hierarchical Namespace' section is also visible, with the 'Enable hierarchical namespace' option unchecked. The 'Access protocols' section shows the 'Enable SFTP' option unchecked. At the bottom, there are buttons for 'Previous', 'Next', and 'Review + create', along with a 'Give feedback' link.

Create a storage account - Microsoft Azure - Google Chrome
portal.azure.com/#create/Microsoft.StorageAccount-ARM

Microsoft Azure Search resources, services, and docs (G+/J) Copilot CS-48019533@cloudsl... CLOUDSLICE (CLOUDSLICEONML...

Home > Create a resource > Marketplace > Storage account >

Create a storage account

Basics **Advanced** Networking Data protection Encryption Tags Review + create

Security

Configure security settings that impact your storage account.

Require secure transfer for REST API operations ☒

Allow enabling anonymous access on individual containers ☒

Enable storage account key access ☒

Default to Microsoft Entra authorization in the Azure portal ☐

Minimum TLS version

Permitted scope for copy operations (preview)

Hierarchical Namespace

Hierarchical namespace, complemented by Data Lake Storage Gen2 endpoint, enables file and directory semantics, accelerates big data analytics workloads, and enables access control lists (ACLs) [Learn more](#)

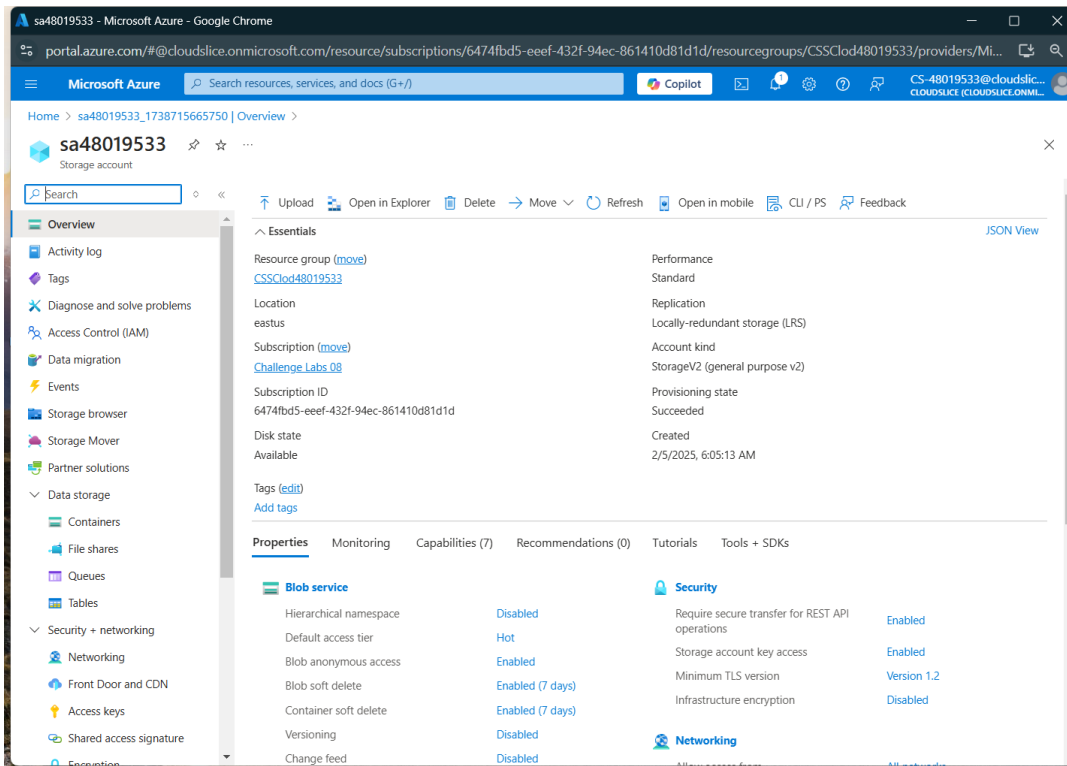
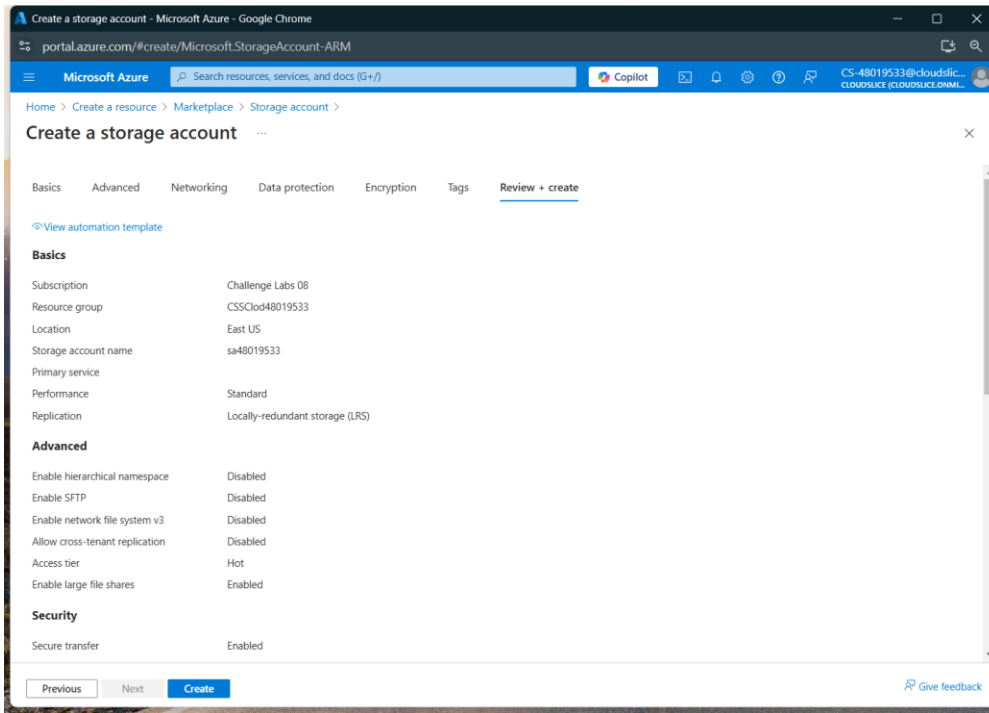
Enable hierarchical namespace ☐

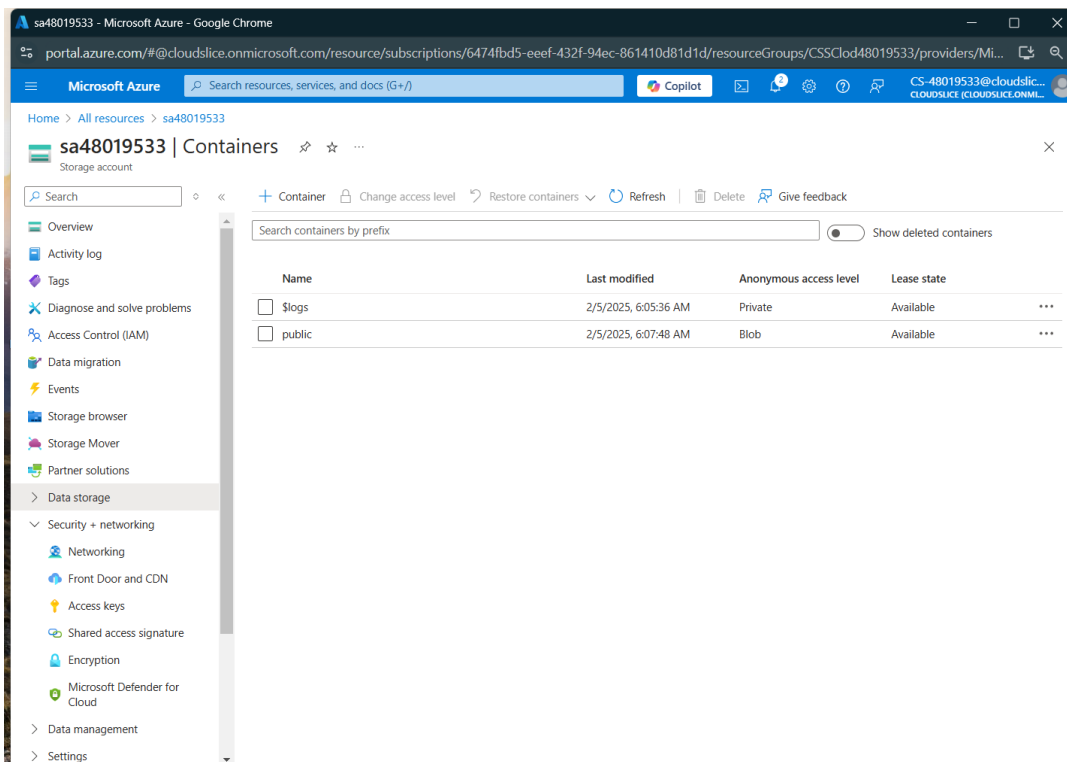
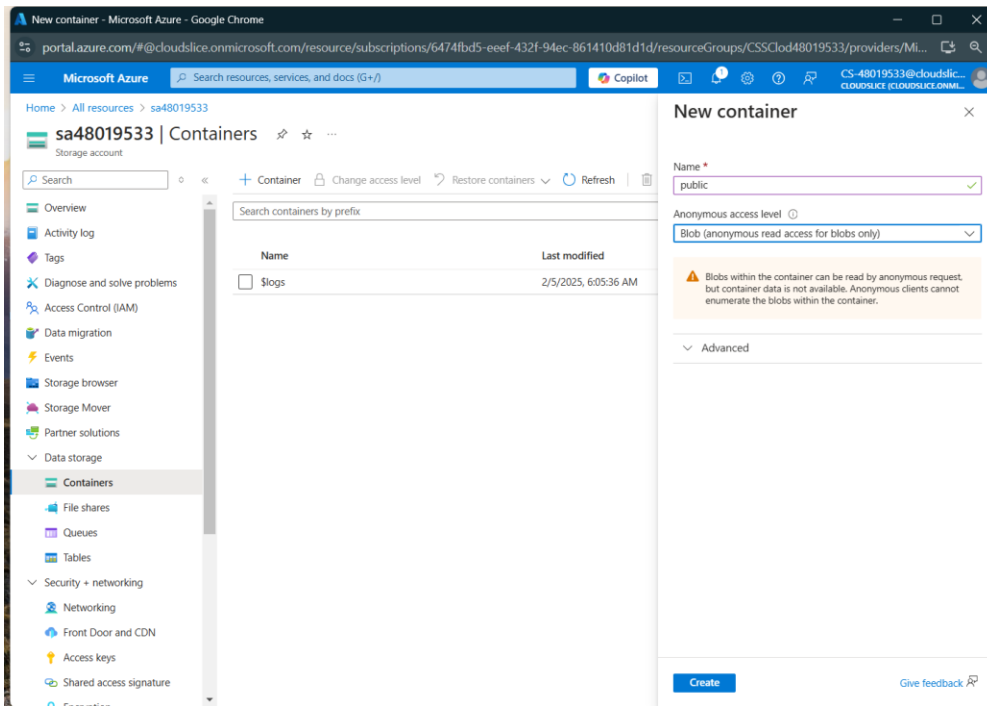
Access protocols

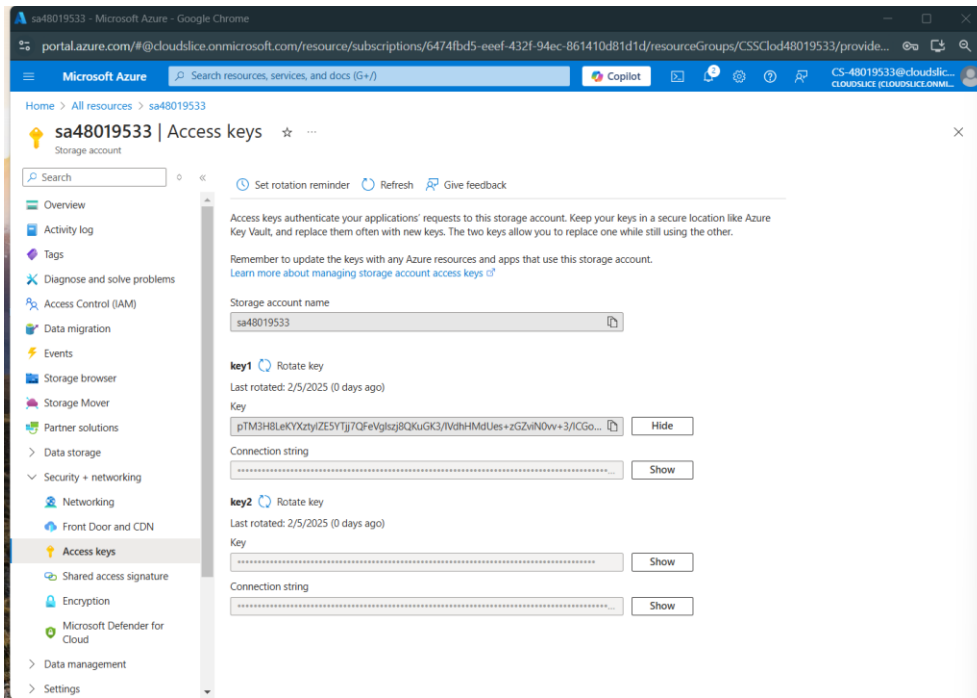
Blob and Data Lake Gen2 endpoints are provisioned by default [Learn more](#)

Enable SFTP ☐

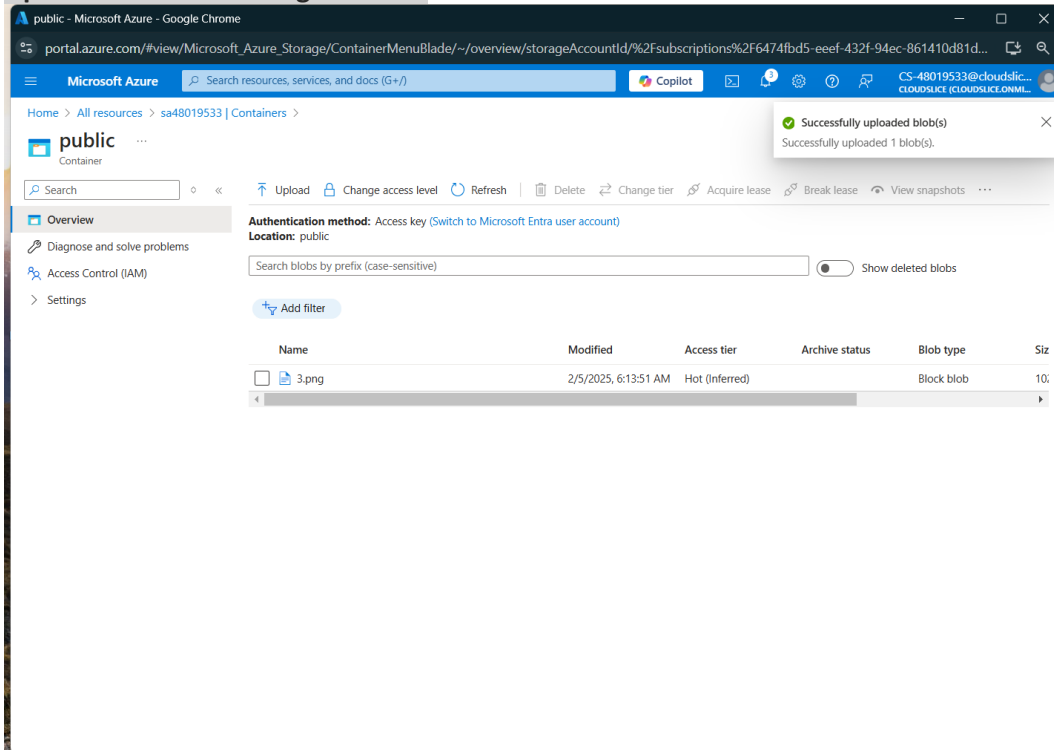
[Give feedback](#)

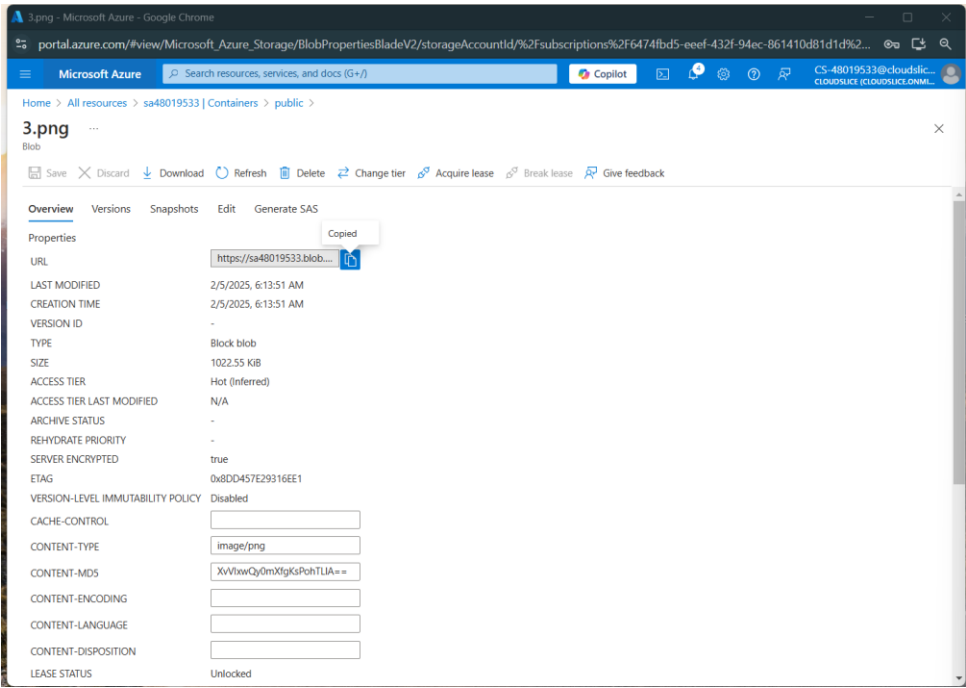
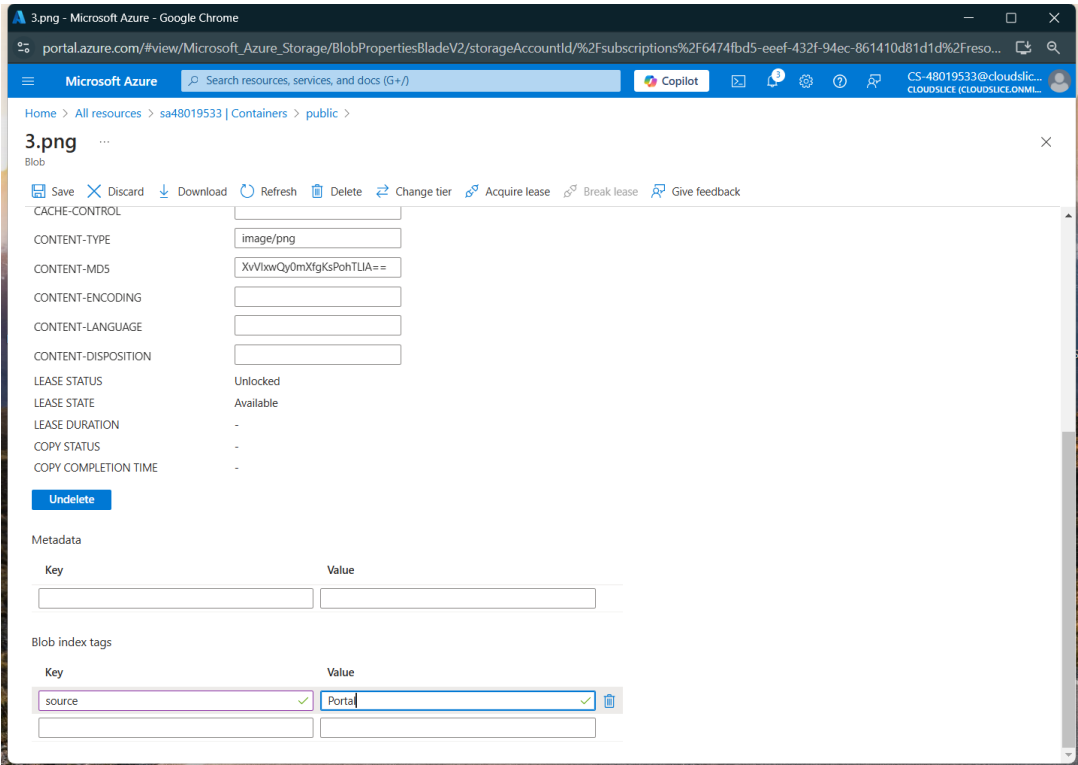




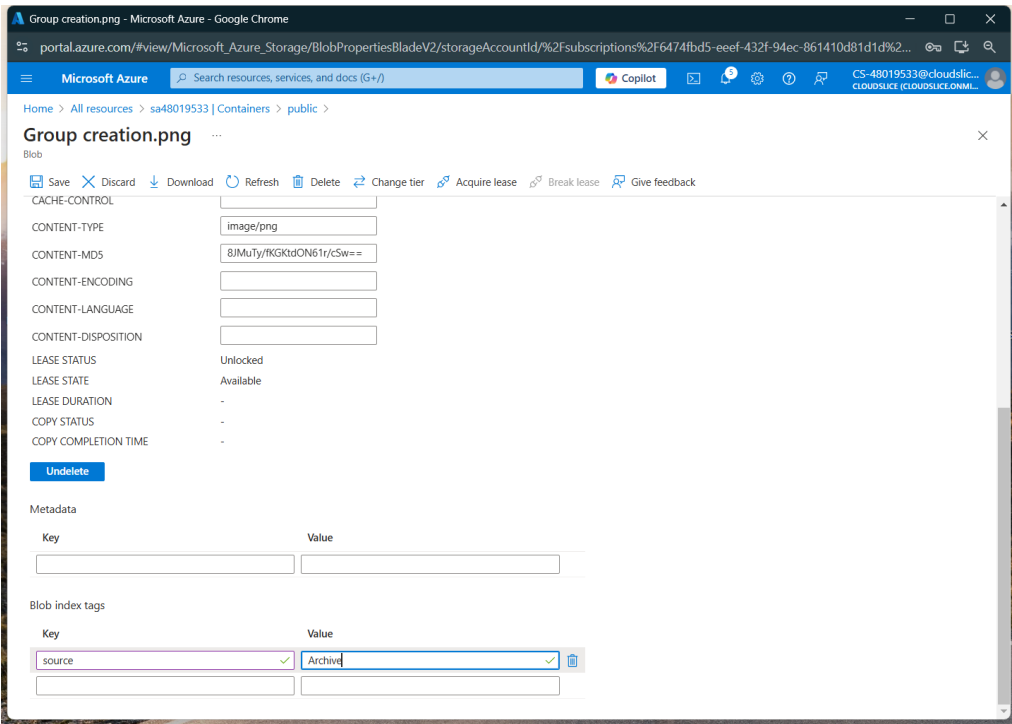
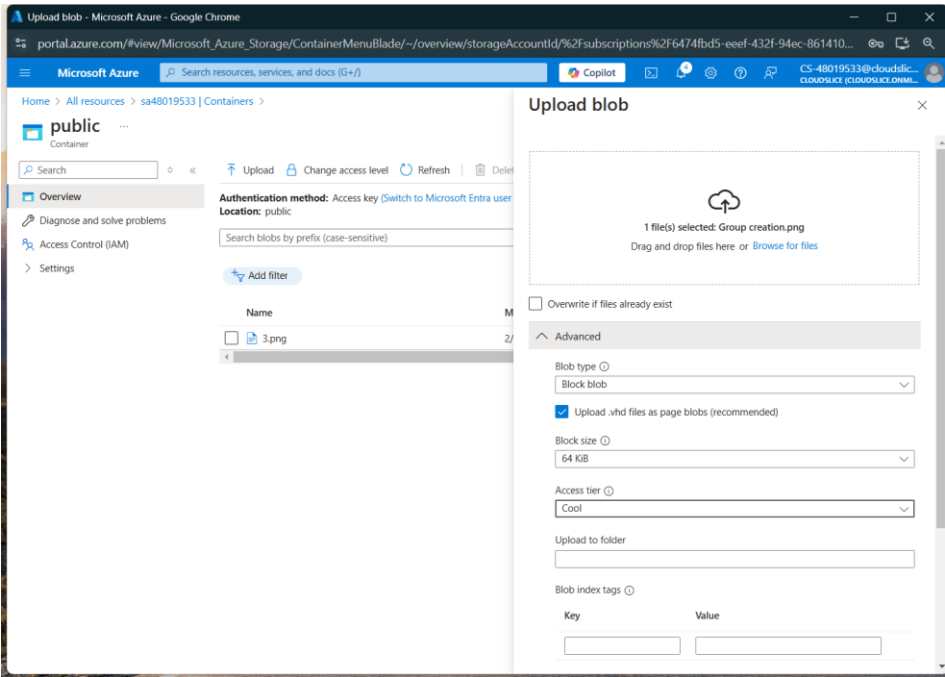


Upload files to the storage account





Test the web application



Group creation.png - Microsoft Azure - Google Chrome

portal.azure.com/#view/Microsoft_Azure_Storage/BlobPropertiesBladeV2/storageAccountId/%2fsubscriptions%2f6474fbd5-eeef-432f-94ec-861410d81d1d%2f...

Microsoft Azure Search resources, services, and docs (G+)

Home > All resources > sa48019533 | Containers > public >

Group creation.png

Blob

Save Discard Download Refresh Delete Change tier Acquire lease Break lease Give feedback

Overview Versions Snapshots Edit Generate SAS

Properties

URL <https://sa48019533.blob.core.windows.net/public/groupcreation.png>

LAST MODIFIED 2/5/2025, 6:20:54 AM

CREATION TIME 2/5/2025, 6:20:54 AM

VERSION ID -

TYPE Block blob

SIZE 15.22 KiB

ACCESS TIER Cool

ACCESS TIER LAST MODIFIED 2/5/2025, 6:20:54 AM

ARCHIVE STATUS -

REHYDRATE PRIORITY -

SERVER ENCRYPTED true

ETAG 0x8DD457F2511CB4D

VERSION-LEVEL IMMUTABILITY POLICY Disabled

CACHE-CONTROL

CONTENT-TYPE image/png

CONTENT-MD5 8JMuTyfKGkdON61r/cSw==

CONTENT-ENCODING

CONTENT-LANGUAGE

CONTENT-DISPOSITION

LEASE STATUS Unlocked

wa48019533 - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/6474fbd5-eeef-432f-94ec-861410d81d1d/resourceGroups/cssclod48019533/providers...

Microsoft Azure Search resources, services, and docs (G+)

Home > All resources >

wa48019533

Web App

Search Browse Stop Swap Restart Delete Refresh Download publish profile Reset publish profile

Click here to access Application Insights for monitoring and profiling for your app.

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Microsoft Defender for Cloud

Events (preview)

Recommended services (preview)

Deployment

Settings

Performance

App Service plan

Development Tools

API

Monitoring

Automation

Support + troubleshooting

Essentials

Resource group (move) [cssclod48019533](#)

Status Running

Location (move) East US

Subscription (move) [Challenge Labs 08](#)

Subscription ID 6474fbd5-eeef-432f-94ec-861410d81d1d

Tags (edit) displayName: Website

Default domain [wa48019533.azurewebsites.net](#)

App Service Plan dataASP (B1: 1)

Operating System Windows

Health Check Not Configured

Properties Monitoring Logs Capabilities Notifications Recommendations

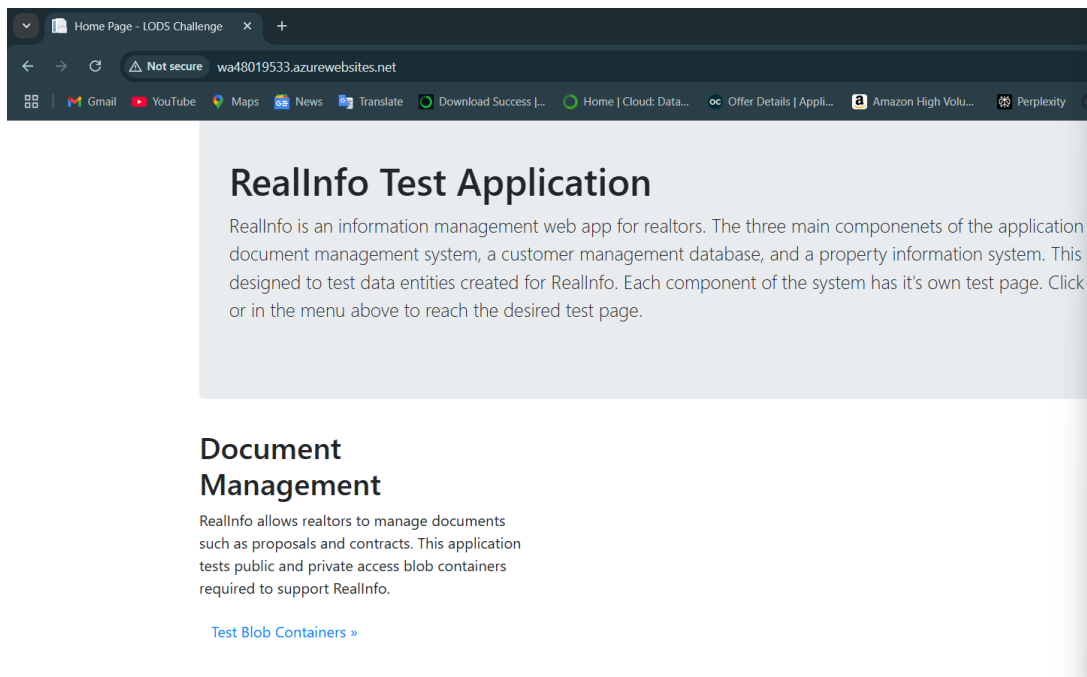
Web app

Name wa48019533

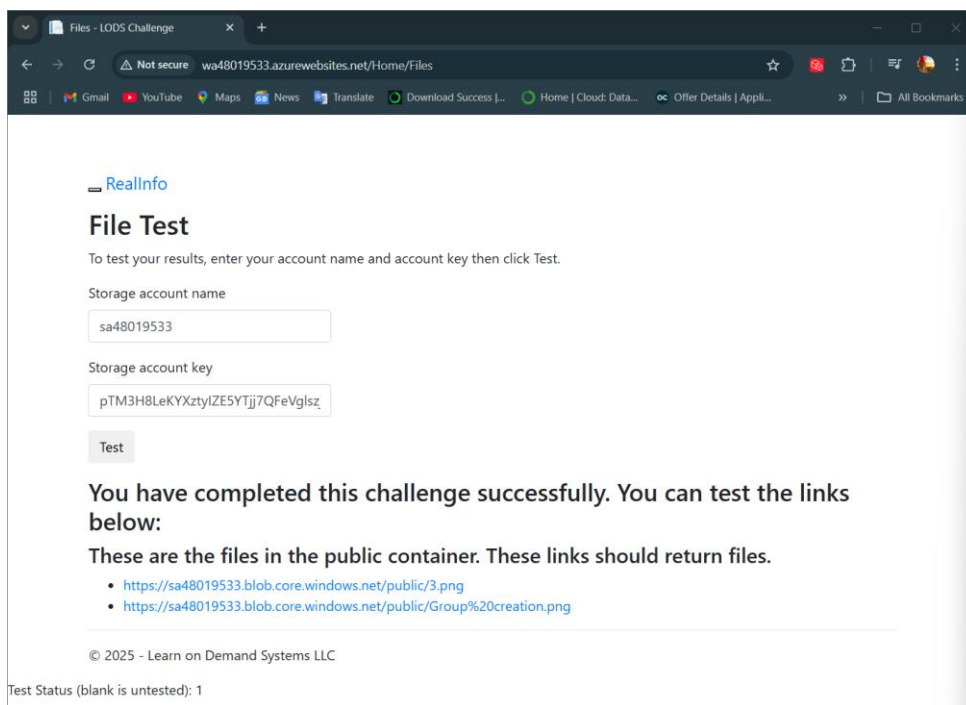
Publishing model Code

Domains

Default domain wa48019533.azurewebsites.net



Test Status (blank is untested):



Test Status (blank is untested): 1

Task 09: Configure Blob Storage with Private Access

Objective:

Create an Azure Storage Account with a private container, upload confidential files, and provide secure access using a Shared Access Signature.

Steps Performed:

- Provisioned a storage account and configured a blob container with no public access.
- Uploaded files into the private container.
- Generated a SAS token with restricted permissions and a defined expiry to securely share access.

Create a storage account that has a private blob container

The screenshot shows the 'Create a storage account' wizard in the Microsoft Azure portal. The browser address bar shows 'portal.azure.com/#create/Microsoft.StorageAccount-ARM'. The page title is 'Create a storage account'. The navigation tabs include Basics, Advanced, Networking, Data protection, Encryption, Tags, and Review + create (which is the active tab). Below the tabs, there is a link to 'View automation template'. The 'Basics' section contains the following configuration details:

Property	Value
Subscription	Challenge Labs 01
Resource group	CSSClod48020058
Location	East US
Storage account name	sa48020058
Primary service	
Performance	Standard
Replication	Locally-redundant storage (LRS)

The 'Advanced' section contains the following configuration details:

Property	Value
Enable hierarchical namespace	Disabled
Enable SFTP	Disabled
Enable network file system v3	Disabled
Allow cross-tenant replication	Disabled
Access tier	Hot
Enable large file shares	Enabled

The 'Security' section contains the following configuration details:

Property	Value
Secure transfer	Enabled

At the bottom of the wizard, there are three buttons: 'Previous', 'Next', and 'Create'. The 'Create' button is highlighted in blue. A 'Give feedback' link is also visible in the bottom right corner.

sa48020058 - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/e305e6bf-4d96-4a4f-a0ec-f2f4d37c75b4/resourcegroups/CSSClod48020058/providers/Mic...

Microsoft Azure Search resources, services, and docs (G+/)

Home > sa48020058_1738717181863 | Overview

sa48020058 Storage account

Search

Upload Open in Explorer Delete Move Refresh Open in mobile CLI / PS Feedback

Overview

Activity log Tags Diagnose and solve problems Access Control (IAM) Data migration Events Storage browser Storage Mover Partner solutions Data storage Containers File shares Queues Tables Security + networking Networking Front Door and CDN Access keys Shared access signature

Essentials

Resource group (move) CSSClod48020058

Location eastus

Subscription (move) Challenge Labs 01

Subscription ID e305e6bf-4d96-4a4f-a0ec-f2f4d37c75b4

Disk state Available

Performance Standard

Replication Locally-redundant storage (LRS)

Account kind StorageV2 (general purpose v2)

Provisioning state Succeeded

Created 2/5/2025, 6:30:11 AM

Tags (edit) Add tags

Properties Monitoring Capabilities (7) Recommendations (0) Tutorials Tools + SDKs

Blob service

Hierarchical namespace Disabled

Default access tier Hot

Blob anonymous access Disabled

Blob soft delete Enabled (7 days)

Container soft delete Enabled (7 days)

Versioning Disabled

Change feed Disabled

Security

Require secure transfer for REST API operations Enabled

Storage account key access Enabled

Minimum TLS version Version 1.2

Infrastructure encryption Disabled

Networking

sa48020058 - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/e305e6bf-4d96-4a4f-a0ec-f2f4d37c75b4/resourcegroups/CSSClod48020058/providers/Mic...

Microsoft Azure Search resources, services, and docs (G+/)

Home > All resources > sa48020058

sa48020058 Containers Storage account

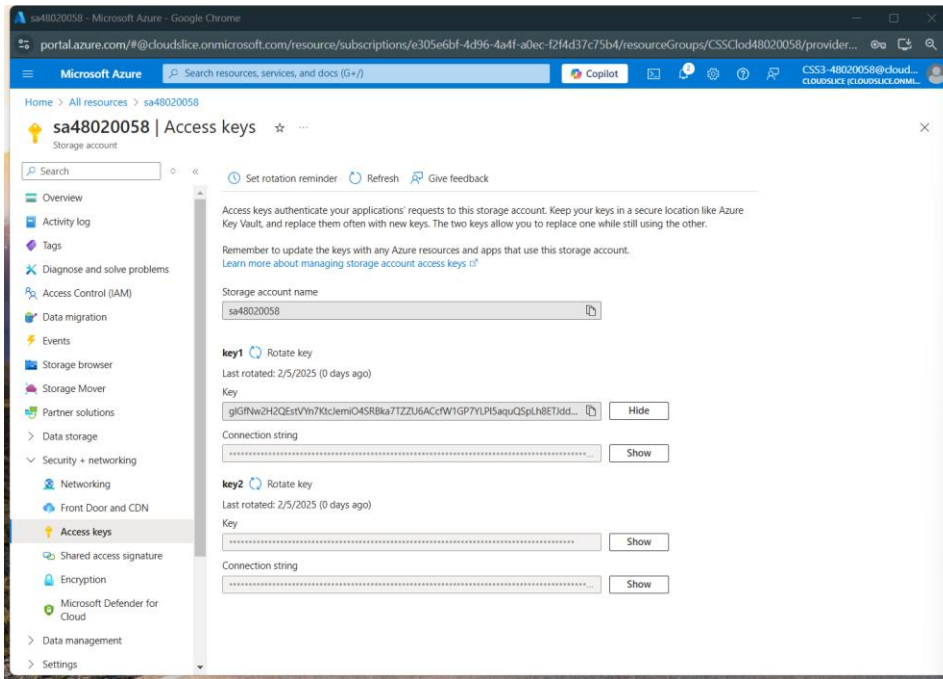
Search

+ Container Change access level Restore containers Refresh Delete Give feedback

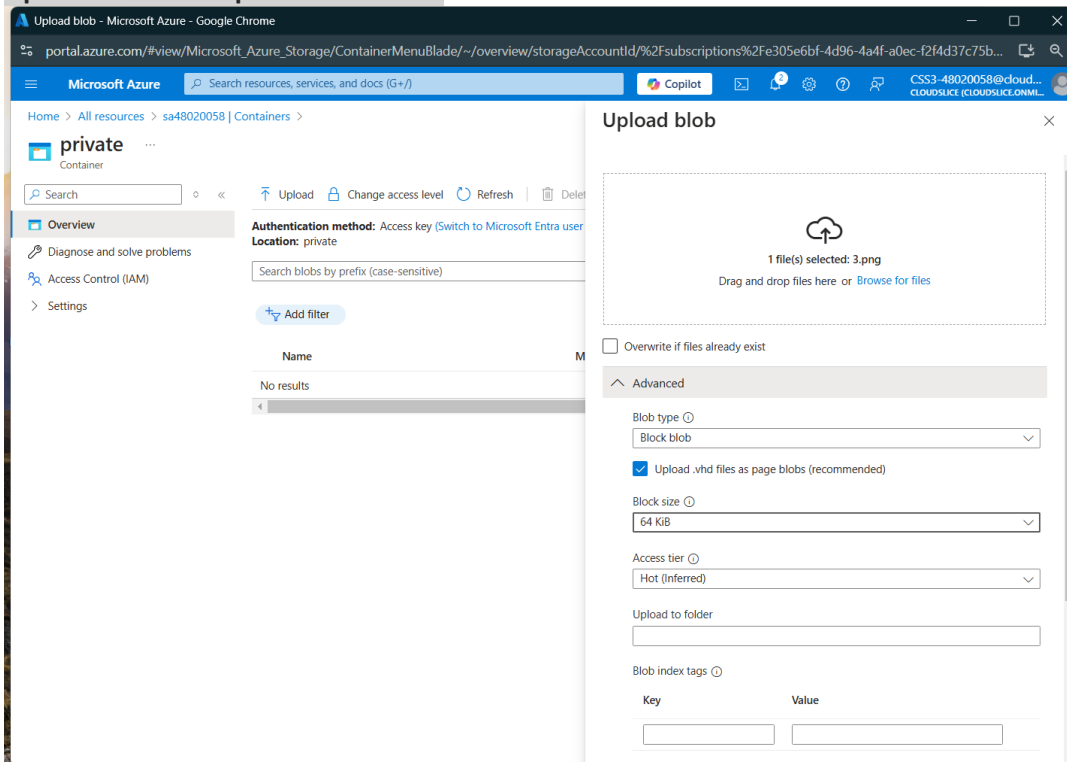
Search containers by prefix Show deleted containers

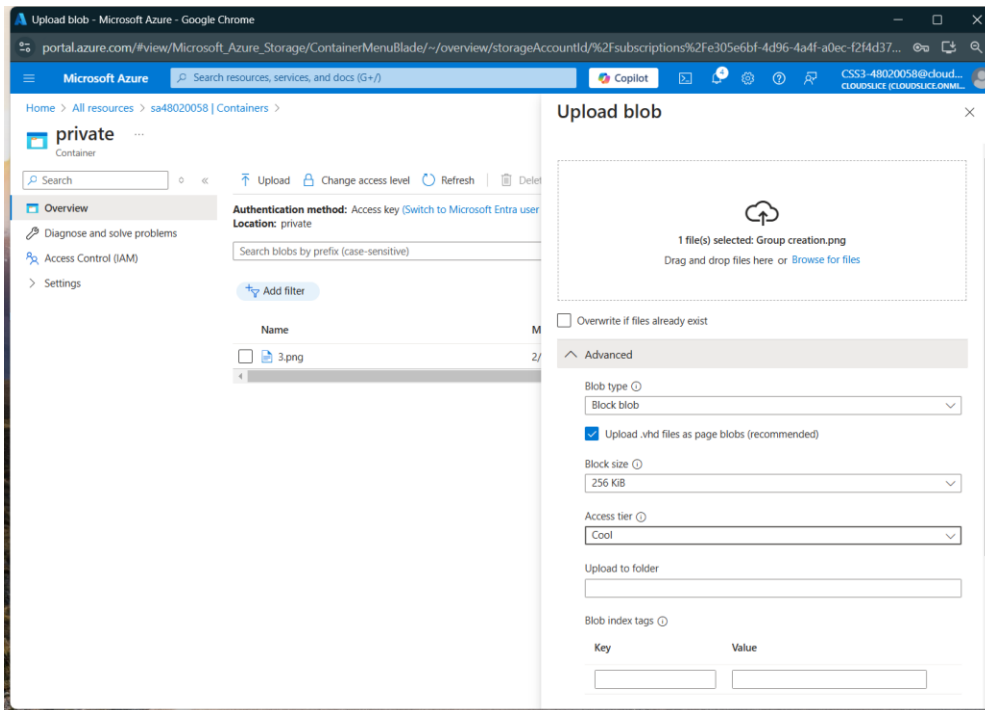
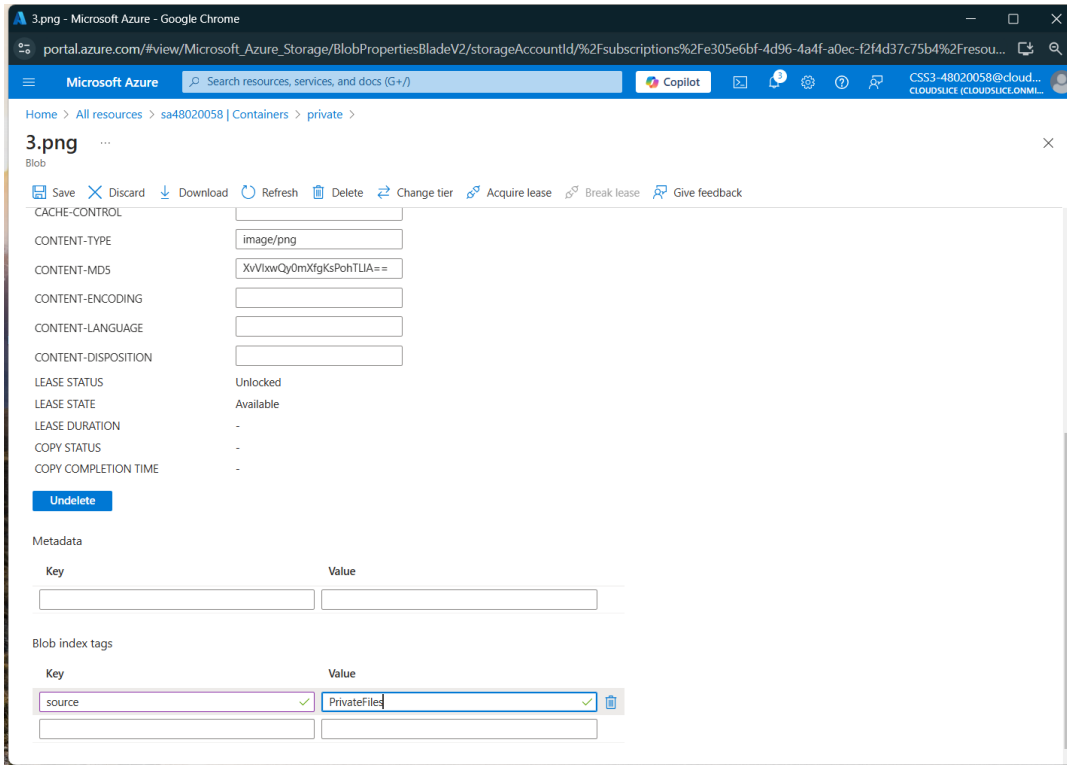
Name	Last modified	Anonymous access level	Lease state	
☐ \$logs	2/5/2025, 6:30:33 AM	Private	Available	...
☐ private	2/5/2025, 6:34:18 AM	Private	Available	...

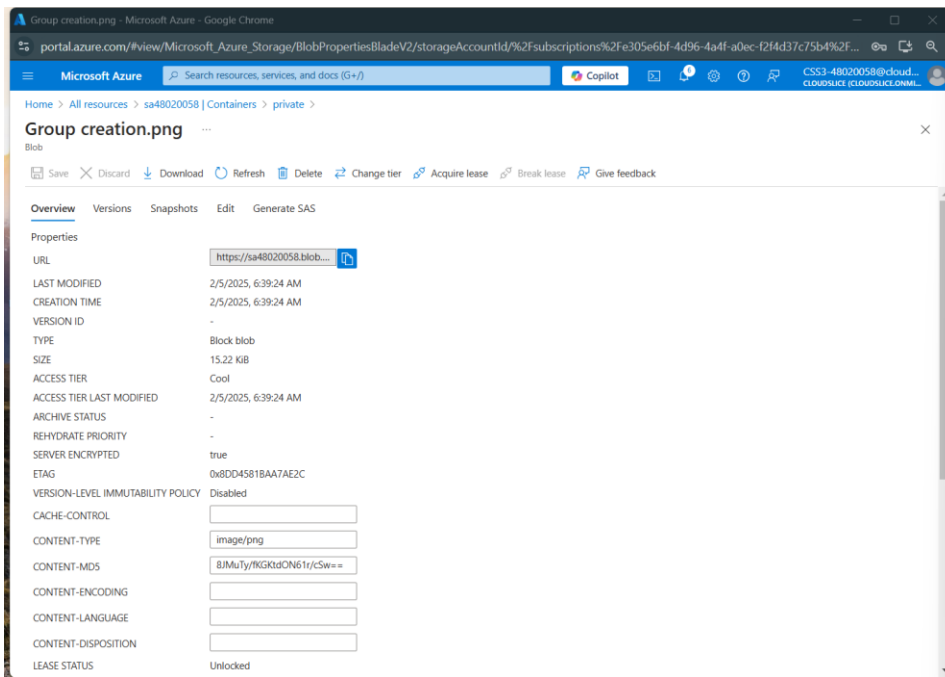
Overview Activity log Tags Diagnose and solve problems Access Control (IAM) Data migration Events Storage browser Storage Mover Partner solutions Data storage Containers File shares Queues Tables Security + networking Networking Front Door and CDN Access keys Shared access signature



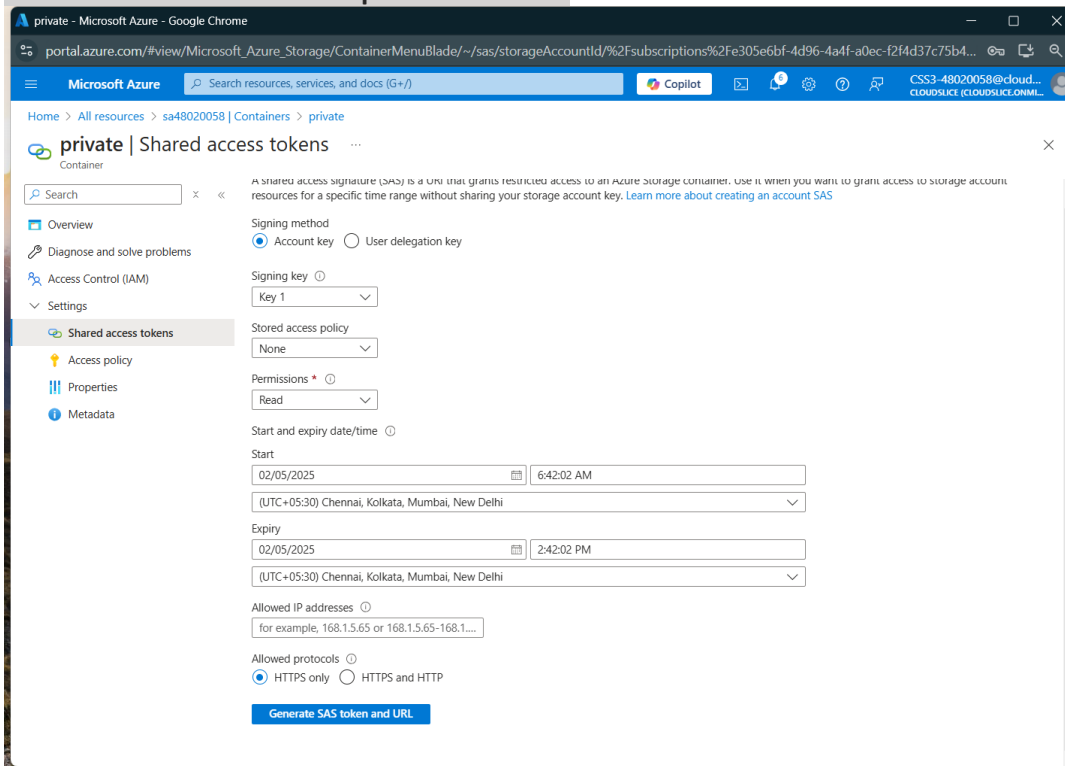
Upload files to the private container

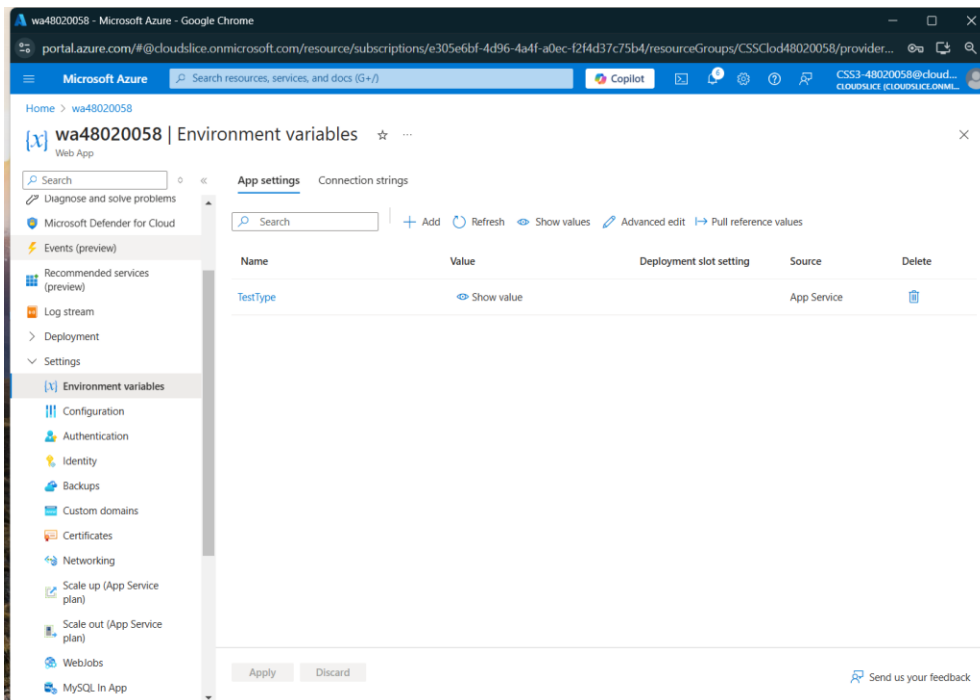
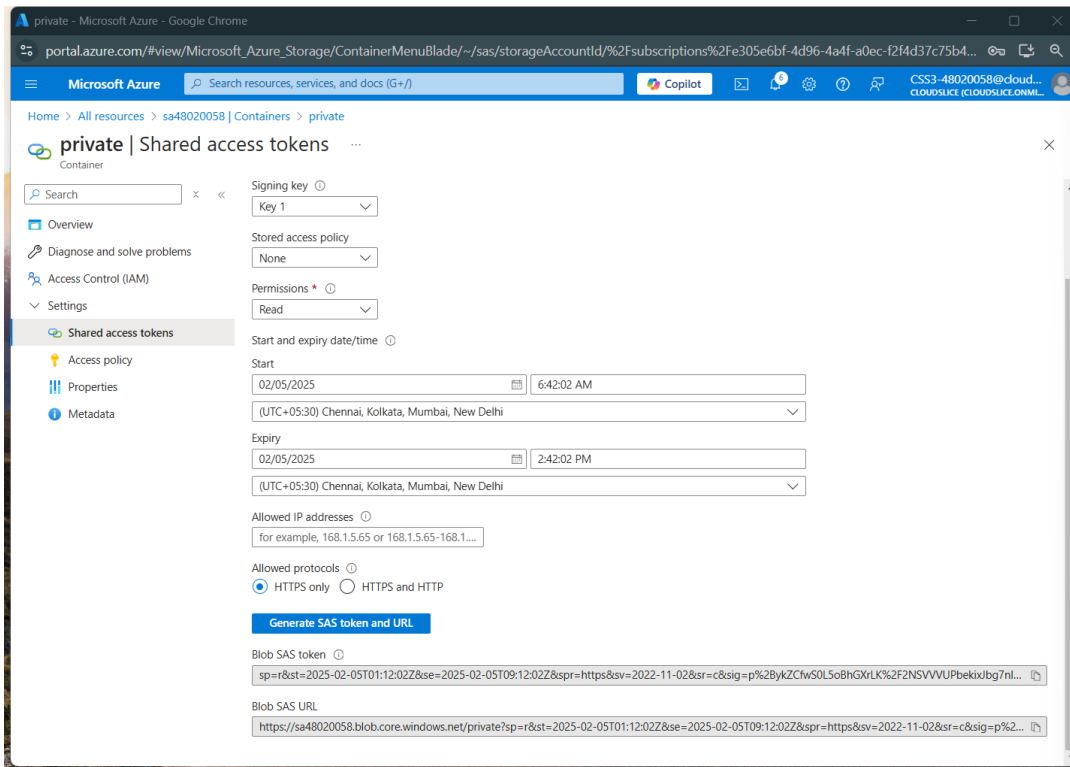


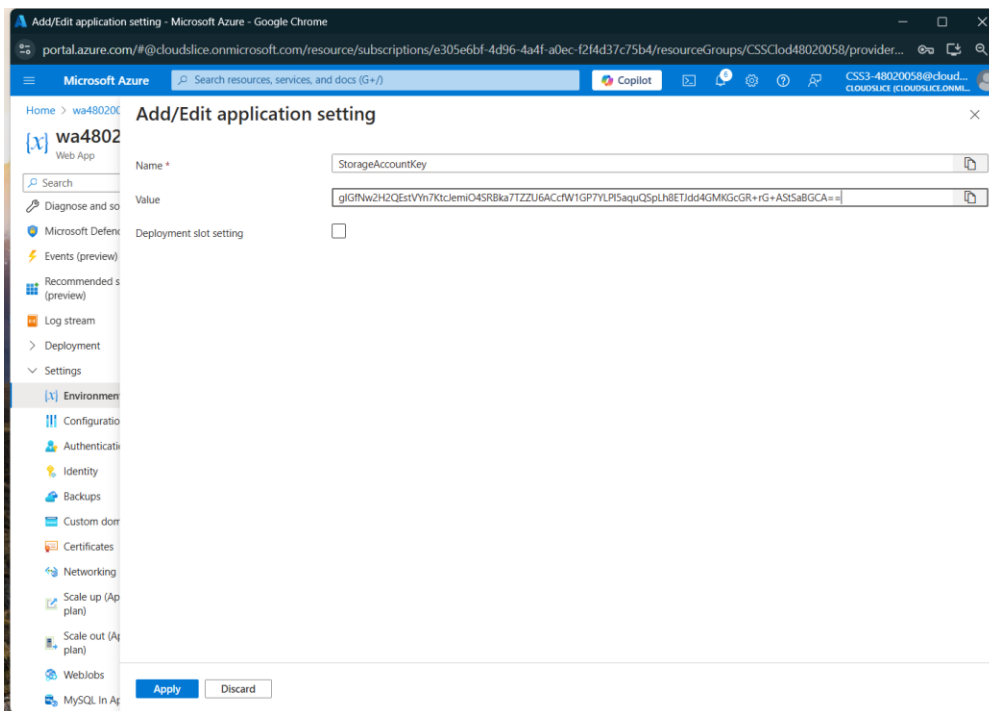
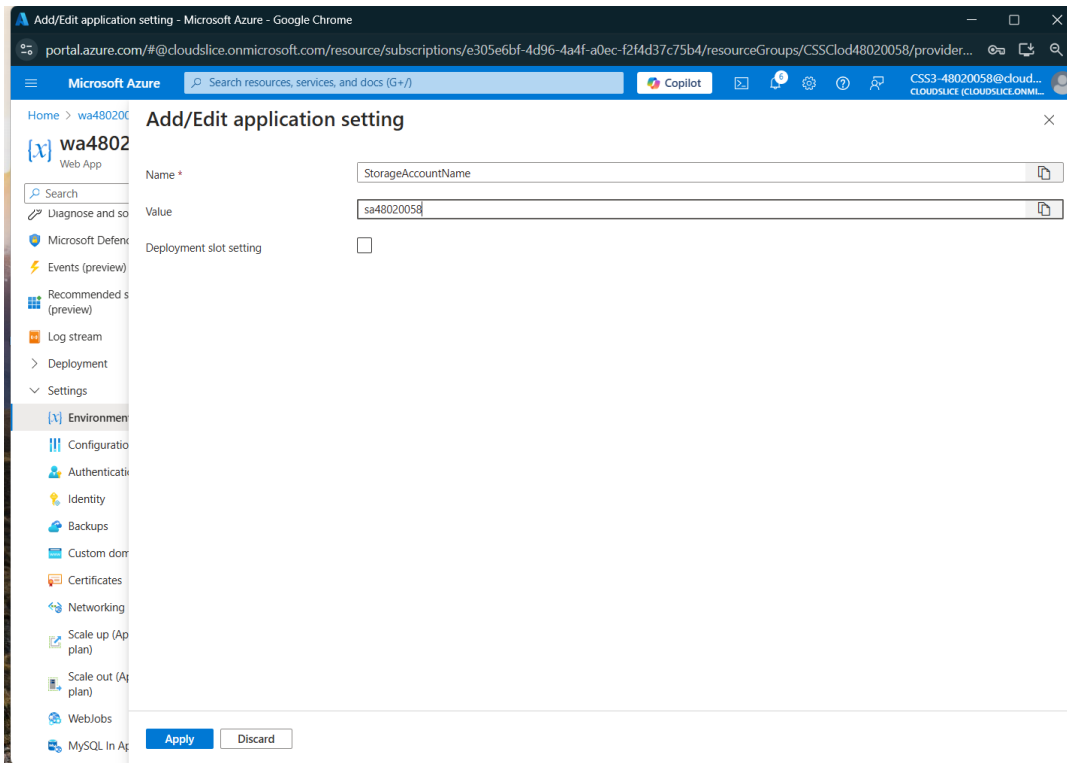


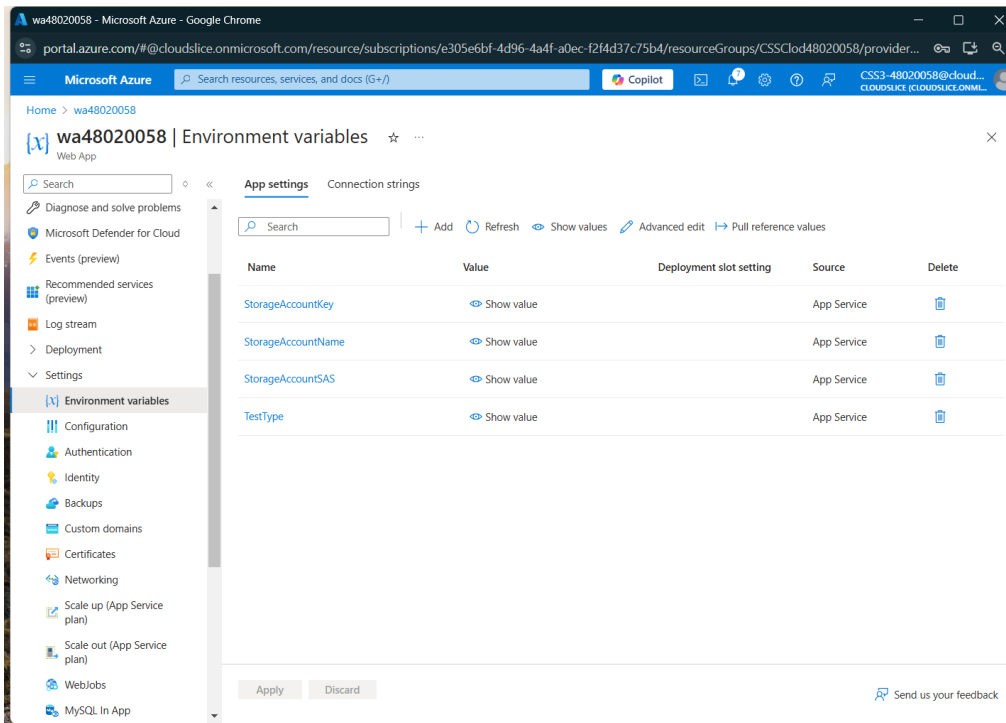
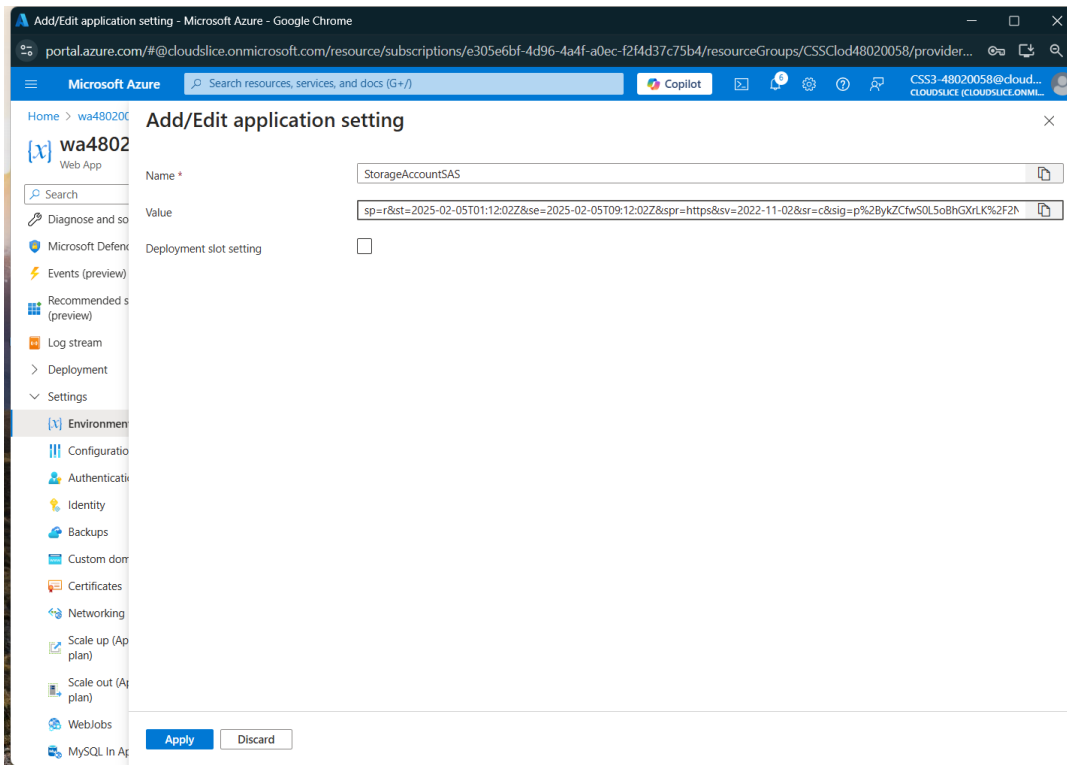


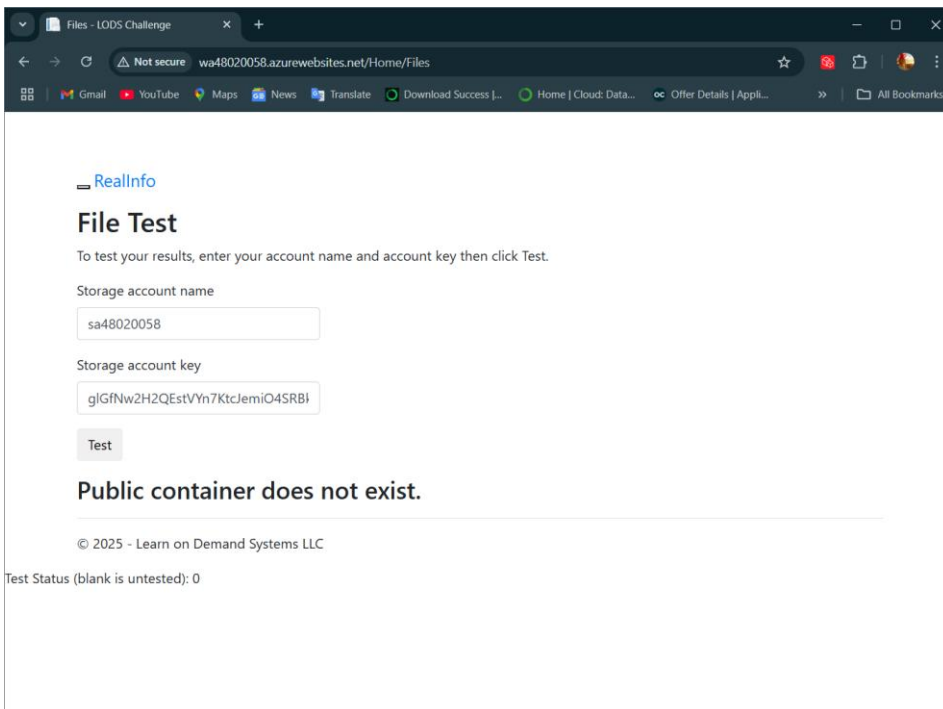
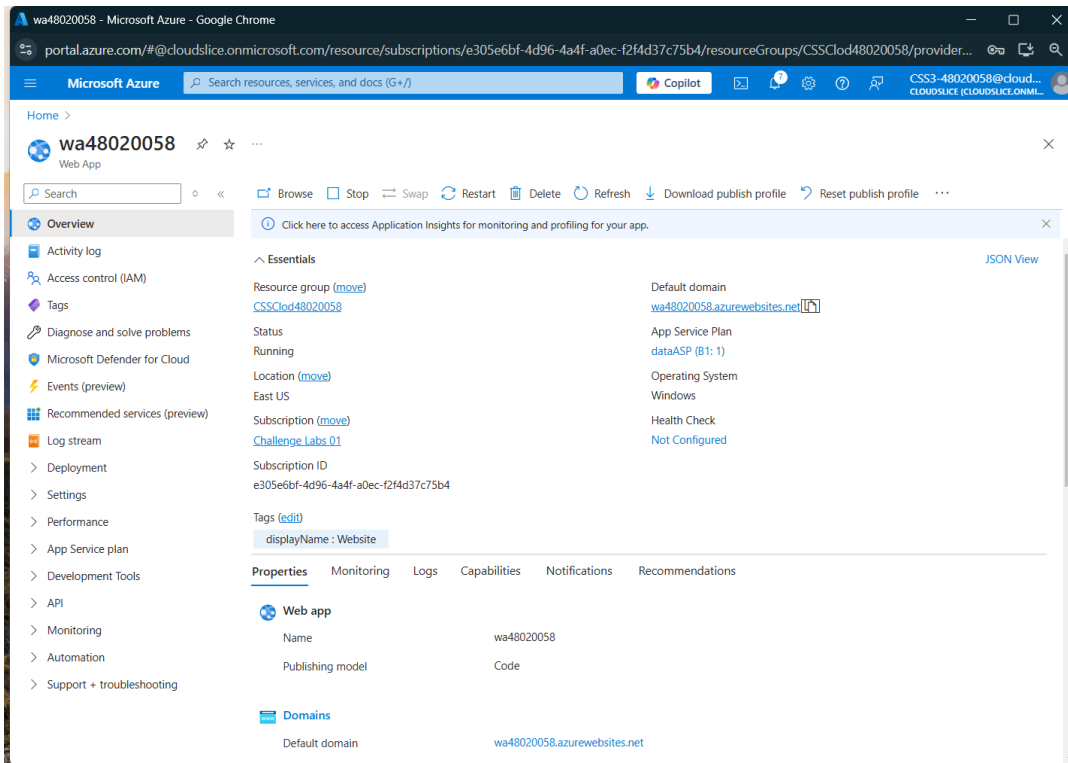
Generate a SAS token for the private container











Outcome:

Successfully verified that blob data remained private and was accessible only via the SAS token, demonstrating secure storage and controlled access management in Azure.

Task10 : Configure a Near Real-Time Metric Alert

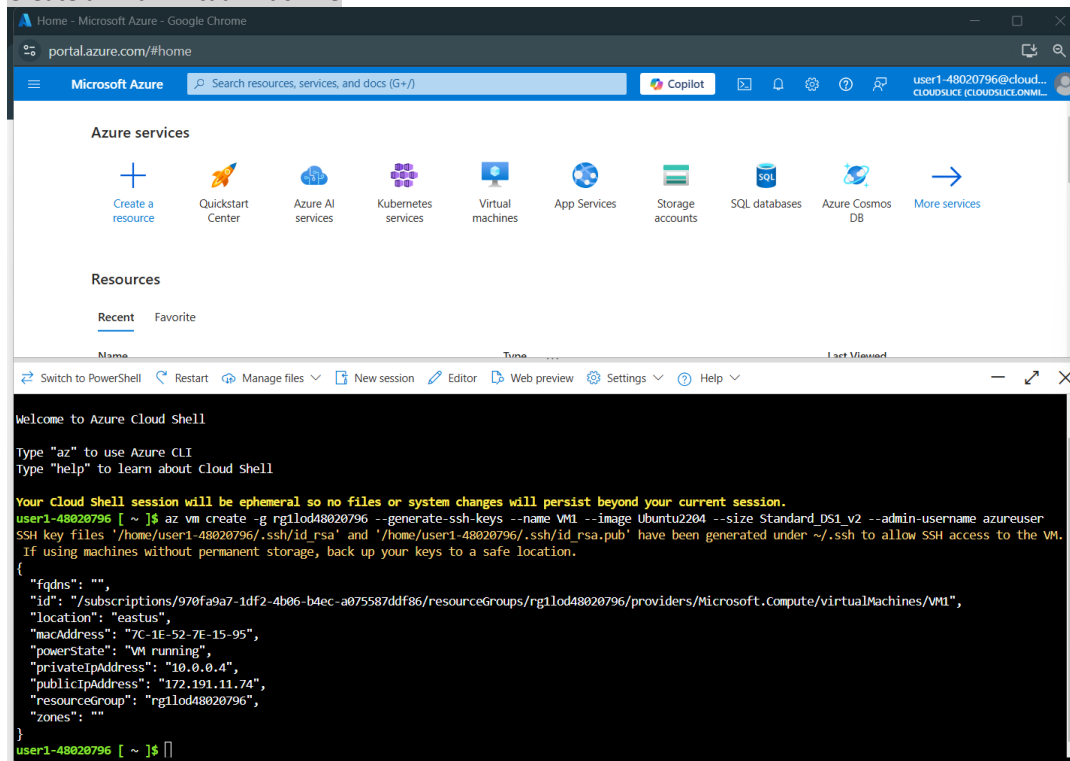
Objective:

Deploy a Linux virtual machine, create an action group for notifications, and configure a near real-time metric alert to monitor performance metrics.

Steps Performed:

- Created a Linux virtual machine to serve as the monitored resource.
- Set up an action group to send email notifications.
- Configured a metric alert to trigger when CPU utilization exceeds 70%.
- Tested the alert by generating CPU load and verified that the alert was received.

Create a Linux virtual machine



The screenshot shows the Azure portal interface in a Google Chrome browser. The top navigation bar includes the Microsoft Azure logo, a search bar, and a Copilot button. The main content area displays 'Azure services' with icons for 'Create a resource', 'Quickstart Center', 'Azure AI services', 'Kubernetes services', 'Virtual machines', 'App Services', 'Storage accounts', 'SQL databases', 'Azure Cosmos DB', and 'More services'. Below this, the 'Resources' section is visible with 'Recent' and 'Favorite' tabs. The bottom part of the image shows the Azure Cloud Shell terminal. It displays a welcome message and instructions. A command is executed to create a Linux VM: `az vm create -g rg1lod48020796 --generate-ssh-keys --name VM1 --image Ubuntu2204 --size Standard_DS1_v2 --admin-username azureuser`. The output shows the VM's details, including its ID, location, MAC address, power state, IP addresses, resource group, and zones.

```
Welcome to Azure Cloud Shell

Type "az" to use Azure CLI
Type "help" to learn about Cloud Shell

Your Cloud Shell session will be ephemeral so no files or system changes will persist beyond your current session.
user1-48020796 [ ~ ]$ az vm create -g rg1lod48020796 --generate-ssh-keys --name VM1 --image Ubuntu2204 --size Standard_DS1_v2 --admin-username azureuser
SSH key files '/home/user1-48020796/.ssh/id_rsa' and '/home/user1-48020796/.ssh/id_rsa.pub' have been generated under ~/.ssh to allow SSH access to the VM.
If using machines without permanent storage, back up your keys to a safe location.
{
  "fqdns": "",
  "id": "/subscriptions/970fa9a7-1df2-4b06-b4ec-a075587ddf86/resourceGroups/rg1lod48020796/providers/Microsoft.Compute/virtualMachines/VM1",
  "location": "eastus",
  "macAddress": "7C-1E-52-7E-15-95",
  "powerState": "VM running",
  "privateIpAddress": "10.0.0.4",
  "publicIpAddress": "172.191.11.74",
  "resourceGroup": "rg1lod48020796",
  "zones": ""
}
user1-48020796 [ ~ ]$
```

```
Home - Microsoft Azure - Google Chrome
portal.azure.com/#home

Microsoft Azure Search resources, services, and docs (G+/) Copilot user1-48020796@cloud... CLOUDSLICE (CLOUDSLICE.ONML...)

Switch to PowerShell Restart Manage files New session Editor Web preview Settings Help

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

azureuser@VM1:~$ sudo apt-get update
Hit:1 http://azure.archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://azure.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu jammy/universe amd64 Packages [14.1 MB]
Get:6 http://azure.archive.ubuntu.com/ubuntu jammy/universe Translation-en [5652 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu jammy/universe amd64 c-n-f Metadata [286 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu jammy/multiverse amd64 Packages [217 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu jammy/multiverse Translation-en [112 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu jammy/multiverse amd64 c-n-f Metadata [8372 B]
Get:11 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [2313 kB]
Get:12 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main Translation-en [386 kB]
Get:13 http://azure.archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packages [2940 kB]
Get:14 http://azure.archive.ubuntu.com/ubuntu jammy-updates/restricted Translation-en [515 kB]
Get:15 http://azure.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1184 kB]
Get:16 http://azure.archive.ubuntu.com/ubuntu jammy-updates/universe Translation-en [290 kB]
Get:17 http://azure.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 c-n-f Metadata [26.4 kB]
Get:18 http://azure.archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 Packages [44.5 kB]
Get:19 http://azure.archive.ubuntu.com/ubuntu jammy-updates/multiverse Translation-en [11.5 kB]
Get:20 http://azure.archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 c-n-f Metadata [440 B]
Get:21 http://azure.archive.ubuntu.com/ubuntu jammy-backports/main amd64 Packages [67.7 kB]
Get:22 http://azure.archive.ubuntu.com/ubuntu jammy-backports/main Translation-en [11.1 kB]
Get:23 http://azure.archive.ubuntu.com/ubuntu jammy-backports/main amd64 c-n-f Metadata [388 B]
Get:24 http://azure.archive.ubuntu.com/ubuntu jammy-backports/restricted amd64 c-n-f Metadata [116 B]
Get:25 http://azure.archive.ubuntu.com/ubuntu jammy-backports/universe amd64 Packages [30.0 kB]
Get:26 http://azure.archive.ubuntu.com/ubuntu jammy-backports/universe Translation-en [16.6 kB]
Get:27 http://azure.archive.ubuntu.com/ubuntu jammy-backports/universe amd64 c-n-f Metadata [672 B]
Get:28 http://azure.archive.ubuntu.com/ubuntu jammy-backports/multiverse amd64 c-n-f Metadata [116 B]
Get:29 http://azure.archive.ubuntu.com/ubuntu jammy-security/main amd64 Packages [2077 kB]
Get:30 http://azure.archive.ubuntu.com/ubuntu jammy-security/main Translation-en [325 kB]
Get:31 http://azure.archive.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [2836 kB]
Get:32 http://azure.archive.ubuntu.com/ubuntu jammy-security/restricted Translation-en [498 kB]
Get:33 http://azure.archive.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [961 kB]
Get:34 http://azure.archive.ubuntu.com/ubuntu jammy-security/universe Translation-en [205 kB]
```

Create an action group

Create action group - Microsoft Azure - Google Chrome

portal.azure.com/#view/Microsoft_Azure_Monitoring_Alerts/CreateActionGroupBlade/subscriptionId/subscriptionName/resourceGroupName/actionGroupName/...

Microsoft Azure Search resources, services, and docs (G+/) Copilot user1-48020796@cloud... CLOUDSLICE (CLOUDSLICE.ONML...)

Home > Monitor > Alerts >

Create action group

Basics Notifications Actions Tags Review + create

An action group invokes a defined set of notifications and actions when an alert is triggered. [Learn more](#)

Project details

Select a subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription

Resource group [Create new](#)

Region

Instance details

Action group name ✓

Display name ✓

The display name is limited to 12 characters

[Review + create](#) [Previous](#) [Next: Notifications >](#)

Email/SMS message/Push/Voice - Microsoft Azure - Google Chrome

portal.azure.com/#view/Microsoft_Azure_Monitoring_Alerts/CreateActionGroupBlade/subscriptionId/subscriptionName/resourceGroupName/actionGroupName/...

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

user1-48020796@cloud...
CLOUDSUICE (CLOUDSUICE.ONML...

Home > Monitor | Alerts >

Create action group ...

BasicsNotificationsActionsTagsReview + create

Choose how to get notified when the action group is triggered. This step is optional.

Notification type ⓘName ⓘSelected ⓘ

Email/SMS message/Push/Voice

Email/SMS message/Push/Voice

Email

Email * ⓘ101562227@georgebrown.ca

SMS (Carrier charges may apply)

Country code ⓘ1

Phone number ⓘ

Azure mobile app notification

Azure account email ⓘ

Voice

Country code ⓘ1

Phone number ⓘ

Enable the common alert schema. [Learn more](#)

YesNo

OK

Review + createPreviousNext: Actions >

Create action group - Microsoft Azure - Google Chrome

portal.azure.com/#view/Microsoft_Azure_Monitoring_Alerts/CreateActionGroupBlade/subscriptionId/subscriptionName/resourceGroupName/actionGroupName/...

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

user1-48020796@cloud...
CLOUDSUICE (CLOUDSUICE.ONML...

Home > Monitor | Alerts >

Create action group ...

BasicsNotificationsActionsTagsReview + create

This is a summary of your action group. Please review to ensure the information is correct and consider [Azure Monitoring Pricing](#) and the [Azure Privacy Statement](#).

Basics

SubscriptionChallenge Labs 05

Resource grouprg1lod48020796

Regionglobal

Action group nameCloud Operations

Display nameCloudOps

Notifications

Notification typeNameSelected

Email/SMS message/Push/VoiceEmail CloudOpsEmail

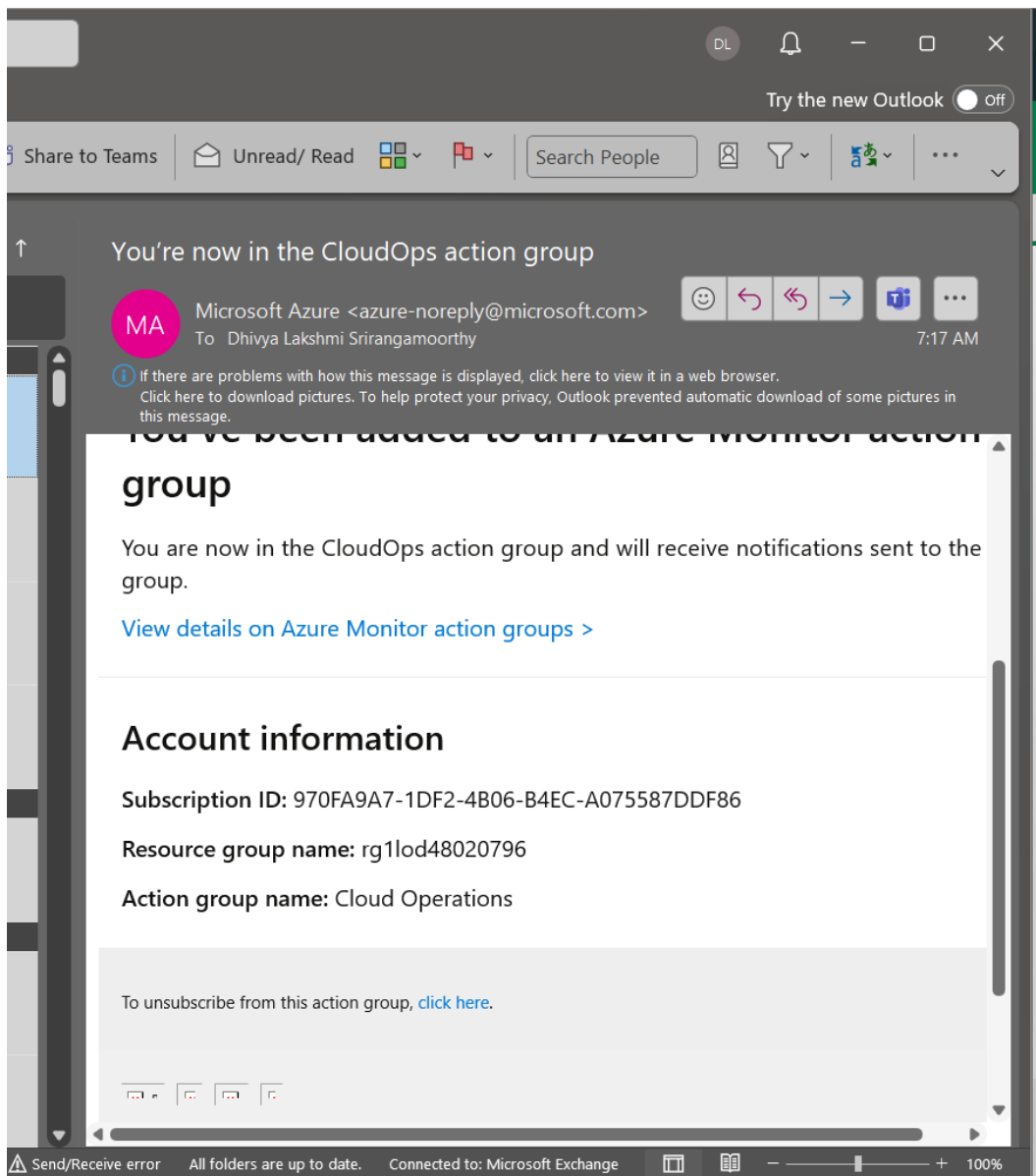
Actions

None

Tags

None

CreatePrevious



Create a virtual machine metric alert

Select a signal - Microsoft Azure - Google Chrome

portal.azure.com/#view/Microsoft_Azure_Monitoring/CreateAlertRuleBlade/scopes~/7%5B%2Fsubscriptions%2F970fa9a7-1df2-4b06-b4ec-a075587ddf86%2Fresour...

Microsoft Azure Search resources, services, and docs (G+)

Copilot

user1-48020796@cloud... CLOUDSLICE (CLOUDSLICE.ONML...

Home > Virtual machines > VM1 | Alerts >

Create an alert rule

Scope Condition Actions Details Tags Review

Configure when the alert rule should trigger by selecting a signal and defining its logic.

Signal name * [See all signals](#)

Select a signal

Signal type: All Signal source: All

Signal name	Signal source
Metrics	
Percentage CPU	Platform metrics

[Review + create](#) [Previous](#) [Next: Actions >](#) [Apply](#) [Cancel](#)

Create an alert rule - Microsoft Azure - Google Chrome

portal.azure.com/#view/Microsoft_Azure_Monitoring/CreateAlertRuleBlade/scopes~/7%5B%2Fsubscriptions%2F970fa9a7-1df2-4b06-b4ec-a075587ddf86%2Fresour...

Microsoft Azure Search resources, services, and docs (G+)

Copilot

user1-48020796@cloud... CLOUDSLICE (CLOUDSLICE.ONML...

Home > Virtual machines > VM1 | Alerts >

Create an alert rule

Configure when the alert rule should trigger by selecting a signal and defining its logic.

Signal name * [See all signals](#)

Alert logic

Alert logic

Threshold type ☒ Static ☐ Dynamic

Aggregation type

Value is

Threshold * %

When to evaluate

Check every

Lookback period

[Add condition](#)

Preview

Preview \$0.10 USD/month

Whenever the average Percentage CPU is greater than 85%

Preview time range: Over the last 6 hours **Time series:** Aggregate

Percentage CPU (Avg): vm1 | 0.32%

[Review + create](#) [Previous](#) [Next: Actions >](#)

Select action groups - Microsoft Azure - Google Chrome

portal.azure.com/#view/Microsoft_Azure_Monitoring/CreateAlertRuleBlade/scopes~//%5B%2Fsubscriptions%2F970fa9a7-1df2-4b06-b4ec-a075587ddf86%2Fresour...

Microsoft AzureSearch resources, services, and docs (G+)

Copilot

user1-48020796@cloud...CLOUDSUICE (CLOUDSUICE.ONML...

Home > Virtual machines > VM1 | Alerts >

Create an alert rule...

ScopeConditionActionsDetailsTagsReview

An action group is a set of actions that can be applied to an alert n

Select actions

☐ Use quick actions
Select one or more

☒ Use action groups
Add an existing ac

☐ None

Action groups

Action group name

No action group selecte

Select action groups

Select up to five action groups to attach to this rule.

+ Create action group

Subscription

Action group name	Resource group	Contains actions	Location
Cloud Operations	rg1lod48020796	1 Email	Global

Review + create

Previous

Next: Details >

Select

Create an alert rule - Microsoft Azure - Google Chrome

portal.azure.com/#view/Microsoft_Azure_Monitoring/CreateAlertRuleBlade/scopes~//%5B%2Fsubscriptions%2F970fa9a7-1df2-4b06-b4ec-a075587ddf86%2Fresour...

Microsoft AzureSearch resources, services, and docs (G+)

Copilot

user1-48020796@cloud...CLOUDSUICE (CLOUDSUICE.ONML...

Home > Virtual machines > VM1 | Alerts >

Create an alert rule...

Signal name

Percentage CPU

Operator

Greater than

Aggregation type

Average

Threshold value

85

Lookback period

5 minutes

Check every

1 minute

Actions

Action group name

Contain actions

Cloud Operations

1 Email

Details

Project details

Subscription

Challenge Labs 05

Resource group

rg1lod48020796

Region

global

Alert rule details

Alert rule name

Percentage CPU greater than 85

Alert rule description

Alert when the average Percentage CPU is greater than 85

Severity

3 - Informational

Enable upon creation

☒

Automatically resolve alerts

☒

Create

Previous

74 | Page

VM1 - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/970fa9a7-1df2-4b06-b4ec-a075587ddf86/resourceGroups/rg1lod48020796/providers/Mic...

Microsoft Azure Search resources, services, and docs (G+/)

user1-48020796@cloud... CLOUDSLICE (CLOUDSLICE.ONMICROSOFT.COM)

Home > Virtual machines > VM1

VM1 | Alerts Virtual machine

Search

View as timeline (preview) Create Set up recommended alerts

New: We value your feedback! Please take a moment to complete a [short survey](#). Your insights will help us improve our services.

Search Add filter More (4)

Total alerts Critical Error Warning Informational Verbose

0 0 0 0 0 0

No grouping

Name	Severity	Affected resource	Alert condition
No alerts found			

Try changing your search or choose a different scope level if you don't see what you're looking for.

Clear filters

Notifications

More events in the activity log → Dismiss all

Alert rule created

Alert rule Percentage CPU greater than 85 successfully created. It might take a few minutes for changes to be shown.

a few seconds ago

Create action group

Action group created successfully

8 minutes ago

Test the near real-time metric alert

VM1 - Microsoft Azure - Google Chrome

portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/970fa9a7-1df2-4b06-b4ec-a075587ddf86/resourceGroups/rg1lod48020796/providers/Mic...

Microsoft Azure Search resources, services, and docs (G+/)

user1-48020796@cloud... CLOUDSLICE (CLOUDSLICE.ONMICROSOFT.COM)

Home > Virtual machines > VM1

VM1 | Alerts Virtual machine

Search

View as timeline (preview) Create Set up recommended alerts Alert rules Action groups Alert processing rules

Switch to PowerShell Restart Manage files New session Editor Web preview Settings Help

```
Last login: Wed Feb 5 01:40:21 2025 from 135.234.203.118
azureuser@VM1:~$ sudo apt-get install stress
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  stress
0 upgraded, 1 newly installed, 0 to remove and 28 not upgraded.
Need to get 18.4 kB of archives.
After this operation, 52.2 kB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu jammy/universe amd64 stress amd64 1.0.5-1 [18.4 kB]
Fetched 18.4 kB in 0s (180 kB/s)
Selecting previously unselected package stress.
(Reading database ... 62674 files and directories currently installed.)
Preparing to unpack .../stress_1.0.5-1_amd64.deb ...
Unpacking stress (1.0.5-1) ...
Setting up stress (1.0.5-1) ...
Processing triggers for man-db (2.10.2-1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
azureuser@VM1:~$
```



```
VM1 - Microsoft Azure - Google Chrome
portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/970fa9a7-1df2-4b06-b4ec-a075587ddf86/resourceGroups/rg1lod48020796/providers/Micr...

Microsoft Azure
Search resources, services, and docs (G+/I) Copilot

Home > Virtual machines > VM1
VM1 | Alerts
Virtual machine

View as timeline (preview) + Create Set up recommended alerts Alert rules Action groups Alert processing rules

Switch to PowerShell Restart Manage files New session Editor Web preview Settings Help

Reading state information... Done
The following NEW packages will be installed:
stress
0 upgraded, 1 newly installed, 0 to remove and 28 not upgraded.
Need to get 18.4 kB of archives.
After this operation, 52.2 kB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu jammy/universe amd64 stress amd64 1.0.5-1 [18.4 kB]
Fetched 18.4 kB in 0s (180 kB/s)
Selecting previously unselected package stress.
(Reading database ... 62674 files and directories currently installed.)
Preparing to unpack .../stress_1.0.5-1_amd64.deb ...
Unpacking stress (1.0.5-1) ...
Setting up stress (1.0.5-1) ...
Processing triggers for man-db (2.10.2-1) ...
Scanning processes...
Scanning linux images...

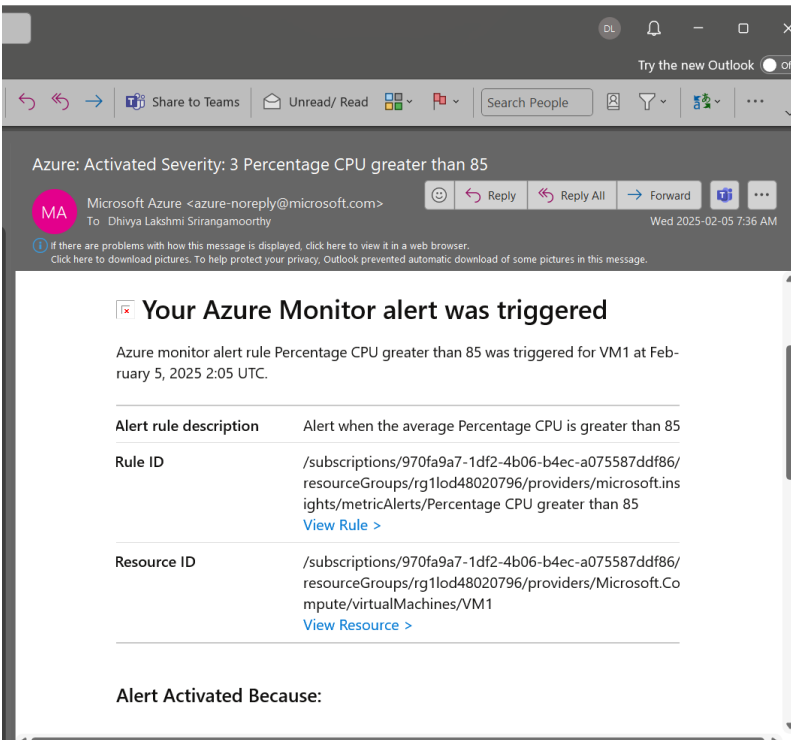
Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
azureuser@VM1:~$ sudo stress --cpu 8 --timeout 480 &
[1] 2308
azureuser@VM1:~$ stress: info: [2310] dispatching hogs: 8 cpu, 0 io, 0 vm, 0 hdd
stress: info: [2310] successful run completed in 480s
[]
```



Outcome:
Successfully implemented near real-time performance monitoring with automated notifications, demonstrating Azure Monitor’s capability to detect and alert resource conditions promptly.